

College Algebra

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2.1 Use a General Strategy to Solve Linear Equations

Learning Objectives

By the end of this section, you will be able to:

- Solve linear equations using a general strategy
- Classify equations
- Solve equations with fraction or decimal coefficients

A **linear equation** is an equation in which the highest power of the variable is 1 (meaning, you won't find a variable with an exponent larger than 1).

Examples: $x + 6 = 10$, $2x + 5 = 20$, $3(x - 8) + 4 = -6x$

Our goal is to solve the equation. A **solution** of an equation is a value of a variable that makes a true statement when substituted into the equation. To solve an equation, we want to "get x alone". We can do this using the following steps.

HOW TO: Solve linear equations using a general strategy.

1. Simplify each side of the equation as much as possible.
 - Use the Distributive Property to remove any parentheses.
 - Combine like terms.
2. Collect all the variable terms on one side of the equation.
 - Use the Addition or Subtraction Property of Equality.
3. Collect all the constant terms on the other side of the equation.
 - Use the Addition or Subtraction Property of Equality.
4. Make the coefficient of the variable term equal to 1.
 - Use the Multiplication or Division Property of Equality.
 - State the solution to the equation.
5. Check the solution.
 - Substitute the solution into the original equation to make sure the result is a true statement.

Example A

Solve: $3(x + 7) - 6 = -9$

$$3(x + 7) - 6 = -9 \quad \text{Distribute 3 into the parentheses}$$

$$3x + 21 - 6 = -9 \quad \text{Combine like terms on the left side (+21 and -6)}$$

$$3x + 15 = -9 \quad \text{Subtract 15 from both sides}$$

$$3x = -24 \quad \text{Divide both sides by 3}$$

$$\boxed{x = -8}$$

Check your answer:

$$3(-8 + 7) - 6 = -9$$

$$3(-1) - 6 = -9$$

$$-3 - 6 = -9$$

$$-9 = -9 \quad \checkmark$$

Example B

Solve: $\frac{2}{3}(3x - 9) + 4 = -6(2x - 5)$

$$\frac{2}{3}(3x - 9) + 4 = -6(2x - 5) \quad \text{Distribute into each set of parentheses}$$

$$2x - 6 + 4 = -12x + 30 \quad \text{Combine like terms on the left side}$$

$$2x - 2 = -12x + 30 \quad \text{Add 12x to both sides to collect variables on the left}$$

$$14x - 2 = 30 \quad \text{Add 2 to both sides to collect constants on the right}$$

$$14x = 32 \quad \text{Divide both sides by 14 to get x alone}$$

$$x = \frac{32}{14} \quad \text{Simplify your answer}$$

$$\boxed{x = \frac{16}{7}}$$

You may check your answer.

Example C

Solve: $4[2 - 3(5x + 9)] = 8(7 - 11x)$

$$4[2 - 3(5x + 9)] = 8(7 - 11x) \quad \text{Work with the innermost parentheses first. Distribute}$$

$$4[2 - 15x - 27] = 8(7 - 11x) \quad \text{Combine like terms in the brackets}$$

$$4[-25 - 15x] = 8(7 - 11x)$$

Distribute on both sides

$$-100 - 60x = 56 - 88x$$

Add $88x$ to both sides to collect variables on the left

$$-100 + 28x = 56$$

Add 100 to both sides to collect constants on right

$$28x = 156$$

Divide both sides by 28 to get x alone

$$x = \frac{156}{28}$$

Simplify your answer

$$\boxed{x = \frac{39}{7}}$$

You may check your answer.

Classifying Equations

All the equations we solved above are called conditional equations. A **conditional equation** is an equation that is true for one or more values of the variable and false for all other values of the variable. For example, in Example A above, the equation $3(x + 7) - 6 = -9$ is only true when we substitute in $x = -8$, and would be false for any other value of x .

Let's see what happens in the next two examples.

Example D

Solve: $3(x - 4) = 3x - 12$

$$3(x - 4) = 3x - 12 \quad \text{Distribute}$$

$$3x - 12 = 3x - 12 \quad \text{Subtract } 3x \text{ on both sides to collect the variables on the left}$$

$$-12 = -12 \quad \text{The variables are eliminated, and the resulting equation is **true**}$$

This means that this equation is true for any value of x . This equation is called an **identity** and its solution is all real numbers.

Example E:

Solve: $7(x + 1) = 7x + 3$

$$7(x + 1) = 7x + 3 \quad \text{Distribute}$$

$$7x + 7 = 7x + 3 \quad \text{Subtract } 7x \text{ on both sides to collect the variables on the left}$$

$$7 = 3$$

The variables are eliminated, and the resulting equation is **false**

This means that this equation is false for any value of x . This equation is called a **contradiction** and it has is no solution.

Solving Equations with Fraction or Decimal Coefficients

For equations with many fractions or decimals, it is often easier to *clear* the fractions/decimals first so we don't need to work with them in each step.

1. To clear the fractions/decimals, multiply both sides of the equation by the LCD of the fractions (or the decimals in their fraction form).
2. Then solve the resulting equation using the steps above.

Example F

Solve: $\frac{1}{2}x = \frac{2}{3}x + 7$

$$\frac{1}{2}x = \frac{2}{3}x + 7$$

Multiply all 3 terms of the equation by the LCD = 6

$$6\left(\frac{1}{2}x\right) = 6\left(\frac{2}{3}x\right) + 6(7)$$

$$3x = 4x + 42$$

Subtract $4x$ on both sides to collect the variables on the left

$$-x = 42$$

Divide both sides by -1 to get x alone

$$\boxed{x = -42}$$

Example G

Solve: $0.05(x + 3) - 0.25x = 5.15$

Notice that the decimals have two decimal places, which means they can be written as fractions over 100: $0.05 = \frac{5}{100}$, $0.25 = \frac{25}{100}$, $5.15 = 5\frac{15}{100}$, so the LCD = 100.

(Decimals with one decimal place would have an LCD of 10, since for example $0.2 = \frac{2}{10}$)

$$0.05(x + 3) - 0.25x = 5.15$$

Multiply all 3 terms of the equation by the LCD = 100

$$100(0.05(x + 3)) - 100(0.25x) = 100(5.15)$$

$$5(x + 3) - 25x = 515$$

Distribute

$$5x + 15 - 25x = 515$$

Combine like terms

$$-20x + 15 = 515$$

Subtract 15 from both sides to get constants on right

$$-20x = 500$$

Divide both sides by -20 to get x alone

$$\boxed{x = 25}$$

Worksheet 2.1 Use a General Strategy to Solve Linear Equations

Solve each equation.

1. $3x + 8 = 2$

7. $4[3 + 6(2x - 5)] = -8$

8. $10x = 2(5x + 3)$

2. $-4x - 11 = 3x + 10$

9. $-3(2x + 4) = -6x - 12$

3. $2(3x - 1) = 5(x + 6)$

10. $\frac{2}{3}x + \frac{1}{4} = \frac{5}{12}$

4. $4(y - 5) + 15 = -9$

11. $\frac{1}{2}(x + 4) = \frac{1}{3}(x - 1)$

5. $\frac{2}{3}(6y - 9) - 1 = y + 8$

12. $\frac{3y-1}{2} + 5 = \frac{1}{4}$

6. $6(2n + 1) - 12 = -3(n - 9)$

13. $0.2x + 1.7 = 0.3$

$$14. 0.05(n + 3) + 0.25 = 1.45$$

2.3 Solve a Formula for a Specific Variable

Learning Objectives

By the end of this section, you will be able to:

- Solve a formula for a specific variable

When you solve an equation you “get x alone”. We can use similar steps to “get a variable alone” in a formula.

Solving an Equation for x $3x + 15 = 27$ Subtract 15 $3x + 15 - 15 = 27 - 15$ $3x = 12$ Divide by 3 $\frac{3x}{3} = \frac{12}{3}$ $x = 4$	vs.	Solving a Formula for x $Ax + By = C$ Subtract By $Ax + By - By = C - By$ $Ax = C - By$ Divide by A $\frac{Ax}{A} = \frac{C - By}{A}$ $x = \frac{C - By}{A}$
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Example A

Solve: $3x - 7y = 8$ for y

$$3x - 7y = 8$$

Subtract $3x$ from both sides

$$3x - 7y - 3x = 8 - 3x$$

$$-7y = 8 - 3x$$

Divide both sides by -7

$$\frac{-7y}{-7} = \frac{8-3x}{-7}$$

$$y = \frac{8-3x}{-7}$$

y is alone

Example B

Solve: $d = rt$ for r

$$d = rt$$

Divide both sides by t

$$\frac{d}{t} = \frac{rt}{t}$$

$$\boxed{\frac{d}{t} = r}$$

r is alone

Example C

Solve: $A = \frac{1}{2}h(b_1 + b_2)$ for b_2

$$A = \frac{1}{2}h(b_1 + b_2) \quad \text{First clear the fraction by multiplying both sides by 2}$$

$$2A = 2\left(\frac{1}{2}h(b_1 + b_2)\right)$$

$$2A = h(b_1 + b_2) \quad \text{Divide both sides by } h$$

$$\frac{2A}{h} = \frac{h(b_1 + b_2)}{h}$$

$$\frac{2A}{h} = b_1 + b_2 \quad \text{Subtract } b_1 \text{ from both sides}$$

$$\frac{2A}{h} - b_1 = b_1 + b_2 - b_1$$

$$\boxed{\frac{2A}{h} - b_1 = b_2}$$

b_2 is alone

Worksheet 2.3 Solve a Formula for a Specific Variable

1. Solve $A = \frac{1}{2}bh$ for h

2. Solve $a + b + c = 180$ for b

3. Solve $P = 2L + 2W$ for W

4. Solve $V = \pi r^2 h$ for h

5. Solve $A = P + Prt$ for r

6. Solve $6x + 2y = 5$ for y

7. Solve $x - 3y = -8$ for y

2.4 Solve Mixture and Uniform Motion Applications

Learning Objectives

By the end of this section, you will be able to:

- Solve coin, ticket, and stamp word problems
- Solve uniform motion applications

Coin, Ticket, and Stamp Word Problems

We will use the following table to help us organize the information given to us. Then, from the last column, “Total Value,” we will create an equation to solve.

	Number	X	Value (\$)	=	Total Value (\$)
Type	Number		Value		Total Value

Example A

Julie has \$3.00 worth of quarters and dimes in her pocket. The number of quarters is two times the number of dimes. How many of each type of coin does Julie have?

Step 1: Fill in the table based on the information given in the problem.

Since we are given more information about the number of quarters than of dimes, start with naming dimes with a variable:

- Dimes = d
- Quarters = $2d$

	Number	×	Value (\$)	=	Total Value (\$)
Type	Number		Value		Total Value
Quarters	$2d$		0.25		

Dimes	d	0.10	
			3.00

Step 2: Multiply each Number $\times$ Value in the table, then create the equation from the last column of the table.

	Number	$\times$	Value (\$)	=	Total Value (\$)
Type	Number		Value		Total Value
Quarters	2d		0.25		0.50d
Dimes	d		0.10		0.10d
					3.00

$$0.50d + 0.10d = 3.00$$

Step 3: Solve the equation.

$$0.50d + 0.10d = 3.00$$

$$0.60d = 3.00$$

$$d = 5$$

Step 4: State your answer.

- Dimes = $d = 5$
- Quarters = $2d = 2(5) = 10$

Julie has 5 dimes and 10 quarters.

Example B

Bobby paid \$14.00 for stamps. The number of 49-cent stamps was four less than two times the number of 35-cent stamps. How many of each stamp did he buy?

Steps 1&2: Fill in the table and create the equation.

Since we are given more information about the number of 49-cent stamps than of 35-cent ones, start with naming the 35-cent stamps with a variable:

- 35-cent = x
- 49-cent = $2x - 4$

	Number	×	Value (\$)	=	Total Value (\$)
Type	Number		Value		Total Value
49-cent	$2x - 4$		0.49		$0.49(2x - 4)$
35-cent	x		0.35		$0.35x$
					14.00

$$0.49(2x - 4) + 0.35x = 14.00$$

Step 3: Solve the equation.

$$0.49(2x - 4) + 0.35x = 14.00$$

$$0.98x - 1.96 + 0.35x = 14.00$$

$$1.33x - 1.96 = 14.00$$

$$1.33x = 15.96$$

$$x = 12$$

Step 4: State your answer.

- 49-cent = $2x - 4 = 2(12) - 4 = 20$
- 35-cent = $x = 12$

Bobby bought twenty 49-cent stamps and twelve 35-cent stamps.

Example C

Molly sold 80 tickets to her school's fundraising event for a total revenue of \$229. Children's tickets cost \$2 each and adult tickets cost \$5 each. How many of each type of ticket did Molly sell?

Steps 1&2: Fill in the table and create the equation.

Since we know *number of children's tickets + number of adult's tickets = 80*

We can solve this equation for number of children's tickets:

$$\text{number of children's tickets} = 80 - \text{number of adult's tickets}$$

Now we have more information about the number of children's tickets, so name the number of adult's tickets with a variable:

- Number of adult's tickets = x
- Number of children's tickets = $80 - x$

	Number	×	Value (\$)	=	Total Value (\$)
Type	Number		Value		Total Value
Adult's	x		5		$5x$
Children's	$80 - x$		2		$2(80 - x)$
					229

$$5x + 2(80 - x) = 229$$

Step 3: Solve the equation.

$$5x + 2(80 - x) = 229$$

$$5x + 160 - 2x = 229$$

$$3x + 160 = 229$$

$$3x = 69$$

$$x = 23$$

Step 4: State your answer.

- Adult's = $x = 23$
- Children's = $80 - x = 80 - 23 = 57$

Molly sold 23 adult's tickets and 57 children's tickets.

Uniform Motion Applications

Uniform motion applications will have us solve problems about objects traveling at constant speeds. For example, two cars traveling at different speeds or two friends walking towards one another.

We will use similar steps as the problems above, and use the following table:

	Rate	×	Time	=	Distance
Object/Person	Rate		Time		Distance

Example D

Adam and his brother Ricky are both driving from their home to the beach. Adam drives 10 miles per hour faster than Ricky, so Adam completes the drive in 2 hours whereas it takes Ricky 2.5 hours. Find each of their driving speeds.

Step 1: Fill in the table based on the information given in the problem.

Since we are given more information about Adam's speed than Ricky's, start with naming Ricky's speed with a variable:

- Ricky's speed = x
- Adam's speed = $x + 10$

	Rate	×	Time	=	Distance
Object/Person	Rate		Time		Distance
Ricky	x		2.5		
Adam	$x + 10$		2		

Step 2: Multiply each Rate $\times$ Time in the table, then create the equation from the last column of the table.

In this problem, since they doing the same drive from home to the beach, their distances will equal each other.

	Rate	×	Time	=	Distance
Object/Person	Rate		Time		Distance
Ricky	x		2.5		$2.5x$
Adam	$x + 10$		2		$2(x + 10)$

$$2.5x = 2(x + 10)$$

Step 3: Solve the equation.

$$2.5x = 2(x + 10)$$

$$2.5x = 2x + 20$$

$$0.5x = 20$$

$$x = 40$$

Step 4: State your answer.

- Ricky's speed = $x = 40$
- Adam's Speed = $x + 10 = 40 + 10 = 50$

Ricky was driving 40 mph and Adam was driving 50 mph.

Example E

Nicole and her friend Rose live 125 miles apart. They decide to meet for lunch at a restaurant between their houses. Nicole drove for one hour and Rose drove for one and a half hours to get to the restaurant. Nicole's average speed was five mph faster than Rose's. What were each of their average speeds?

Step 1: Fill in the table based on the information given in the problem.

Since we are given more information about Nicole's speed than Rose's, start with naming Rose's speed with a variable:

- Rose's speed = x
- Nicole's speed = $x + 5$

	Rate	×	Time	=	Distance
Object/Person	Rate		Time		Distance
Rose	x		1.5		
Nicole	$x + 5$		1		
					125

Step 2: Multiply each Rate $\times$ Time in the table, then create the equation from the last column of the table.

In this problem, the distance Rose traveled plus the distance Nicole traveled must total 125 miles.

	Rate	×	Time	=	Distance
Object/Person	Rate		Time		Distance
Rose	x		1.5		$1.5x$
Nicole	$x + 5$		1		$1(x + 5)$
					125

$$1.5x + 1(x + 5) = 125$$

Step 3: Solve the equation.

$$1.5x + 1(x + 5) = 125$$

$$1.5x + x + 5 = 125$$

$$2.5x + 5 = 125$$

$$2.5x = 120$$

$$x = 48$$

Step 4: State your answer.

- Rose's speed = $x = 48$
- Nicole's Speed = $x + 5 = 48 + 5 = 53$

Rose was driving 48 mph and Nicole was driving 53 mph.

Worksheet 2.4 Solve Mixture and Uniform Motion Applications

1. Andy has \$22.25 worth of quarters and nickels in his piggy bank. The number of nickels is five more than three times the number of quarters. How many of each type of coin does he have?
2. Sam paid \$27.34 for stamps. The number of 73-cent stamps was two less than four times the number of 68-cent stamps. How many of each stamp did he buy?
3. On a certain day, a museum sold 120 tickets for a total revenue of \$2160. Children's tickets cost \$8 each and adult tickets cost \$23 each. How many of each type of ticket did the museum sell?
4. Cassie and Dylan are riding their bikes the total length of their city park's bike trail. Cassie rides 2 miles per hour faster than Dylan, so Cassie completes the drive in 3 hours whereas it takes Dylan 4 hours. Find each of their driving speeds.
5. Two trains leave the train station in New York at the same time. One travels north towards Boston and the other travels south towards Washington D.C. The train traveling north travels at 80 mph and the train traveling south travels at 75 mph. How long will it take until they are 310 miles apart?
6. Two friends who live 260 miles apart decide to meet at a restaurant between their houses for dinner. Tom drives for one and a half hours and Ryan drives for 2 hours. Tom drives 10 mph faster than Ryan. Find each of their speeds.

2.5 Solve Linear Inequalities

Learning Objectives:

In this section, you will:

- Graph inequalities on the number line
- Solve linear inequalities
- Translate words to an inequality and solve
- Solve applications with linear inequalities

Any problem containing any of the following symbols is an inequality: $<$, $>$, $\leq$, $\geq$

Since problems like this give multiple solutions in an interval, we can use a number line or interval notation to write the solution.

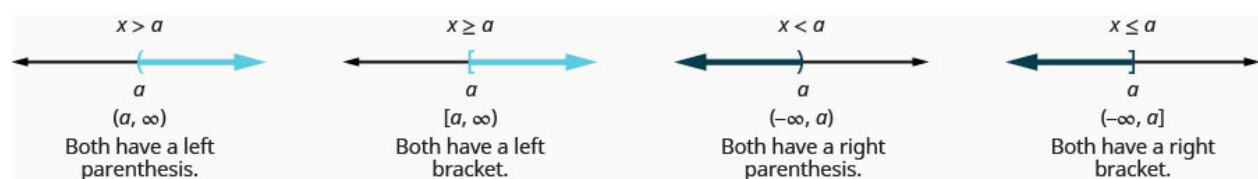
In order to work with inequalities, we use some conventions

- $<$ or $>$ are considered open and use $($ or $)$. The value listed is not included in the solution
- $\leq$ or $\geq$ are considered closed and use $[$ or $]$. The value listed is included in the solution

For an interval, we write “lower limit, upper limit” with the appropriate parenthesis or bracket. One useful way to think about it is the lower limit is the floor, the upper limit is the ceiling. You are above the floor and below the ceiling.

If we are only given one limiting value, that means the other end goes to positive or negative infinity. Since infinity is not a number but an idea, we use a parenthesis.

Inequalities, Number Lines, and Interval Notation



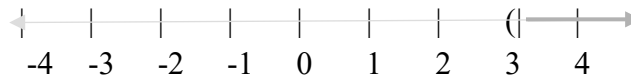
Point: Always read the inequality from the variable

Example A

Draw on a number line and write with interval notation: $x > 3$

For the interval, we see we need to be larger than 3, meaning that must be the floor/lower bound. The 3 is not included, so we use a parenthesis. Since we are never told how far up to go, it can go to infinity.

$$(3, \infty)$$

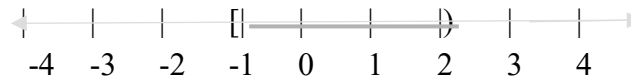


Example B

Draw on a number line and write with interval notation: $-1 \leq x < 2$

For the interval, we see we have to be at -1 or above, but not get bigger than 2. This makes the lower limit -1 and the upper limit 2. Since the lower limit includes the or equal to, it will get a bracket. The upper limit does not have an equal to, so it will get a parenthesis.

$$[-1, 2)$$



A linear inequality is just like a linear equation, except it has an inequality symbol instead of an equal sign. That means to solve linear inequalities, we solve them similar to equalities for one piece:

If you multiply both sides by a negative or divide both sides by a negative, we flip the sign.

Example C

Solve the inequality, graph the solution on the number line, and write the solution in interval notation:
 $2x + 3 > 7$

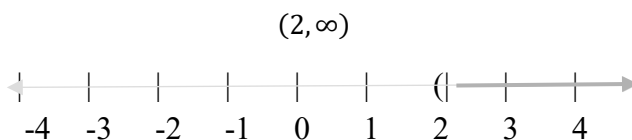
$$2x > 4$$

Subtract 4 from both sides of the inequality

$$x > 2$$

Divide by sides by two

Since we are told to be above 2, that would be the floor/lower limit. We are not able to include the 2 since there is no “or equal to”, so we use a parenthesis. We are not given an upper limit, so it goes to infinity

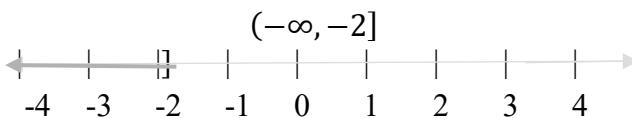


Example D

Solve the inequality, graph the solution on the number line, and write the solution in interval notation:
 $3(x - 2) \leq 8x + 4$

$3x - 6 \geq 8x + 4$	Distribute the left side
$-5x \geq 10$	Subtract $8x$ from both sides and add 6 to both sides
$x \leq -2$	Divide by -5. Since we divided by a negative, we flip the sign.

Since we are told to not get above -2, that would be the ceiling/upper limit. We are able to include the 2 since there is an “or equal to”, so we use a bracket. We are not given an lower limit, so it goes to negative infinity



Translate to an inequality and solve

When translating words to math, we have to know the key phrases to look for. Much like how “is” means equal, we have phrases for inequalities.

$>$	$\geq$	$<$	$\leq$
is greater than	is greater than or equal to	is less than	is less than or equal to
is more than	is at least	is smaller than	is at most
is larger than	is no less than	has fewer than	is no more than
exceeds	is the minimum	is lower than	is the maximum

Example E

Translate to an inequality:

- 2 times an x is more than the product of 3 and 4 more than a number x

$$-2x > 3(x + 4)$$

- 7 subtracted from a number n is at most 12

$$n - 7 \leq 12$$

Example F

Translate to an inequality and solve:

Spencer has 408 movies in his collection. In order to store them, he found shelves that hold 15 DVDs each. How many shelves will he need to buy?

x = Shelves

Define the variable

$$15x \geq 408$$

Write the equation, noting that each shelf holds 15 DVD's, so their product is the total storage and it must hold at least the total amount of DVD's.

$$x \geq 27.2 = 28$$

Round up to the next whole number

Worksheet 2.5: Solve Linear Inequalities

Graph the inequality:

1) $x < 3$

2) $x \geq -1$

3) $-3 < x \leq 0$

Solve the inequality, writing the solution in interval notation:

4) $2x > 5$

8) $7(x + 4) > 30$

12) $7 < -(-k - 3) + 2$

5) $x + 4 \leq 6$

9) $-3 < \frac{-x}{2}$

13) $-(x + 1) - 4x > x - 3$

6) $-4p < 12$

10) $1 - \frac{3}{2}x \geq x - 4$

14) $20 - 2x > 2(x + 2) + 4x$

7) $1 - n \geq 7 + 3n$

11) $3 - 2(x - 4) > 1$

15) $5(x + 1) \geq 8(1 - x)$

Translate to an inequality and solve, writing in interval notation:

16) Seven more than x is more than 12

17) The product of 5 and v is no more than the difference of v and 6

18) 15 added to x is at least -15

19) A taxi charges a flat rate of \$2.10 and \$0.65 per mile. If Anna only has a \$20 bill, how far can she get?

20) Selling sunglasses earns Erin \$12 a pair. If her kiosk has a rental price of \$230 a week, how many sunglasses must she sell to pay her rent?

21) If it takes Brittany 13 minutes to produce a tray of 8 cookies, when will her order of 112 cookies be completed.

2.6 Solve Compound Inequalities

Learning Objectives:

In this section, you will:

- Solve compound inequalities with “and”
- Solve compound inequalities with “or”
- Solve applications with compound inequalities

Continuing our work with inequalities, we now want to cover problems that have multiple conditions.

Compound Inequality

Compound Inequality

A **compound inequality** is made up of two inequalities connected by the word “and” or the word “or.”

We use “and” and “or” in the same way we do in the English language. And represents needing to meet both conditions. Or represents wanting any conditions to be met.

And is an intersection of parts. Or is a union of separate parts.

For a compound inequality:

- And will be one region that is shared between the inequalities. The solution needs to meet every criterion from the inequalities.
- OR will be two separate regions that are not shared between the inequalities. The solution only needs to meet one criterion from the inequalities. We will use $\cup$ between the intervals to represent union, or joining the regions together for a solution.

Example A

Draw on a number line and write with interval notation: $x > 1$ and $x < 4$

For the interval, we need to be above 1, so that is the lower limit. We also need to be below 4, so that is the upper limit.

(1,4)



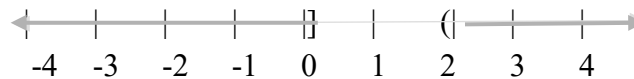
-4 -3 -2 -1 0 1 2 3 4

Example B

Draw on a number line and write with interval notation: $x > 2$ or $x \leq 0$

For the interval, notice how the two regions are separated. We want to be bigger than the large number, and smaller than the small number. This means we will have two separate intervals.

$$(-\infty, 0] \cup (2, \infty)$$



To solve a compound problem, we can either solve each part separately, or if joined, we use our normal method of solving linear inequalities remembering the main rule of math:

What we do to one side of the equation/inequality, we do to the other.

That idea holds no matter how many sides the inequality has.

Example C

Solve the inequality, graph the solution on the number line, and write the solution in interval notation:

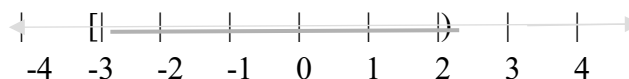
$$-3 \leq 2x + 3 < 7$$

$$-6 \leq 2x < 4 \quad \text{Subtract 3 from all side}$$

$$-3 \leq x < 2 \quad \text{Divide all sides by 2}$$

Since we are told to be at or above -3, that would be the floor/lower limit. We are told to not get above 2, so that is the ceiling/upper limit.

$$[-3, 2)$$



Example D

Solve the inequality, graph the solution on the number line, and write the solution in interval notation:

$$-\frac{1}{2}x > 2 \text{ or } 3x - 1 > 2$$

$$x < -4$$

Multiply the first inequality by -2

$$3x > 3$$

Add one to both sides of the second inequality

$$x > 1$$

Divide the second inequality by 3

$$x < -4 \text{ or } x > 1$$

Write both solutions together

Each of these is a sperate statement. The first needs to be under -4, so that is the upper limit with no lower limit given. The second needs to be above 1, so that is a lower limit with no upper limit given. Neither have an "or equal to", so we use a parentheses for both.

$$(-\infty, -4] \cup (1, \infty)$$



Example E

Translate to an inequality: Rasheeda is redoing her kitchen. She is willing to spend between \$730 and \$920 on the countertops. If the quarts countertop is \$54 per square foot, how much square footage can she afford?

$$x = \text{square footage}$$

Define your variable

$$730 \leq 54x \leq 920$$

Since the question relates to money, we set the cost for the countertop, \$54 per square foot and x square feet, between the two dollar figures.

$$13.5 \leq x \leq 17$$

Divide all sides of the inequality by 54

$$[13.5, 17]$$

Write in interval notation

7 subtracted from a number n is at most 12

$$n - 7 \leq 12$$

Worksheet 2.6: Solve Compound Inequalities

Solve the compound inequality with “and”:

1) $x > 2$ and $x < 3$

2) $6x - 3 < 9$ and $2x + 7 \geq 3$

3) $3x < 15$ and $-2x < 4$

4) $4x - 7 < 9$ and $5x + 8 \geq 3$

5) $2(3x + 1) \leq 20$ and $4(x - 1) < 2$

6) $r + 8 \geq 9$ and $\frac{x}{7} \leq 1$

7) $-3 \leq \frac{p}{2} < 0$

8) $-6 < 9 + 3y < 7$

Solve the compound inequality with “or”:

9) $6 - n > n$ or $3n - 1 < n + 13$

10) $3x + 2 < -2 + 7x$ or $8x - 4 \leq 3x - 4$

11) $4x - 5 > 7$ or $4x - 5 < -9$

12) $5m - 5 > 7m$ or $3m + 5 \leq m - 1$

13) $6 + 7c < 6c - 5$ or $3c - 7 < 5 + 6c$

14) $2x + 7 \geq 27$ or $3 + 3x \leq 30$

Translate to an inequality and solve, writing in interval notation:

15) Six more than a number is either more than ten or less than four

16) Nine less than a number is no more than seven and is no less than 4.

17) The sum of five times a number and 3 is between zero and eighteen.

18) Mike was deciding on a tip to give. He believes he should leave between 10% and 20%.
If his bill was \$24, how much tip should he leave?

2.7 Solve Absolute Value Inequalities

Learning Objectives:

In this section, you will:

- Solve absolute value equations
- Solve absolute value inequalities with “less than”
- Solve absolute value inequalities with “greater than”
- Solve applications with absolute value

Absolute Value

Recall: An absolute value is the distance a number is from zero on the number line and is always a number that is greater than or equal to zero.

Our goal is to solve equations and inequalities with absolute values in them. We will start with equations than move to inequalities.

Absolute Value Equations

For any algebraic expression, u , and any positive real number, a ,

if $|u| = a$
then $u = -a$ or $u = a$

Remember that an absolute value cannot be a negative number.

Solving an absolute value equation means turning the equation into two separate equations and solving it.

Solve absolute value equations.

1. Step 1. Isolate the absolute value expression.
2. Step 2. Write the equivalent equations.
3. Step 3. Solve each equation.
4. Step 4. Check each solution.

Example A

Solve: $|2x| = 10$

$$2x = 10 \text{ or } 2x = -10$$

$$x = 5 \text{ or } x = -5$$

$$x^{\frac{1}{2}} = \sqrt{x}$$

Write the equivalent equations

Solve each equation

The denominator of 2 means square root.

Example B

Solve: $3|2x - 1| = 6$

$$|2x - 1| = 2$$

$$2x - 1 = 2 \text{ or } 2x - 1 = -2$$

$$x = \frac{3}{2} \text{ or } x = -\frac{1}{2}$$

Isolate the absolute value

Write the equivalent equations

Solve each equation

Equations with Two Absolute Values
For any algebraic expressions, u and v ,
if $|u| = |v|$
then $u = v$ or $u = -v$

Example C

Write with a rational exponent $|5x + 2| = |7x - 8|$

$$5x + 2 = 7x - 8 \text{ or } 5x + 2 = -(7x - 8)$$

Write the equivalent equations

$$5x + 2 = 7x - 8 \text{ or } 5x + 2 = -7x + 8$$

$$-2x = -10 \quad \text{or} \quad 12x = 6$$

Solve each equation

$$x = 5 \quad \text{or} \quad x = \frac{1}{2}$$

Solving Absolute Value Inequalities

Remember that when solving an inequality, the only real difference is if we multiply or divide by a negative, we switch the sign. Since rewriting an absolute value means that one expression gets a negative, we will need to switch the sign in that inequality. The only other consideration is if the solutions can be one region like an “and” or if they are two separate regions like “or”.

Absolute Value Inequalities with $<$ or $\leq$

For any algebraic expression, u , and any positive real number, a ,

if $|u| < a$ then $-a < u < a$

if $|u| \leq a$ then $-a \leq u \leq a$

Absolute Value Inequalities with $>$ or $\geq$

For any algebraic expression, u , and any positive real number, a ,

if $|u| > a$ then $u < -a$ or $u > a$

if $|u| \geq a$ then $u \leq -a$ or $u \geq a$

It is one shared interval when the absolute value starts less than the expression, and it is separate intervals when the absolute value starts greater than the expression. As always, the “or equal to” part does not change the process, just if we can include endpoints.

Example D

Simplify $|3x - 1| + 2 < 7$

$$|3x - 1| < 5$$

$$-5 < 3x - 1 < 5$$

$$-4 < 3x < 6$$

$$-\frac{4}{3} < x < 2$$

$$\left(-\frac{4}{3}, 2\right)$$

Isolate the absolute value.

Write the equivalent inequalities.

Solve

Write in interval notation.

Example E

Simplify $|7x + 2| \geq 16$

$$7x + 2 \leq -16 \text{ or } 7x + 2 \geq 16$$

Write the equivalent inequalities

$$7x \leq -18 \text{ or } 7x \geq 14$$

$$x \leq -\frac{18}{7} \quad \text{or} \quad x \geq 2$$

Solve

$$\left(-\infty, -\frac{18}{7}\right] \cup [2, \infty)$$

Write in interval notation

Worksheet 2.7: Solve Absolute Value Inequalities

Solve:

1) $|-2x| = 10$

4) $|c + 8| + 5 = 2$

7) $|5x + 4| = |7x - 7|$

2) $|2r + 3| = 7$

5) $5 - 8|2x| = -75$

8) $|x| = |2x + 1|$

3) $|1 - a| - 3 = 5$

6) $|1 - 5n| - 7 = 57$

9) $|8n + 3| = |1 - 2n|$

Solve the absolute inequality and write in interval notation:

10) $|-5x| \leq 35$

11) $|10 + 4b| < 14$

12) $|7n + 4| \geq 74$

13) $|x| - 2 > 5$

14) $-|n - 8| + 10 \geq 22$

15) $8 + |4x - 7| \leq 17$

16) $3|4x - 1| < 9$

17) $-4|3x - 1| \leq 8$

18) $2|c - 9| + 6 > 6$

Solve:

19) A Dairy farm supplies a cheese maker with milk. Normal production is 1000 gallons a day, but the cheesemaker's needs can vary by up to 200 gallons. What is the minimum and maximum milk the cheese maker will request?

20) In electronics, a resistor is allowed a 5% difference than its stated resistance and still complies with federal law. If a resistor is stated s 50 ohms, what range of resistances could it have if it passes inspection?

3.1 Graph Linear Equations in Two Variables

Learning Objectives:

In this section, you will:

- Graph a linear equation by plotting points.
- Graph vertical and horizontal lines.
- Graph a line using the intercepts.

Graph a Linear Equation by Plotting Points:

Step 1. Find three points whose coordinates are solutions to the equation. Organize them in a table.

Step 2. Plot the points in a rectangular coordinate system. Check that the points line up. If they do not, carefully check your work.

Step 3. Draw the line through the three points. Extend the line to fill the grid and put arrows on both ends of the line.

Example A

Graph the equation $y = 3x - 1$ by plotting points.

Solution:

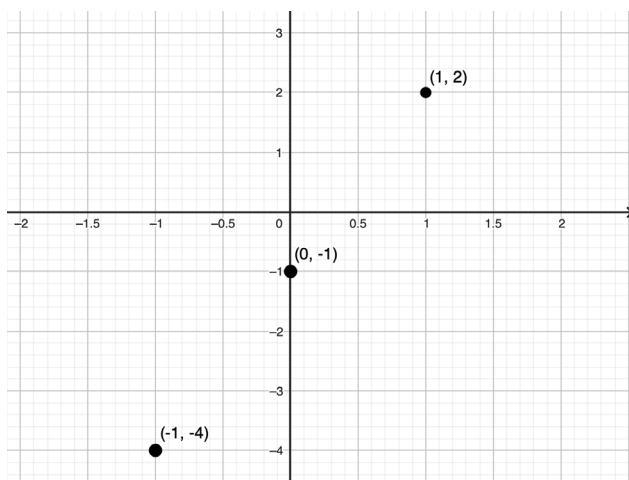
Step 1: You can choose any values for x ; however, you want to select numbers that are easy to compute the arithmetic in the equation. In this case, let x be the values of $-1, 0$, and 1 .

$x = -1$		$(0, -1)$	$y = 2$
$y = 3(-1) - 1$	$x = 0$		$(1, 2)$
$y = -3 - 1$	$y = 3(0) - 1$	$x = -1$	
$y = -4$	$y = 0 - 1$	$y = 3(1) - 1$	
$(-1, -4)$	$y = -1$	$y = 3 - 1$	

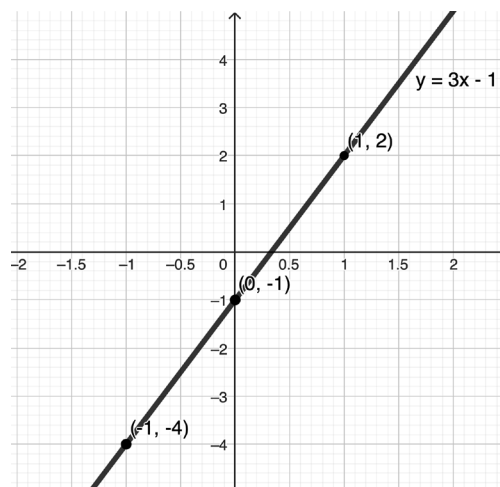
$y = 3x - 1$		
x	y	(x, y)
-1	-4	$(-1, -4)$
0	-1	$(0, -1)$

1	2	(1, 2)
---	---	--------

Step 2: Plot: $(-1, -4)$, $(0, -1)$, $(1, 2)$



Step 3: This line is the graph of $y = 3x - 1$.



Example B

Graph the equation $y = -\frac{1}{4}x + 3$ by plotting points.

Solution:

Step 1: You can choose any values for x ; however, you want to select numbers that are easy to compute the arithmetic in the equation. In this case, let x be the values of $-4, 0$, and 4 .

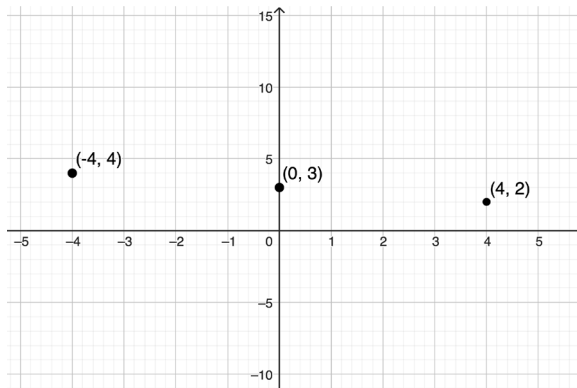
$$\begin{aligned} x &= -4 \\ y &= -\frac{1}{4}(-4) + 3 \\ y &= 1 + 3 \\ y &= 4 \\ (-4, 4) \end{aligned}$$

$$\begin{aligned} x &= 0 \\ y &= -\frac{1}{4}(0) + 3 \\ y &= 0 + 3 \\ y &= 3 \\ (0, 3) \end{aligned}$$

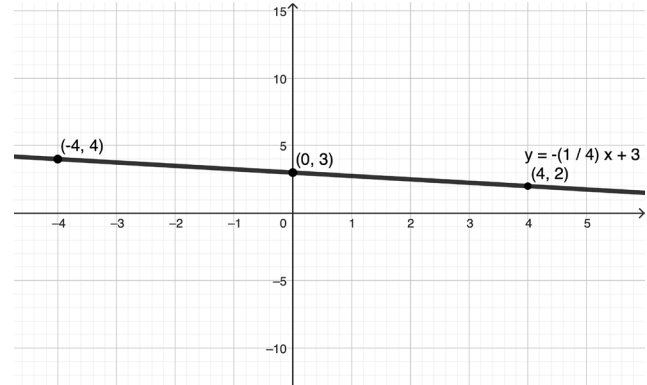
$$\begin{aligned} x &= 4 \\ y &= -\frac{1}{4}(4) + 3 \\ y &= -1 + 3 \\ y &= 2 \\ (4, 2) \end{aligned}$$

$y = 3x - 1$		
x	y	(x, y)
-4	4	$(-4, 4)$
0	3	$(0, 3)$
4	2	$(4, 2)$

Step 2: Plot: $(-4, 4)$, $(0, 3)$, $(4, 2)$



Step 3: This line is the graph of $y = -\frac{1}{4}x + 3$.



Graph Vertical and Horizontal Lines:

A **vertical line** is the graph of an equation of the form $x = a$. The line passes through the x -axis at $(a, 0)$.

A **horizontal line** is the graph of an equation of the form $y = b$. The line passes through the y -axis at $(0, b)$.

Example C

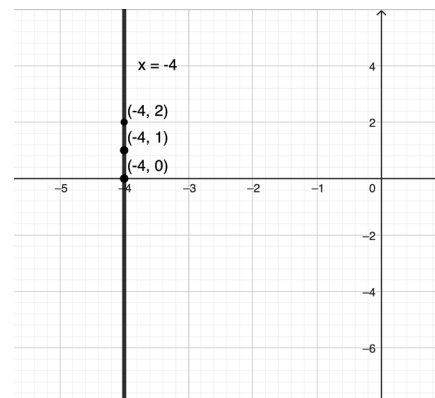
Graph a. $x = -4$ and b. $y = 2$

Solution:

Graph $x = -4$:

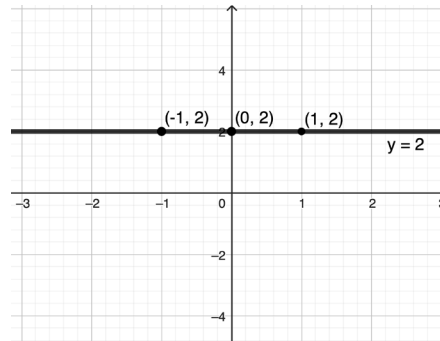
Since x is always equal to -4 , then we can create a table where x is always -4 and put in any values of y .

$x = -4$		
x	y	(x, y)
-4	0	$(-4, 0)$
-4	1	$(-4, 1)$
-4	2	$(-4, 2)$



Graph $y = 2$: Since y is always equal to 2 , then we can create a table where y is always 2 and put in any values of x .

$y = 2$		
x	y	(x, y)
-1	2	$(-1, 2)$
0	2	$(0, 2)$
1	2	$(1, 2)$



Graph a Line Using the Intercepts:

Step 1: Find the x - and y -intercepts of the line

- Let $y = 0$ and solve for x .
- Let $x = 0$ and solve for y .

Step 2: Find a third solution to the equation.

Step 3: Plot the three points and check that the line up.

Step 4: Draw the line.

Example D

Graph $2x - 4y = 16$ using the intercepts.

Solution:

Step 1. Find the x - and y - intercepts of the line.

Let $y = 0$ and solve for x .

$$2x - 4(\mathbf{0}) = 16$$

$$2x = 16$$

$$x = 8$$

The x -intercept is $(8, 0)$.

Let $x = 0$ and solve for y .

$$2(\mathbf{0}) - 4y = 16$$

$$-4y = 16$$

$$y = -4$$

The y -intercept is $(0, -4)$.

Step 2. Find another solution to the equation.

We'll use $x = 4$

$$2(4) - 4y = 16$$

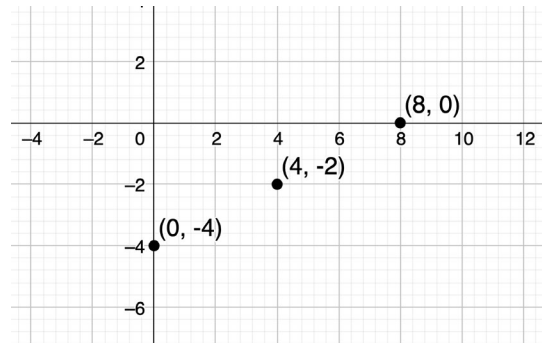
$$8 - 4y = 16$$

$$-4y = 8$$

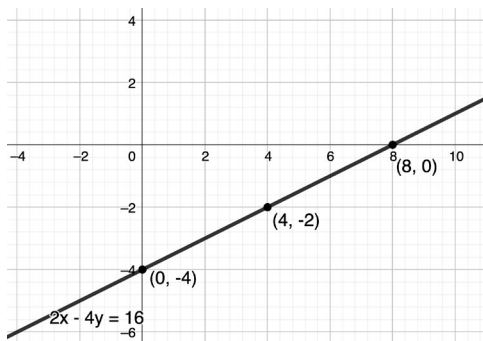
$$y = -2$$

Step 3. Plot the three points. Check that the points line up.

x	y	(x, y)
0	-4	$(0, -4)$
4	-2	$(4, -2)$
8	0	$(8, 0)$



Step 4. Draw the line.



Worksheet 3.1: Graph Linear Equations in Two Variables

Graph a Linear Equation by Plotting Points: In the following exercises, graph by plotting points.

1) $y = x - 4$

4) $y = \frac{2}{3}x - 2$

7) $y = -\frac{3}{4}x + 3$

2) $y = -3x + 5$

5) $x = 5$

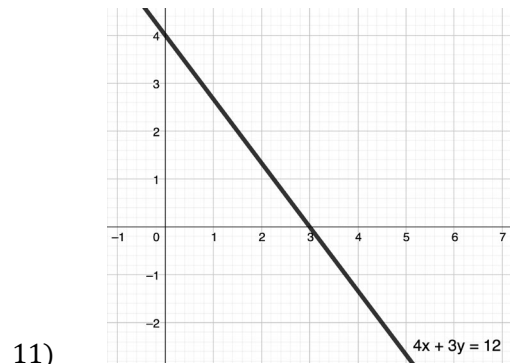
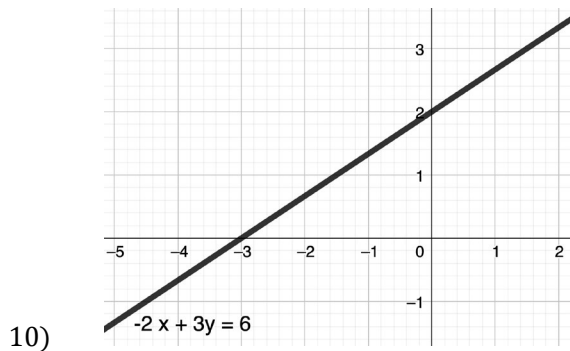
8) $x + 3 = 0$

3) $y = \frac{1}{5}x + 3$

6) $y = -3$

9) $3x - 5y = 15$

Find x - and y -Intercepts: In the following exercises, find the x - and y -intercepts of each graph.



In the following exercises, find the x - and y -intercepts for each equation.

12) $-2x + 5y = -25$

13) $y = -\frac{3}{4}x - 9$

Graph a Line Using the Intercepts: In the following exercises, graph using the intercepts

14) $y = \frac{2}{3}x + 3$

15) $5x - 10y = 15$

16) $y = 3x$

3.2 Slope of a Line

Learning Objectives:

In this section, you will:

- Find the slope of a line.
- Graph a line given a point and the slope.
- Graph a line using its slope and intercept.

Slope of a Line between two points:

The slope of a line between two points (x_1, y_1) and (x_2, y_2) is $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x}$.

Slope of Horizontal and Vertical Line:

The slope of a horizontal line, $y = b$, is 0.

The slope of vertical line, $x = a$, is undefined.

Example A

Use the slope formula to find the slope of the line give each pair of order pairs.

a) $(-4, 2)$ and $(2, -5)$

b) $(-2, 1)$ and $(-2, -3)$

$$m = \frac{-3 - 1}{-2 + 2} = \frac{-4}{0}$$

$$m = \text{Undefined}$$

Solution: Label the points:

$$\begin{pmatrix} x_1, y_1 \\ -4, 2 \end{pmatrix} \begin{pmatrix} x_2, y_2 \\ 2, -5 \end{pmatrix}$$

Solution: Label the points:

$$\begin{pmatrix} x_1, y_1 \\ -2, 1 \end{pmatrix} \begin{pmatrix} x_2, y_2 \\ -2, -3 \end{pmatrix}$$

Substitute the values.

$$m = \frac{2 - (-4)}{-5 - 2}$$

Simplify:

$$m = \frac{2 + 4}{-5 - 2} = \frac{6}{-7}$$

Substitute the values.

$$m = \frac{-3 - 1}{-2 - (-2)}$$

Simplify:

c) $(0, -6)$ and $(4, -6)$

Solution: Label the points:

$$\begin{pmatrix} x_1, y_1 \\ 0, -6 \end{pmatrix} \begin{pmatrix} x_2, y_2 \\ 4, -6 \end{pmatrix}$$

Substitute the values.

$$m = \frac{-6 - (-6)}{4 - 0}$$

Simplify:

$$m = \frac{-6 + 6}{4 - 0} = \frac{0}{4} =$$

Graph a Line Given a Point and Slope:

Step 1. Plot the given point (*y-intercept*).

Step 2. Use the slope formula $m = \frac{\text{rise}}{\text{run}}$ to identify the rise and run.

Step 3. Starting at the given point, count out the rise and run to mark the second point.

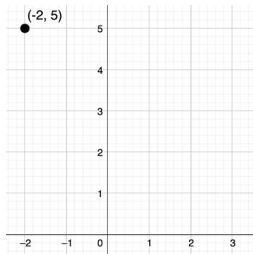
Step 4. Connect the points with a line.

Example B

Graph the line passing through the point $(-2, 5)$ whose slope is $m = -\frac{2}{3}$.

Solution:

Step 1. Plot the given point at $(-2, 5)$.



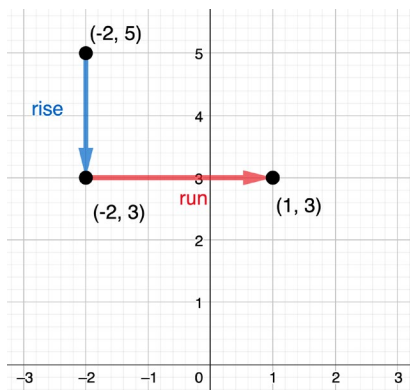
Step 2. Identify the rise and the run from the slope m .

$$m = -\frac{2}{3} = \frac{-2}{3} = \frac{\text{rise}}{\text{run}}$$

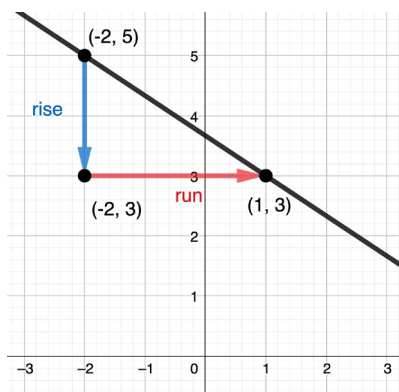
$$\text{rise} = -2$$

$$\text{run} = 3$$

Step 3. Start at $(-2, 5)$ and count the rise and the run. Down 2 units, right 3 units



Step 4. Connect the two points with a line.



Graph a Line Using its Slope and Intercept:

Slope Intercept Form of an Equation of a Line: The slope- intercept form of an equation of a line with slope m and y -intercept, $(0, b)$ is $y = mx + b$.

Example C

Identify the slope and y -intercept of the line from the equation

$$1) \ y = \frac{3}{5}x - 4$$

Solution: Since the equation is in slope-intercept form, we can identify the slope and y -intercept.

$m = \frac{3}{5}$ and y -intercept is $(0, -4)$.

$$2) \quad 2x + 6y = 12$$

Solution: Since this equation is not given in slope-intercept form, we must solve for y and put it into slope-intercept form.

$$2x + 6y = 12 \quad \text{solve for } y$$

$$6y = -2x + 12 \quad \text{subtract } 2x \text{ from each side}$$

$$\frac{6y}{6} = \frac{-2x+12}{6} \quad \text{divide both sides by 6}$$

$$y = -\frac{1}{3}x + 2 \quad \text{simplify}$$

$$m = -\frac{1}{3} \text{ and } y\text{-intercept is } (0, 2).$$

Example D

Graph the line of the equation $3x + 4y = 16$ using its slope and y -intercept.

Solution: Since this equation is not given in slope-intercept form, we must solve for y and put it into slope-intercept form.

$$3x + 4y = 16 \quad \text{solve for } y$$

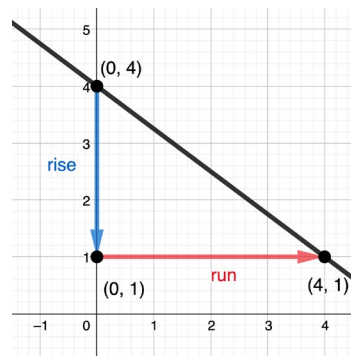
$$4y = -3x + 16 \quad \text{subtract } 3x \text{ from each side}$$

$$\frac{4y}{4} = \frac{-3x+16}{4} \quad \text{divide both sides by 4}$$

$$y = -\frac{3}{4}x + 4 \quad \text{simplify}$$

$$m = -\frac{3}{4} \text{ and } y\text{-intercept is } (0, 4).$$

Plot the y -intercept $(0, 4)$, and identify the rise over the run. *rise -3, run 4*



Strategy for Choosing the Most Convenient Method to Graph a Line:

Consider the form of the equation.

- If it only has one variable, it is a vertical or horizontal line.
 - $x = a$ is a vertical line passing through the x -axis at a .
 - $y = b$ is a horizontal line passing through the y -axis at b .
- If y is isolated on one side of the equation, in the form $y = mx + b$, graph by using the slope and y -intercept.
 - Identify the slope and y -intercept and then graph.
- If the equation is of the form $Ax + By = C$, find the intercepts.
 - Find the x - and y -intercepts, a third point, and then graph.

Example E

Determine the most convenient method to graph each line:

1) $y = -6$

2) $-4x + 12y = 24$

3) $x = \frac{5}{2}$

4) $y = \frac{5}{6}x - 4$

Solution:

- a) This equation has only one variable, y . Its graph is a horizontal line crossing the y -axis at -6 .
- b) This equation is the form $Ax + By = C$. The easiest way to graph it will be to find the intercepts and one more point.
- c) There is only one variable, x , the graph is a vertical line crossing the x -axis at $\frac{5}{2}$.
- d) Since this equation is in $y = mx + b$ form, it will be easiest to graph this line by using the slope and y -intercepts.

Graph and Interpret Applications of Slope-Intercept:

Example F

The equation $S = 2000y + 50000$ is an employee's annual salary S after y years from joining the company.

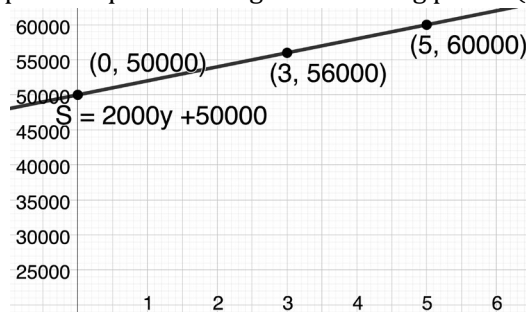
- a) Find and interpret the slope and the initial salary of the employee.
- b) Find the employee's annual salary after 5 years.
- c) Graph the equation.

Solution:

- a) The slope of the employee's annual salary equation is $m = 2000$, means that the employee's annual salary increases \$2000 dollars per year since joining the company. The initial salary of the employee is \$50,000.
- b) Find S when $y = 5$. $S = 2000(5) + 50000$
 $S = 10000 + 50000$ Simplify
 $S = 60000$ Simplify

The employee's annual salary after 5 years since joining the company is \$60,000 dollars.

- c) Graph the equation using the following points $(0, 50000)$ y -intercept, and $(5, 60000)$.



Use Slopes to Identify Parallel and Perpendicular Lines:

Two lines that have the same slope are called **parallel lines**.

Two lines that are **perpendicular lines** have negative reciprocals slope of each other.

Example G

Use slopes to determine if the lines are *parallel*, *perpendicular*, or *neither*.

a) $-4x + 2y = 6$ and $2x - y = -1$

Solution: Solve each equation for y by rewriting it in slope-intercept form.

$$-4x + 2y = 6$$

$$2y = 4x + 6 \quad \text{add } 4x \text{ to both sides}$$

$$\frac{2y}{2} = \frac{4x+6}{2} \quad \text{divides both sides by } 2$$

$$y = 2x + 3$$

The slope for the first equation is $m = 2$ and the y -intercept is $(0, 3)$.

$$2x - y = -1$$

$$-y = -2x - 1 \quad \text{subtract } 2x \text{ to both sides}$$

$$\frac{-y}{-1} = \frac{-2x-1}{-1} \quad \text{divides both sides by } -1$$

$$y = 2x + 1$$

The slope for the second equation is $m = 2$ and the y -intercept is $(0, 1)$.

The lines have the same slope and different y -intercepts and so they are parallel.

b) $3x - y = -2$ and $x + 3y = 15$

Solution: Solve each equation for y by rewriting it in slope-intercept form.

$$3x - y = -2$$

$$-y = -3x - 2 \quad \text{subtract } 3x \text{ to both sides}$$

$$\frac{-y}{-1} = \frac{-3x-2}{-1} \quad \text{divides both sides by } -1$$

$$y = 3x + 2$$

The slope for the first equation is $m = 3$.

$$x + 3y = 15$$

$$3y = -x + 15 \quad \text{subtract } x \text{ to both sides}$$

$$\frac{3y}{3} = \frac{-x+15}{3} \quad \text{divides both sides by } 3$$

$$y = -\frac{1}{3}x + 5$$

The slope for the second equation is $m = -\frac{1}{3}$.

The slopes are negative reciprocals of each other, so the lines are perpendicular.

c) $4x + y = 1$ and $4x - y = 3$

Solution: Solve each equation for y by rewriting it in slope-intercept form.

$$4x + y = 1$$

$$y = -4x + 1 \quad \text{subtract } 4x \text{ to both sides}$$

The slope for the first equation is $m = -4$.

$$4x - y = 3$$

$$-y = -4x + 3 \quad \text{subtract } 4x \text{ to both sides}$$

$$\frac{-y}{-1} = \frac{-4x+3}{-1} \quad \text{divides both sides by } -1$$

$$y = 4x - 3$$

The slope for the second equation is $m = 4$.

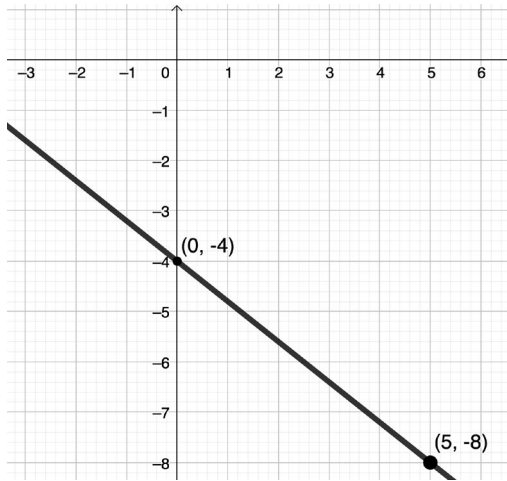
The slopes are not the same nor they are negative reciprocals of each other, so the lines are neither parallel nor perpendicular.

Worksheet 3.2: Slope of a Line

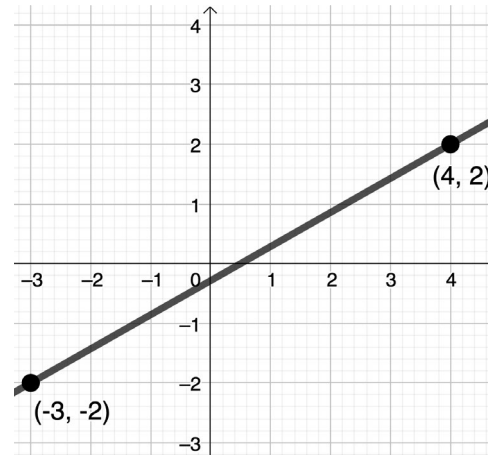
Find the Slope of a line:

In the following exercises, find the slope of each line shown.

1)



2)



In the following exercises, use the slope formula to find the slope of line between each pair of points.

3) $(3, 5), (7, 11)$

5) $(0, -5), (-6, 0)$

7) $(-4, 13), (5, -5)$

4) $(-4, 6), (2, 0)$

6) $(-6, 8), (-6, -5)$

8) $(-6, 7), (-3, 7)$

Graph a Line Given a Point and the Slope:

In the following exercises, graph each line with the given point and slope.

9) $(-1, 2); m = \frac{3}{4}$

10) $(4, 3); m = \text{undefined}$

11) $(2, 5); m = 0$

In the following exercises, Identify the slope and y -intercept of each line.

12) $y - 4 = 0$

13) $8x + 7y = 14$

14) $6x - 3y = 24$

In the following exercises, graph the line of each equation using its slope and y -intercept.

15) $y = -\frac{3}{5}x - 4$

16) $5x - 8y = 48$

17) $4x - 16y = 64$

Choose the Most Convenient Method to Graph a Line:

In the following exercises, determine the most convenient method to graph each line.

18) $x + 7 = 0$

19) $y = -\frac{7}{9}x - 5$

20) $3x - 7y = 21$

Graph and Interpret Applications of Slope-Intercept:

21) The equation $T = -0.5x + 100$ models the temperature T of a substance in x year.

- a) Find and interpret the slope of the temperature change.
- b) Determine the initial temperature.
- c) Find the temperature after 8 years.
- d) Graph the equation.

22) A car rental company uses the following model to calculate the rental cost: $C = 25d + 55$, where C represent the rental cost and d represent the number of days rented.

- a) Find and interpret the slope of the rental cost.
- b) Determine the initial rental cost for 1 day.
- c) Find the rental cost for renting the car for 4 days.
- d) Graph the equation.

Use Slopes to Identify Parallel and Perpendicular Lines:

In the following exercises, use slopes and y -intercepts to determine if the lines are parallel, perpendicular, or neither.

23) $-3x + y = 2$ and $x - 3y = 5$

26) $y = 2x + 3$ and $-x + 2y = 6$

24) $-4x + y = -4$ and $8x - 2y = -10$

27) $y = \frac{3}{4}x - 7$ and $-6x + 8y = 16$

25) $x - 2 = 0$ and $y = 5$

28) $-4x + 5y = 15$ and $5x - 4y = 12$

3.3 Find the Equation of a Line

Learning Objectives:

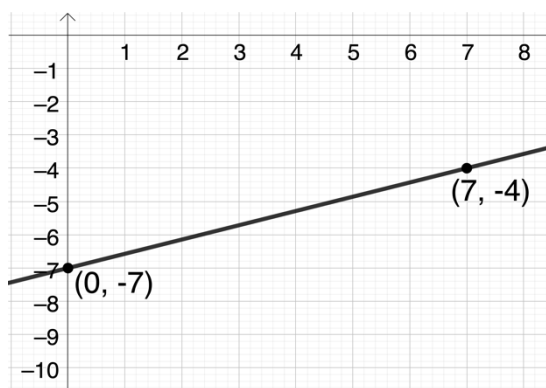
In this section, you will:

- Find an equation of the line given the slope and y -intercept.
- Find an equation of the line given the slope and a point.
- Find an equation of the line given two points.
- Find an equation of a line parallel to a given line.
- Find an equation of a line perpendicular to a given line.

Find an Equation of the Line Given the Slope and y -Intercept:

Example A

Find the equation of the line shown in the graph.



Solution: We need to find the slope and y -intercept of the line from the graph so we can substitute the needed values into the slope-intercept form.

The graph indicates the y -intercept: $(0, -7)$.

The slope between the two points $(0, -7)$ and $(7, -4)$ is $m = \frac{-4 - (-7)}{7 - 0} = \frac{3}{7}$.

Substitute the values of $m = \frac{3}{7}$ and $b = -7$ into $y = mx + b$. The equation of the line is $y = \frac{3}{7}x - 7$.

Point-slope Form of an Equation of a Line: The **point-slope form** of an equation of a line with slope m and containing the point (x_1, y_1) is $y - y_1 = m(x - x_1)$

Find an Equation of the Line Given the Slope and a Point:

Step 1. Identify the slope.

Step 2. Identify the point.

Step 3. Substitute the values into the point-slope from, $y - y_1 = m(x - x_1)$.

Step 4. Write the equation in slope-intercept form.

Example B

Find an equation of a line with slope $m = -\frac{2}{3}$ that contains the point $(9, -5)$. Write the equation in slope-intercept form.

Solution: The slope $m = -\frac{2}{3}$ is given. The point $(9, -5)$ is given, let $x_1 = 9$ and $y_1 = -5$.

Substitute the values into the point-slope form, $y - y_1 = m(x - x_1)$

$$y - (-5) = -\frac{2}{3}(x - 9) \quad \text{Simplify}$$

$$y + 5 = -\frac{2}{3}x + 6 \quad \text{Subtract 5 to both sides}$$

$$y = -\frac{2}{3}x + 1 \quad \text{Write the equation in slope-intercept form}$$

Find an Equation of the Line Given Two Points:

Step 1. Find the slope using the given points. $m = \frac{y_2 - y_1}{x_2 - x_1}$

Step 2. Choose one point.

Step 3. Substitute the values into the point-slope from, $y - y_1 = m(x - x_1)$.

Step 4. Write the equation in slope-intercept form.

Example C

Find an equation of a line that contains the points $(-2, -5)$ and $(3, 6)$. Write the equation in slope-intercept form.

Solution: Find the slope of the line through $(-2, -5)$ and $(3, 6)$.

$$m = \frac{6 - (-5)}{3 - (-2)} = \frac{6 + 5}{3 + 2} = \frac{11}{5}$$

Choose either point. $(3, 6)$

Substitute the values into the point-slope form. $y - 6 = \frac{11}{5}(x - 3)$

Simplify.

$y - 6 = \frac{11}{5}x - \frac{33}{5}$	Distribute $\frac{11}{5}$
$y = \frac{11}{5}x - \frac{33}{5} + 6$	Add 6 to both sides
$y = \frac{11}{5}x - \frac{33}{5} + \frac{30}{5}$	Combine like terms by obtaining a common denominator
$y = \frac{11}{5}x - \frac{3}{5}$	Write the equation in slope-intercept form

Find an Equation of a Line Parallel to a Given Line:

Step 1. Find the slope of the given line.

Step 2. Find the slope of the parallel line.

Step 3. Identify the point.

Step 4. Substitute the values into the point-slope form, $y - y_1 = m(x - x_1)$.

Step 5. Write the equation in slope-intercept form.

Example D

Find an equation of a line parallel to $3x + 6y = 12$ that contains the point $(2, 3)$. Write the equation in slope-intercept form.

Solution: Find the slope of the given line.

$$3x + 6y = 12 \quad \text{Put it in slope-intercept form}$$

$$6y = -3x + 12 \quad \text{Subtract } 3x \text{ to both sides}$$

$$\frac{6y}{6} = \frac{-3x+12}{6} \quad \text{Divide both sides by 6}$$

$$y = -\frac{1}{2}x + 2 \quad \text{The slope is } m = -\frac{1}{2}$$

Parallel lines have the same slope.

$$\text{Hence } m = -\frac{1}{2}$$

$$\text{The given point is } (-2, 1)$$

Substitute the values into the point-slope form.

$$y - 1 = -\frac{1}{2}(x - (-2))$$

$$y - 1 = -\frac{1}{2}(x + 2) \quad \text{Simplify}$$

$$y - 1 = -\frac{1}{2}x - 1 \quad \text{Distribute } -\frac{1}{2}$$

$$y = -\frac{1}{2}x \quad \text{Add 1 to both sides}$$

Find an Equation of a Line Perpendicular to a Given Line:

Step 1. Find the slope of the given line.

Step 2. Find the slope of the perpendicular line.

Step 3. Identify the point.

Step 4. Substitute the values into the point-slope form, $y - y_1 = m(x - x_1)$.

Step 5. Write the equation in slope-intercept form.

Example E

Find an equation of a line that is perpendicular to $4x + 5y = -10$ that contains the point $(3, 4)$. Write the equation in slope-intercept form.

Solution: Find the slope of the given line.

$4x + 5y = -10$ Put it in slope-intercept form

$5y = -4x - 10$ Subtract $4x$ to both sides

$\frac{5y}{5} = \frac{-4x-10}{5}$ Divide both sides by 5

$y = -\frac{4}{5}x - 2$ The slope is $m = -\frac{4}{5}$

The slopes of perpendicular lines are negative reciprocals.

Hence $m = \frac{5}{4}$

The given point is $(3, 4)$

Substitute the values into the point-slope form.

$y - 4 = \frac{5}{4}(x - 3)$

$y - 4 = \frac{5}{4}(x - 3)$ Simplify

$y - 4 = \frac{5}{4}x - \frac{15}{4}$ Distribute $\frac{5}{4}$

$y = \frac{5}{4}x - \frac{15}{4} + 4$ Add 4 to both sides

$y = \frac{5}{4}x - \frac{15}{4} + \frac{16}{4}$ Common denominator

$y = \frac{5}{4}x + \frac{1}{4}$ Simplify

Worksheet 3.3: Find the Equation of a Line

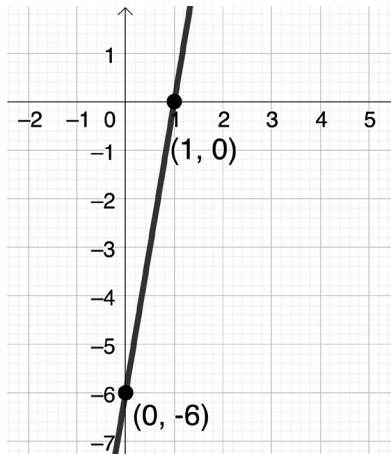
Find an Equation of the Line Given the Slope and y-Intercept:

In the following exercises, find the equation of a line with given slope and y -intercept. Write the equation in slope-intercept form.

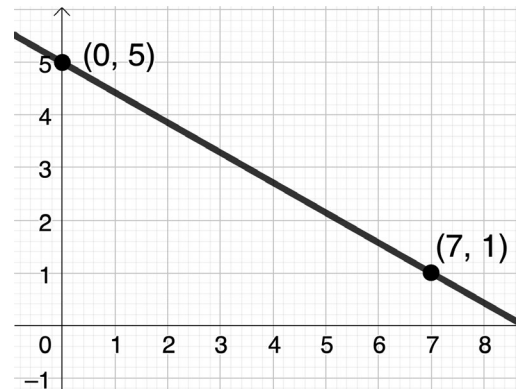
1) Slope $-\frac{3}{7}$ and y -intercept $(0, -3)$

2) Slope 3 and y -intercept $(0, 5)$

3)



4)



Find an Equation of the Line Given the Slope and a Point:

In the following exercises, find the equation of a line with given slope and containing the given point. Write the equation in slope-intercept form.

5) $m = 3$, point $(4, -6)$

7) Horizontal line containing $(-3, -9)$

6) $m = -\frac{6}{7}$, point $(14, 3)$

8) $m = \frac{4}{5}$, point $(-2, 4)$

Find an Equation of the Line Given Two Points:

In the following exercises, find the equation of a line containing the given points. Write the equation in slope-intercept form.

9) $(-2, -2)$ and $(6, 2)$

10) $(-3, 5)$ and $(3, 1)$

11) $(5, -6)$ and $(5, 5)$

12) $(-5, 2)$ and $(3, 2)$

13) $(8, 6)$ and $(4, -3)$

14) $(-7, -4)$ and $(5, 6)$

Find an Equation of a Line Parallel to a Given Line:

In the following exercises, find the equation of a line parallel to the given line and contains the given point. Write the equation in slope-intercept form.

15) line $y = -\frac{3}{2}x + 1$, point $(0, 5)$

16) line $y = \frac{1}{2}x - 1$, point $(2, 3)$

17) line $y = \frac{3}{4}x - 2$, point $(-2, 7)$

18) line $5x + 2y = 6$, point $(4, 6)$

Find an Equation of a Line Perpendicular to a Given Line:

In the following exercises, find the equation of a line perpendicular to the given line and contains the given point. Write the equation in slope-intercept form.

19) line $y = 2x + 3$, point $(4, 5)$

20) line $3x + y = 2$, point $(-1, 4)$

21) line $4x - y = 6$, point $(3, -2)$

22) line $-5x + 2y = -8$, point $(1, -3)$

3.5 Relations and Functions

Learning Objectives:

In this section, you will:

- Find the domain and range of a relation.
- Determine if a relation is a function.
- Find the value of a function.

Find the Domain and Range of a Relation:

A **relation** is any set of ordered pairs, (x, y) . All the x -values in the ordered pairs together make up the **domain**. All the y -values in the ordered pairs together make up the **range**.

Example A

For the relation $\{(1, 3), (2, 5), (3, 7), (4, 9), (5, 11)\}$

- Find the domain of the relation.
- Find the range of the relation.

Solution:

- The domain is the set of all x -values of the relation; hence the domain is $\{1, 2, 3, 4, 5\}$
- The range is the set of all y -values of the relation; hence the range is $\{3, 5, 7, 9, 11\}$

Determine if a Relation is a Function:

A **function** is a relation that assigns to each element in its domain exactly one element in the range.

Example B

Use the set of ordered pairs to (i) determine whether the relation is a function (ii) find the domain or the relation (iii) find the range of the relation.

- $\{(-12, 14), (-9, 10), (-4, -3), (0, 5), (3 - 3), (6, 10), (8, 15)\}$
- $\{(1, -5), (-5, 8), (0, -8), (1, 11), (-3, 0), (-10, -5), (7 - 3)\}$

Solution:

- a) (i) Each x -value is matched with only one y -value. So this relation is a function.
 (ii) The domain is $\{-12, -9, -4, 0, 3, 6, 8\}$.
 (iii) The range is $\{-3, 5, 10, 14, 15\}$
- b) (i) The x -value 1 is matched with two y -values, both -5 and 11 . So this relation is not a function.
 (ii) The domain is $\{-10, -5, -3, 0, 1, 7\}$.
 (iii) The range is $\{-8, -5, -3, 0, 8, 11\}$

In algebra, more often than not, functions will be represented by an equation. It is easiest to see if the equation is a function when it is solved for y . If each value of x results in one value of y , then the equation defines a function.

Example C

Determine whether each equation is a function. Assume x is the independent variable.

- a) $3x + y = -4$ b) $y = 2x^2 - 5$ c) $4x + y^2 = 9$

Solution:

- a) $3x + y = -4$, For each value of x , we multiply it by -3 and then subtract 4 to get the y -value. $y = -3x - 4$. For example, if $x = 2$ then $y = -3(2) - 4$ after simplifying $y = -10$. It would work similarly for any value of x . Since each value of x , corresponds to only one value of y the equation defines a function.
- b) $y = 2x^2 - 5$, For each value of x , we square it and multiply by 2 and then subtract 5 to get the y -value. For example, if $x = 1$ then $y = 2(1)^2 - 5$ after simplifying $y = -3$. It would work similarly for any value of x . Since each value of x , corresponds to only one value of y the equation defines a function.
- c) $4x + y^2 = 9$, First isolate the y term: $y^2 = -4x + 9$. Then take the square root to both sides: $y = \pm\sqrt{-4x + 9}$. For example, if $x = 1$ then $y = \pm\sqrt{-4(1) + 9}$. After simplifying $y = \sqrt{5}$ or $y = -\sqrt{5}$. It would work similarly for any value of x . Since each value of x , does not correspond to only one value of y the equation does not defines a function.

Find the Value of a Function: For the function $y = f(x)$

f is the name of the function

x is the domain value

$f(x)$ is the range value y corresponding to the value x

We read $f(x)$ as f of x or the value of f at x .

We call x the independent variable as it can be any value in the domain. We call y the dependent variable as its value depends on x .

Example D

For the function $f(x) = -3x^2 + 5x - 2$, evaluate the function.

a) $f(2)$

b) $f(c)$

c) $f(x + 1)$

Solutions:

a) To evaluate $f(2)$, substitute **2** for x .

$$f(\mathbf{2}) = -3(\mathbf{2})^2 + 5(\mathbf{2}) - 2$$

$$f(2) = -3(4) + 10 - 2 \quad \text{Simplify}$$

$$f(2) = -12 + 10 - 2$$

$$f(2) = -4$$

b) To evaluate $f(c)$, substitute **c** for x .

$$f(\mathbf{c}) = -3(\mathbf{c})^2 + 5(\mathbf{c}) - 2$$

$$f(c) = -3c^2 + 5c - 2 \quad \text{Simplify}$$

c) To evaluate $f(x + 1)$, substitute **$x + 1$** for x .

$$f(\mathbf{x + 1}) = -3(\mathbf{x + 1})^2 + 5(\mathbf{x + 1}) - 2 \quad \text{Simplify}$$

$$f(\mathbf{x + 1}) = -3(x^2 + 2x + 1) + 5x + 5 - 2$$

$$f(\mathbf{x + 1}) = -3x^2 - 6x - 3 + 5x + 5$$

$$f(\mathbf{x + 1}) = -3x^2 - x$$

Worksheet 3.5: Relations and Functions

Find the Domain and Range of a Relation:

In the following exercises, for each relation a) find the domain of the relation, b) find the range of the relation.

1) $\{(4, -4), (5, 0), (7, 8), (9, 10), (11, 8)\}$

2) $\{(-5, 6), (-2, 1), (3, 9), (-5, -2), (0, 8)\}$

Determine if a Relation is a Function:

In the following exercises, use the set of ordered pairs to a) determine whether the relation is a function, b) find the domain of the relation, c) find the range of the relation.

3) $\{(-6, -2), (-3, -1), (0, 0), (-3, 5), (4, 3)\}$

4) $\{(-8, 4), (-5, 0), (4, 5), (6, 5), (8, 7)\}$

In the following exercises, determine whether each equation is a function.

5) $y = 3x + 2$

7) $y^2 = 4x$

9) $x = 5y - y^2$

6) $x^2 + y^2 = 36$

8) $y = \sqrt{x + 9}$

10) $y = 2x^2 - 3x + 5$

Find the value of a Function:

In the following exercises, evaluate the function: a) $f(-3)$; b) $f(s)$; c) $f(s - 1)$

11) $f(x) = -2x + 7$

13) $f(x) = x^2 - 2x - 2$

12) $f(x) = \frac{2}{3}x - 3$

14) $f(x) = -3x^2 + 5x - 4$

Applications: In the following exercise, solve.

- 15) The monthly revenue of a bookstore is modeled by the function $R(x) = 12x + 500$, where $R(x)$ represents the total monthly revenue and x is the number of books sold.
- a) Identify the independent and dependent variables.
 - b) Calculate $R(0)$ and explain its meaning.
 - c) Calculate $R(100)$ and explain its meaning.
- 16) The monthly profit of a coffee shop is modeled by the function $P(x) = 3.5x - 2500$, where $P(x)$ represents the total monthly profit and x is the number of cups of coffee sold.
- a) Determine the independent and dependent variables.
 - b) Calculate $P(0)$ and explain its meaning.
 - c) Calculate $P(3000)$ and explain its meaning.

3.6 Graphs of Functions

Learning Objectives:

In this section, you will:

- Use the vertical line test.
- Identify graphs of basic functions.
- Read information from a graph of a function.

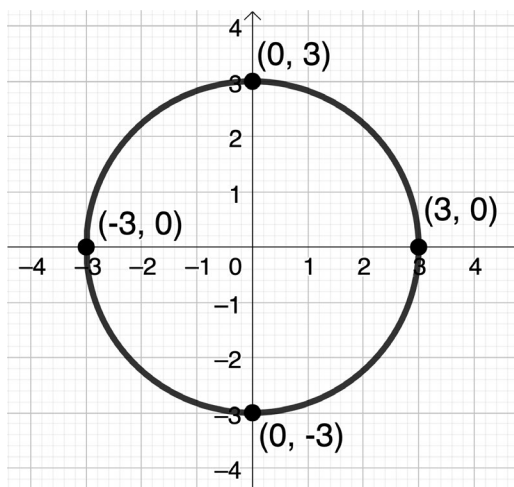
Vertical Line Test

Vertical Line Test: A set of points in a rectangular coordinate system is the graph of a function if every vertical line intersects the graph in at most one point. If any vertical line intersects the graph in more than one point, the graph does not represent a function.

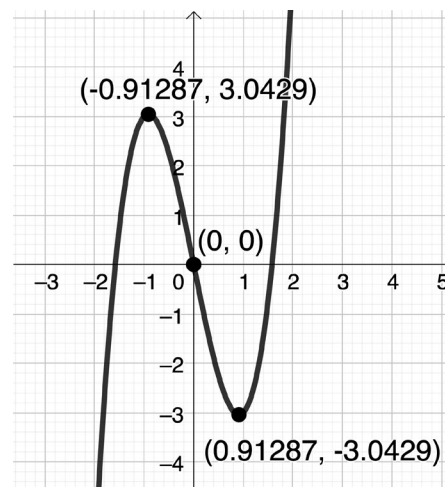
Example A

Determine whether each graph is the graph of a function.

a)

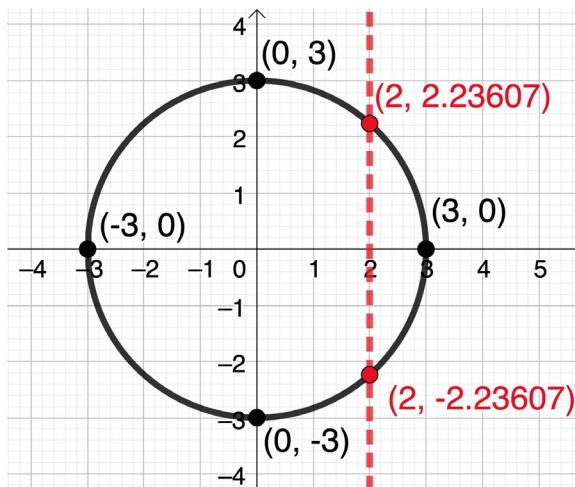


b)

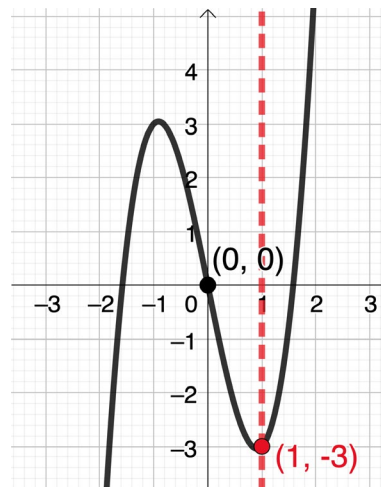


Solution:

- a) One of the vertical lines shown on the graph, intersects it in two points. This graph does not represent a function.



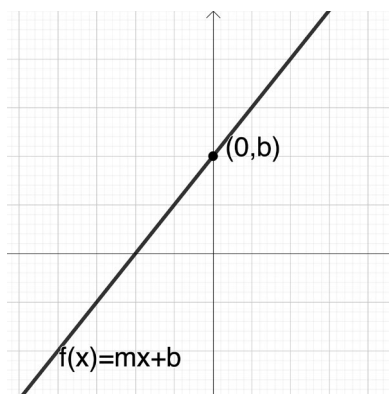
- b) Since any vertical line intersects the graph at most one point, then graph is the graph of a function.



Identify Graphs of Basic Functions:

The graph of a function is the graph of all its ordered pairs, (x, y) or using function notation, $(x, f(x))$ where $y = f(x)$.

Linear Function



$$f(x) = mx + b$$

m, b : all real numbers

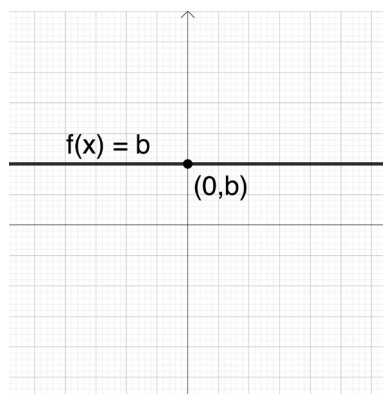
m : slope of the line

b : y -intercept

Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

Constant Function



$$f(x) = b$$

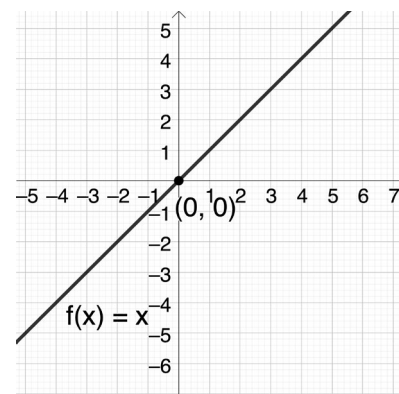
b : any real number

b : y -intercept

Domain: $(-\infty, \infty)$

Range: $\{b\}$

Identity Function



$$f(x) = x$$

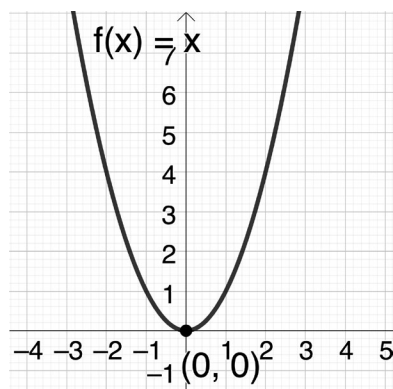
$m = 1$

$b = 0$

Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

Square Function

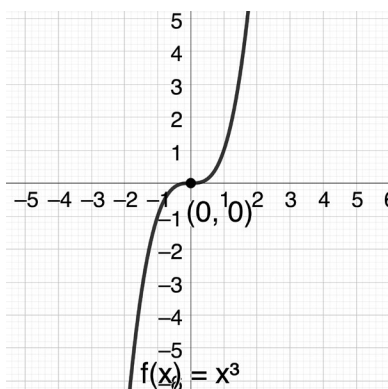


$$f(x) = x^2$$

Domain: $(-\infty, \infty)$

Range: $[0, \infty)$

Cube Function

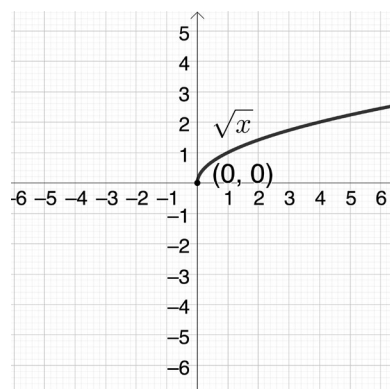


$$f(x) = x^3$$

Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

Square Root Function

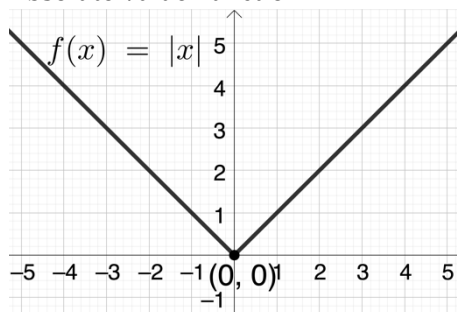


$$f(x) = \sqrt{x}$$

Domain: $[0, \infty)$

Range: $[0, \infty)$

Absolute Value Function



$$f(x) = |x|$$

Domain: $(-\infty, \infty)$

Range: $[0, \infty)$

Example B

Graph each function and state the domain and range using interval notation.

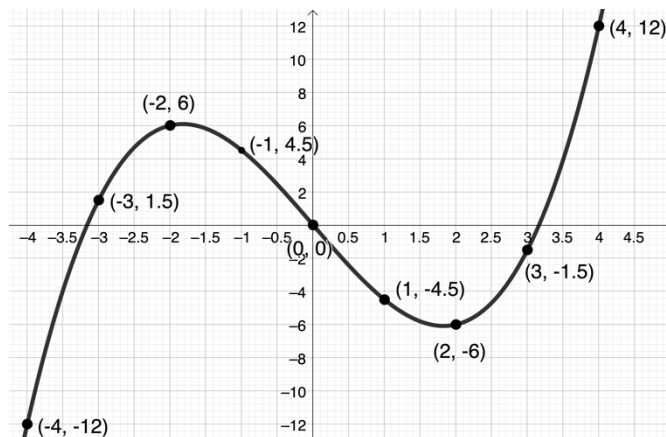
a) $f(x) = \frac{1}{2}x^3 - 5x$

b) $g(x) = \sqrt{x} - 4$

c) $h(x) = |x + 3| - 2$

Solution:

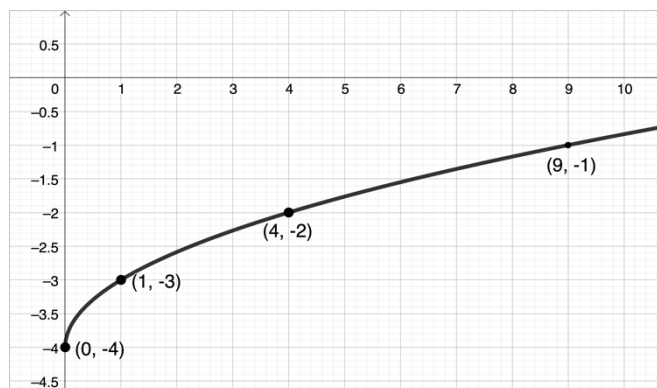
a) We choose x -values. We substitute them in and then create a chart.



x	$f(x)$ $= \frac{1}{2}x^3 - 5x$	$(x, f(x))$
-4	-12	$(-4, -12)$
-3	$3/2$	$(-3, 3/2)$
-2	6	$(-2, 6)$
-1	$9/2$	$(-1, 9/2)$
0	0	$(0, 0)$
1	$-9/2$	$(1, -9/2)$
2	-6	$(2, -6)$
3	$-3/2$	$(3, -3/2)$
4	12	$(4, 12)$

The domain of $f(x)$ is $(-\infty, \infty)$, and the range of $f(x)$ is $(-\infty, \infty)$.

b) We choose x -values. We substitute them in and then create a chart.

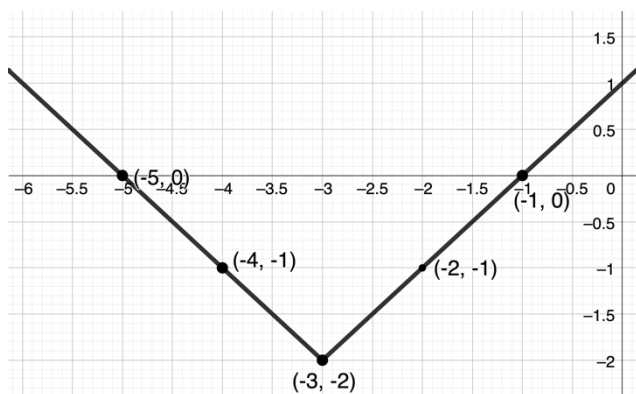


x	$g(x)$ $= \sqrt{x} - 4$	$(x, g(x))$
0	-4	$(0, -4)$
1	-3	$(1, -3)$
4	-2	$(4, -2)$
9	-1	$(9, -1)$

0	0	$(0, -4)$
1	-3	$(1, -3)$
4	-2	$(4, -2)$
9	-1	$(9, -1)$

The domain of $f(x)$ is $[0, \infty)$, and the range of $f(x)$ is $[-4, \infty)$.

c) We choose x -values. We substitute them in and then create a chart.



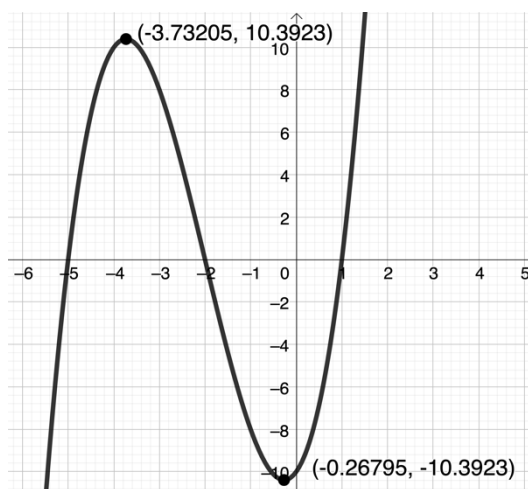
x	$h(x) = x + 3 - 2$	$(x, h(x))$
-5	0	$(-5, 0)$
-4	-1	$(-4, -1)$
-3	-2	$(-3, -2)$
-2	-1	$(-2, -1)$
-1	0	$(-1, 0)$

The domain of $f(x)$ is $(-\infty, \infty)$, and the range of $f(x)$ is $[-2, \infty)$.

Read Information from a Graph of a Function:

Example C

Use the graph of the function to find the indicated values.



Find $f(-3)$.

Find $f(-1)$.

Find the x -intercepts.

Find the y -intercept.

Find the domain. Write it in interval notation.

Find the range. Write it in interval notation.

Solution:

When $x = -3$, the function crosses the y -axis at 8. So, $f(-3) = 8$.

When $x = -1$, the function crosses the y -axis at -8 . So, $f(-1) = -8$.

The x -intercepts occur when $y = 0$. So the x -intercepts occur when $f(x) = 0$. The x -intercepts are $(-5, 0)$, $(-2, 0)$, and $(1, 0)$.

The y -intercept occur when $x = 0$. So the x -intercepts occur when $f(0)$. The y -intercept is $(0, -10)$.

This function has a value for all values of x . therefore, the domain in interval notation is $(-\infty, \infty)$.

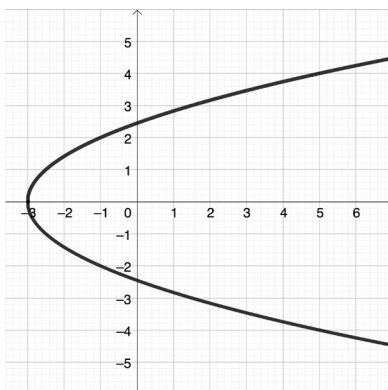
This function values, or y -values go for all values of x . therefore, the range in interval notation is $(-\infty, \infty)$.

Worksheet 3.6: Graphs of Functions

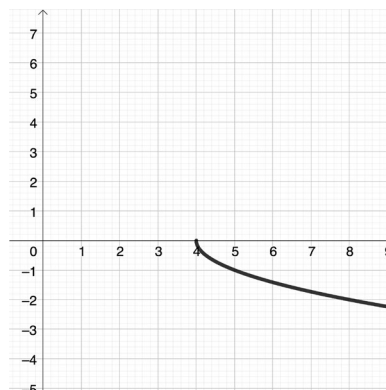
Use the Vertical Line Test:

In the following exercises, determine whether each graph is the graph of a function.

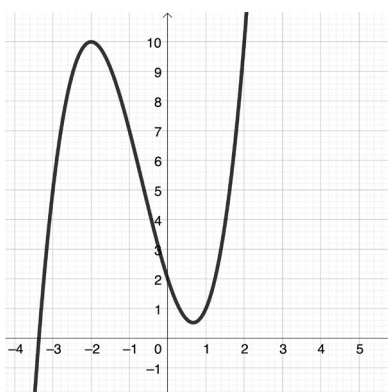
1)



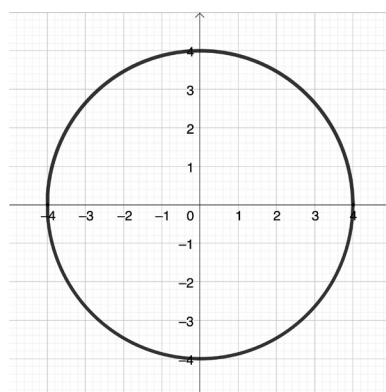
3)



2)



4)



Identify Graphs of Basic Functions:

In the following exercises, a) graph each function, b) state its domain and range using interval notation.

5) $f(x) = \frac{2}{5}x - 4$

6) $f(x) = -5$

7) $f(x) = -3x^2$

8) $f(x) = x^2 + 2x - 15$

9) $f(x) = 3\sqrt{x}$

10) $f(x) = \sqrt{x+3} - 2$

11) $f(x) = -3|x|$

12) $f(x) = |x - 4| + 2$

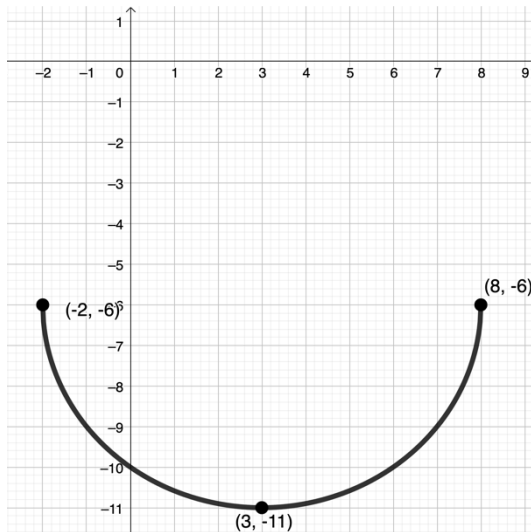
13) $f(x) = -2x^3$

14) $f(x) = x^3 - x^2 - 8x + 12$

Read Information from a Graph of a Function:

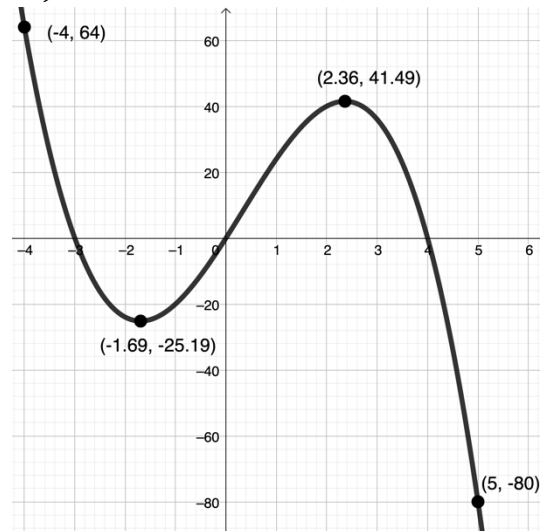
In the following exercises, use the graph of the function to find the indicated values.

15)



- Find $f(-1)$.
- Find $f(x) = -10$.
- Find the x -intercepts.
- Find the y -intercept.
- Find the domain. Write it in interval notation.
- Find the range. Write it in interval notation.

16)



- Find $f(-4)$.
- Find $f(2.36)$.
- Find the x -intercepts.
- Find the y -intercept.
- Find the domain. Write it in interval notation.
- Find the range. Write it in interval notation.

4.1 Solving Systems of Linear Equations of Two Variables

Learning Objectives:

In this section, you will:

- Determine whether an ordered pair is a solution of a system of equations
- Solve a system of linear equations by graphing
- Solve a system of equations by substitution
- Solve a system of equations by elimination
- Choose the most convenient method to solve a system of linear equations

System of Linear Equations

When two or more linear equations are grouped, they form a **system of linear equations**.

We will deal with systems of two equations and two variables that can be written as:

$$\begin{cases} Ax + By = C \\ Dx + Ey = F \end{cases}$$

Point: The variables in a system of equations all represent the same value, i.e. the x is the same x in all equations and the y is the same y for all equations, etc.

Solutions of a System of Equations

The **solutions of a system of equations** are the values of the variables that make *all* the equations true. A solution of a system of two linear equations is represented by an ordered pair (x, y) .

To know if an ordered pair is a solution, we plug the values in for the variables. It must work in every equation to solve the system.

Example A) Determine if the ordered pair is a solution to the system: $\begin{cases} x + 2y = 2 \\ x - 2y = 6 \end{cases}$

1) (4,-1) $4 + 2(-1) = 2$ $4 - 2(-1) = 6$	The ordered pair is a solution to the first equation The ordered pair is a solution to the second equation (4,-1) is a solution to the system
2) (8,1) $8 + 2(1) \neq 2$ $8 - 2(1) = 6$	The ordered pair is not a solution to the first equation The ordered pair is a solution to the second equation (8,1) is not a solution to the system

It is important to remember that we have three possible solutions for a system of equations:

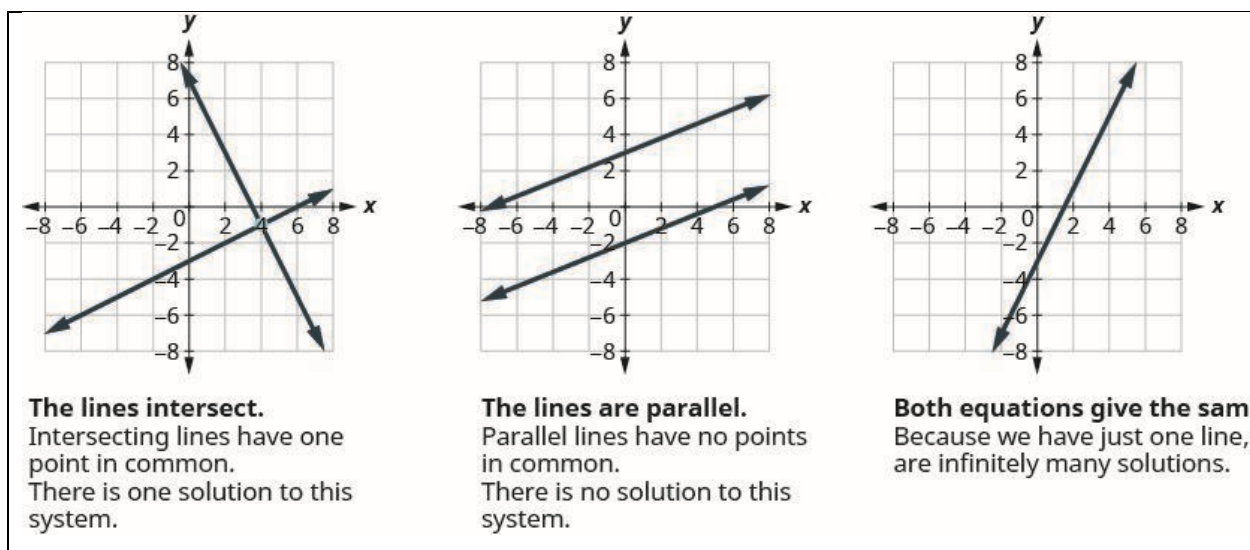
- The system has one solution (x, y)
- The system has no solutions
- The system has infinitely many solutions of the form $(x, f(x))$

We will consider the no solutions or infinitely many solutions both as special cases

For each solution method we will discuss, in addition to learning the method, we must be able to identify which of the three possible solutions we have found. We will learn three methods for solving a system of equations: graphing, substitution, and addition.

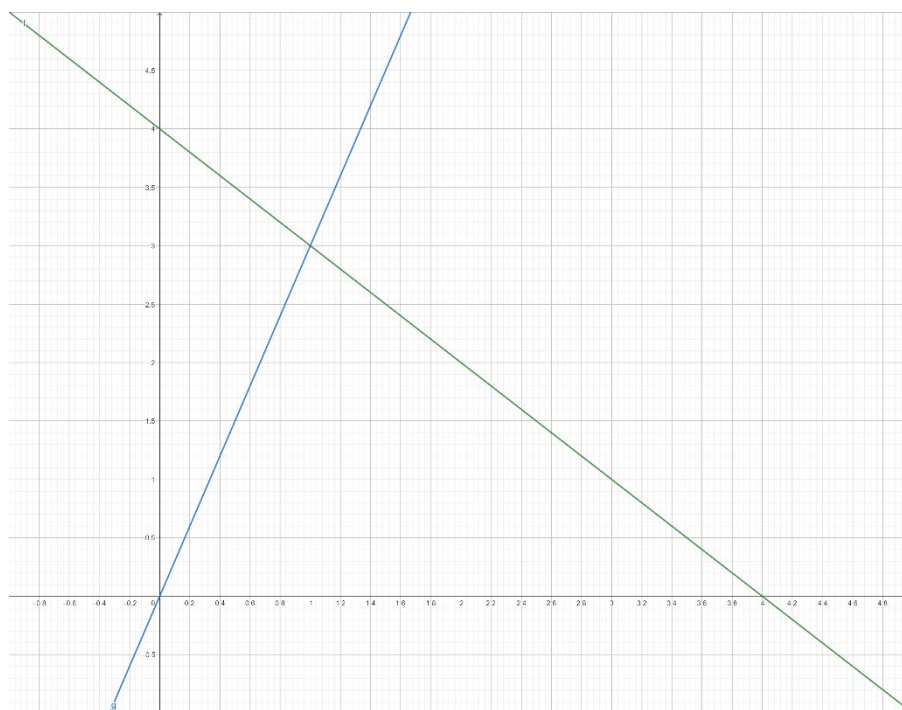
Solve a System of Linear Equations by Graphing

To solve by graphing, we just graph the equations together. If we think about what a solution means, its that the same ordered pair can go into each equation. That means graphically, we need the same ordered pair to show up in each equation, so we are looking for intersections of lines.



Example B: Solve by graphing: $\begin{cases} x + y = 4 \\ y = 3x \end{cases}$

As we can see in the graph, the two equations have a point of intersection of (1,3), so that is the solution.



The issue with graphing for a solution are twofold. One, its possible the intersection does not occur at an integer value, making it difficult to determine the exact value. Two, its possible the intersection, if it has one, could be very far from the origin. Two lines may look parallel, but their graphs intersect hundreds of

values down. We simply do not graph that large. Graphing works best when we can easily graph the equations, and the solution is a small integer value.

In general, we use either of the following two methods to solve our problems and will discuss how to know which to choose after we learn them.

For the substitution and the addition method, we need to be able to spot when a system has no solution or has infinity many solution.

Solving a Linear System of Equations by Substitution.

The goal of the method is to reduce from two variables down to one. In order to do this, we will substitute one solved variable into the other equation. This method is best used when one of the variables has a coefficient of 1, to make the solving easier. If there is a coefficient of 1, we make that the variable to solve for in the first step listed below.

Steps for substitution method

1. Solve one of the equations for either variable
2. Substitute the expression from Step 1 into the other equation
3. Solve the resulting equation
4. Substitute the solution in Step 3 into either of the original equations to find the other variable
5. Write the solution as an ordered pair
6. Check that the ordered pair is a solution to both original equations

Example C: Solve the system by substitution: $\begin{cases} 2x + y = 7 \\ x - 2y = 6 \end{cases}$

$x = 2y + 6$	Solve the second equation for x
$2(2y + 6) + y = 7$ $4y + 12 + y = 7$ $5y = -5$ $y = -1$	Substitute the expression into equation one and solve for y
$x = 2(-1) + 6 = 4$	Substitute the solution into step 1 to find x
$(4, -1)$	Write the solution as an ordered pair

The special cases can be seen because the variable from Step (3) will cancel out, leaving just numerical values.

1. If the system has no solution, the numbers will not equal one another, making a false statement.
2. If the system has infinitely many solutions, the numbers will be equal, making a true statement

Example D: Solve the system by substitution: $\begin{cases} 5x - y = -10 \\ y = 5x \end{cases}$

$$5x - 5x = -10$$

Since the second equation is already solved for y , we substitute it into the first equation.

$$0 \neq -10$$

Since the result is false, there is no solution

Solving a Linear System of Equations by Addition.

The goal of the addition method is to eliminate a variable by adding multiples of the two equations together. Essentially, we want one variable to have the same coefficient with opposite signs, that way when we add the equations together, that variable will cancel out.

Steps for Addition method

1. Write both equations in standard form. If any coefficients are fractions, clear them
2. Pick one variable and multiply either/both equations by a constant to make those variable have the same coefficient with opposite signs
3. Add the equations resulting from Step 2 to eliminate one variable
4. Solve for the remaining variable
5. Substitute the solution from Step 4 into one of the original equation to solve for the other variable
6. Write the solution as an ordered pair
7. Check that the ordered pair is a solution to both the original equations

Point: To clear fractions, we multiply the equation by the least common denominator. If both equations have fractions, we repeat this process for the other equation. Be sure when clearing that you multiply every term on both the left and right by the least common denominator.

Example E: Solve the system by substitution: $\begin{cases} 2x - y = 5 \\ 5x + 2y = -28 \end{cases}$

$\begin{array}{r} 2(2x-y=5) \\ 5x+2y=-28 \end{array}$	Multiply the first equation by 2
$\begin{array}{r} 4x-2y=10 \\ 5x+2y=-28 \end{array}$	Simplify
$\begin{array}{r} 4x-2y=10 \\ + \quad 5x+2y=-28 \end{array}$	Add the two equations
$9x = -18$	
$x = -2$	Solve the resulting equation for x
$2(-2) - y = 5$	Substitute the solution for x into the first equation to find y
$y = -9$	
$(-2, -9)$	Write the solution as an ordered pair

Example F: Solve the system by substitution: $\begin{cases} 3x + 2y = 4 \\ 9x + 6y = 12 \end{cases}$

$\begin{array}{r} -3(3x+2y=4) \\ 9x+6y=12 \end{array}$	Multiply the first equation by -3
$\begin{array}{r} -9x-6y=-12 \\ 9x+6y=12 \end{array}$	Simplify
$\begin{array}{r} -9x-6y=-12 \\ + \quad 9x+6y=12 \end{array}$	Add the two equations
$0 = 0$	Solve the resulting equation for x

Since the result is a true statement without variables, we know there are infinitely many solutions. We write the solution as an ordered pair where the x value is the solved for x value.

$$\left(x, -\frac{3}{2}x + 2 \right)$$

Point: Knowing which method to use depends on the problem. In general, we only graph if we also want to see the picture. Though important to be able to do, the other two methods are better. We use substitution when either an equation is already solved for a variable, or there is at least one coefficient of 1. We use the addition method any other time.

Worksheet 4.1: Solving Systems of Linear Equations of Two Variables

Verify if the ordered pair is a solution to the system: $\{4x - 5y = 3 \quad x + 2y = 4\}$

1. $(2,1)$

2. $(6,-1)$

3. $(7,5)$

Solve the system by graphing:

4. $\{y = 2x + 2 \quad y = x - 1\}$

6. $\{x + y = 2 \quad 2x + 2y = 6\}$

5. $\{y = -\frac{5}{3}x + 3 \quad 3y - x = -9\}$

Solve the system by substitution:

7. $\{2x - y = 4 \quad 2x + y = 8\}$

9. $\{y = 2x + 2 \quad x = \frac{y}{2} - 1\}$

8. $\{2x + 3y = 4 \quad x - 5y = -11\}$

Solve the system by addition:

10. $\{3x + 5y = -11 \quad x - 2y = 11\}$

12. $\{8x + 14y = 4 \quad -6x - 7y = -10\}$

11. $\{\frac{x}{3} + \frac{y}{6} = 3 \quad 2x - y = 1\}$

4.2 Solve Applications with System of Equations

Learning Objectives:

In this section, you will:

- Solve direct translation applications
- Solve geometry applications
- Solve uniform motion applications

Anytime we are solving a word problem, we can follow a general process to help us through:

Solve applications with systems of equations.

- Step 1. **Read** the problem. Make sure all the words and ideas are understood.
- Step 2. **Identify** what we are looking for.
- Step 3. **Name** what we are looking for. Choose variables to represent those quantities.
- Step 4. **Translate** into a system of equations.
- Step 5. **Solve** the system of equations using good algebra techniques.
- Step 6. **Check** the answer in the problem and make sure it makes sense.
- Step 7. **Answer** the question with a complete sentence.

It is often helpful to read the problem once and just try to identify what they want you to find. Do not worry about any of the numbers, context, or other ideas to start. Then read the problem again, trying to see how they want you to get the information they are asking about. Be sure to always read full sentences and not try to just pluck the values out of the equation. Beyond that, it can be helpful sometimes to draw a picture to visualize the scenario.

Example A: The sum of two numbers is 40. One number is 6 more than the other. Find the numbers.

We start by declaring the variables,

X = 1st number
Y = 2nd number

Each of the first two sentences gives us a relation so we write them as equations:

$$\{x + y = 40 \quad x = y + 6$$

Since the second equation is already solved for x , we will use substitution and substitute the expression from equation two into the first and solve:

$$\frac{y + 6 + y = 40}{2y = 34}$$

$$y = 17$$

We can now use what y is equal to into the second equation to find x :

$$x = 17 + 6 = 23$$

So the numbers are 17 and 23

Point: The value of an item multiplied by how many of the item we have gives us the total that item accounts for. It's the same reason you know that three quarters is seventy-five cents. You know how many you have and you know what they are worth.

Example B: The school that Joe attends is selling seats to a musical performance. On the first day of sales the school sold 4 normal seats and 1 premium seat for a total of \$46. The school took in \$52 on the second day by selling 3 normal seats and 2 premium seats. Find the price of each seat.

We start by declaring the variables,

X = normal seat

Y = premium seat

The second and third sentences both tell us how much money was made from selling various amounts of seats.

$$\begin{cases} 4x + y = 46 \\ 3x + 2y = 52 \end{cases}$$

For this problem, we will use the addition method. We are going to match the y variables:

$$\frac{-2(4x + y = 46)}{3x + 2y = 52}$$

$$\frac{-8x - 2y = -92}{3x + 2y = 52}$$

$$+ \frac{-8x - 2y = -92}{3x + 2y = 52}$$

$$-5x = -40$$

$$x = 8$$

Since we know the cost of the normal seat, we can now find the cost of the premium seat:

$$4(8) + y = 46$$

$$y = 14$$

So, the normal seats are \$8 and the premium seats are \$14.

Example C: Blake has 30 feet of mesh to make a chicken enclosure with. If the length needs to be twice as much as the width less three, find each dimension.

We start by declaring the variables,

X= width

Y= length

The first sentence gives the total amount to work with, keeping in mind it would be a perimeter, while the second sentence specifies each variable to each other.

$$\{2x + 2y = 30 \quad y = 2x - 3$$

For this problem we will use the substitution method. To do so we just plug the expression for y from the second equation into the first equation:

$$\frac{2x + 2(2x - 3) = 30}{6x = 36}$$

$$x=6$$

Since we know the width, we can now find the length:

$$y = 2(6) - 3 = 9$$

So, the width is 6 feet and the length is 9 feet.

Solve Uniform Motion Problems

For uniform motion, we will consider problems consisting of covering a distance D based on moving at a rate r for a time period t . The equation is $D = rt$

Example D: Julio left an hour after his brother Juan from their house. Knowing his brother is traveling at 50 mph, Julio decides to drive 60mph to catch him. How long till Julio catches his brother.

We start by declaring the variables,

X= Julio's time

Y= Juan's time

	Rate	Time	Distance
Julio	60	x	60x
Juan	50	y	50y

We know the total distance for each brother is the same, since they stop when Julio catches up. We also know that Juan has a 1-hour head start, so his time is 1 more than Julio's.

$$\begin{cases} 60x = 50y \\ y = x + 1 \end{cases}$$

For this problem we will use the substitution method. To do so we just plug the expression for y from the second equation into the first equation:

$$\begin{aligned} 60x &= 50(x + 1) \\ 60x &= 50x + 50 \end{aligned}$$

$$\begin{aligned} 10x &= 50 \\ x &= 5 \end{aligned}$$

Since we know Julio's time, we can now find Juan's:

$$y = 5 + 1 = 6$$

So, it takes Julio 5 hours to catch his brother, and in that time they have driven 300 miles.

Worksheet 4.2: Solve Applications with System of Equations

Solve the word problem:

- 1) Find the value of two numbers if their sum is 12 and their difference is 4.
- 2) If one number is three times larger than the other, and they sum up to 40, what are the two numbers.
- 3) A large pizza at Panucci's costs \$10 plus \$0.50 per topping. A large pizza at Cygnoid's is \$8 plus \$0.75 per topping. How many toppings from each pizzeria do you need if your want to spend the same amount of money.
- 4) A local convenience store sells water for \$2 and sodas for \$3. If yesterday they made \$156 and sold 8 more waters than sodas, how many of each did they sell?
- 5) The length of a rectangle is 4cm longer than the width. The perimeter is 80 cm. Find the length and the width.
- 6) A boat traveled 210 miles downstream and back. The trip downstream took 10 hours. The trip back took 70 hours. What is the speed of the boat and the speed of the current?
- 7) For a road trip, Steve and Josh take turns driving. If they both agree to drive 55 mph but Steve drives for twice as long as Josh, how long did each drive for their 605 mile journey?
- 8) A farmer needs to enclose a space along a river for his cows to graze. Since the cows will not cross the river, he only needs to enclose three sides. If he has 550 feet of fence and the length opposite the river needs to be 10 feet longer than the width, find the size of the enclosure.

5.2 Properties of Exponents and Scientific Notation

Learning Objectives

By the end of this section, you will be able to:

- Simplify expressions using the properties for exponents
- Use the definition of a negative exponent

Summary of Exponent Properties

If a and b are real numbers, and m and n are integers, then

Property	Description	Example
Product Property	$a^m \cdot a^n = a^{m+n}$	$x^3 \cdot x^4 = x^7$
Power Property	$(a^m)^n = a^{m \cdot n}$	$(x^3)^4 = x^{12}$
Product to a Power	$(ab)^m = a^m b^m$	$(2x)^3 = 2^3 x^3 = 8x^3$
Quotient Property	$\frac{a^m}{a^n} = a^{m-n}, a \neq 0$	$\frac{x^8}{x^3} = x^5$
Zero Exponent Property	$a^0 = 1, a \neq 0$	$x^0 = 1$
Quotient to a Power Property	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}, b \neq 0$	$\left(\frac{x}{y}\right)^2 = \frac{x^2}{y^2}$
Properties of Negative Exponents	$a^{-n} = \frac{1}{a^n}$ and $\frac{1}{a^{-n}} = a^n$	$x^{-3} = \frac{1}{x^3}$ $\frac{1}{x^{-5}} = x^5$
Quotient to a Negative Exponent	$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$	$\left(\frac{x}{y}\right)^{-4} = \left(\frac{y}{x}\right)^4$

We can use these properties to **simplify** exponential expressions.

An exponential expression is fully simplified when:

- 1) Parentheses are removed
- 2) Same bases are combined
- 3) The expression doesn't contain any negative exponents

Example A: Simplify

$$(5x^3y^7)^2(2x^5y)^4$$

$$(5x^3y^7)^2(2x^5y)^4$$

Remove parentheses using the Product to a Power Property

$$5^2x^6y^{14} \cdot 2^4x^{20}y^4$$

$$25x^6y^{14} \cdot 16x^{20}y^4 \quad \text{Combine same bases using the Product Property}$$

$$25 \cdot 16 \cdot x^6 \cdot x^{20} \cdot y^{14} \cdot y^4$$

$$\boxed{400x^{26}y^{18}}$$

Example B

$$\text{Simplify } \frac{(x^4)^3(x^{-5})^2}{(x^6)^{-2}}$$

$$\frac{(x^4)^3(x^{-5})^2}{(x^6)^{-2}}$$

Remove parenthesis using the Power Property

$$\frac{x^{12}x^{-10}}{x^{-12}}$$

Combine same bases: Apply the Product Property in the numerator

$$\frac{x^2}{x^{-12}}$$

Combine same bases: Use the Quotient Property

$$\boxed{x^{14}}$$

Example C

$$\text{Simplify } \left(\frac{3x^{-3}y}{x^5y^8}\right)^4$$

$$\left(\frac{3x^{-3}y}{x^5y^8}\right)^4$$

Remove parenthesis using the Product & Quotient to a Power Properties

$$\frac{3^4x^{-12}y^4}{x^{20}y^{32}}$$

Combine same bases: Use the Quotient Property

$$81x^{-32}y^{-28}$$

Simplify negative exponents

$$\boxed{\frac{81}{x^{32}y^{28}}}$$

Worksheet 5.2 Properties of Exponents and Scientific Notation

1. $2x^5 \cdot 4x \cdot x^3$

6. $\frac{5x^{-3}}{y^{-6}}$

2. $\frac{y^4}{y^7}$

7. $(m^{-2}n^5)^{-3}(m^8n)^4$

3. $\frac{a^{12}b^4}{a^7b^8}$

8. $\left(\frac{3x^6y^{-2}}{2z^{-9}}\right)^4$

4. -3^0

9. $\frac{(t^{-5})^4(t^3)^6}{(t^{-1})^{-7}}$

5. $\left(\frac{2}{5}\right)^{-3}$

10. $\left(\frac{3x^{-4}y^2}{z}\right)^{-2}\left(\frac{4x^3y}{z^{-6}}\right)^4$

6.1 Greatest Common Factor and Factor by Grouping

Learning Objectives: In this section, you will:

- Find the greatest common factor of two or more expressions.
- Factor the greatest common factor from a polynomial.
- Factor by grouping.

Find the Greatest Common Factor of Two or More Expressions:

Step 1. Factor each coefficient into primes. Write all variables with exponents in expanded form.

Step 2. List all factors-matching common factors in a column. In each column, circle the common factors.

Step 3. Bring down the common factors that all expressions share.

Step 4. Multiply the factors.

Example A: Find the greatest common factor of $42x^4$, $21x^2$, $49x^3$

Solution: Factor each coefficient into primes and write the variables with exponents in expanded form.

$$\begin{aligned}42x^4 &= 2 \cdot 3 \cdot 7 \cdot x \cdot x \cdot x \cdot x \\21x^2 &= 3 \cdot 7 \cdot x \cdot x \\49x^3 &= 7 \cdot 7 \cdot x \cdot x \cdot x \\GCF &= 7 \cdot x \cdot x \text{ The common factors are in red.}\end{aligned}$$

The GCF of $42x^4$, $21x^2$, $49x^3$ is $7x^2$.

Factor the Greatest Common Factor from a Polynomial:

Step 1. Find the GCF of all the terms of the polynomial.

Step 2. Rewrite each term as a product using the GCF.

Step 3. Use the “reverse” Distributive Property to factor the expression.

Step 4. Check by multiplying the factors.

Example B: Factor $16x^5y^4 - 8x^3y^3 - 24x^2y$

Solution: Find GCF of $16x^5y^4$, $-8x^3y^3$, and $-24x^2y$

$$\begin{aligned}16x^5y^4 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot x \cdot x \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \\-8x^3y^3 &= -2 \cdot 2 \cdot 2 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \\-24x^2y &= -2 \cdot 2 \cdot 2 \cdot 3 \cdot x \cdot x \cdot y \\GCF &= 2 \cdot 2 \cdot 2 \cdot x \cdot x \cdot y\end{aligned}$$

$$GCF = 8x^2y$$

Rewrite each term using the GCF, $8x^2y$

$$8x^2y \cdot 2x^3y^3 - 8x^2y \cdot xy^2 - 8x^2y \cdot 3$$

Factor the GCF. $8x^2y(2x^3y^3 - xy^2 - 3)$

Factor by Grouping:

Step 1. Group terms with common factors.

Step 2. Factor out the common factor in each group.

Step 3. Factor the common factor from the expression.

Step 4. Check by multiplying the factors.

Example C) Factor by grouping: $3x^2y + 6xy - 2x^2 - 4x$

Solution:

$$\begin{aligned} & x[3xy + 6y - 2x - 4] \quad \text{factor out GCF from all four terms} \\ & \quad x[3xy + 6y - 2x - 4] \quad \text{Separate into two parts} \\ & x[3y(x + 2) - 2(x + 2)] \quad \text{Factor the GCF in each group} \\ & \quad x[(3y - 2)(x + 2)] \quad \text{Factor out the common factor} \end{aligned}$$

Worksheet 6.1: Greatest Common Factor and Factor by Grouping

Find the Greatest Common Factor of Two or More Expressions: In the following exercises, find the greatest common factor.

1) $66x^4, 88x^3y$

3) $96p^5q^6, 120p^4q^3, 168p^2q^2$

2) $78m^8, 39m^5, 52m^3$

4) $60r^9s^8, 105r^8s^8, 165r^8s^6$

Factor the Greatest Common Factor from a Polynomial: In the following exercises, factor the greatest common factor from each polynomial.

5) $56p - 24$

8) $35p^4q^3 - 65p^2q^2 + 55p^3q$

6) $4a^3 + 8a^2b + 12ab^2$

9) $5x(2x - 3) - 4(2x - 3)$

7) $6m^2n^3 + 9m^3n^2 + 27mn^4$

10) $8r^2(3r + 2s) - 9r(3r + 2s)$

Factor by Grouping: In the following exercises, factor by grouping.

11) $x^3 + x^2 + x + 1$

15) $42p^2 - 7p - 18pq + 3q$

12) $x^2 + 4x - x - 4$

16) $4m^2 - 8 + m^2n - 2n$

13) $4x^2 - 12xy + 3x - 9y$

17) $20a^2 - 25a - 16a + 20$

14) $5x^2 - 10xy + 15x - 30y$

18) $56p^2q + 55q - 35p^2 - 88q^2$

6.2 Factor Trinomials

Learning Objectives:

In this section, you will:

- Factor trinomial of the form $x^2 + bx + c$.
- Factor trinomial of the form $ax^2 + bx + c$ using AC method.
- Factor using substitution.

Factor Trinomials of the Form $x^2 + bx + c$:

Step 1. Write the factors as two binomials with first term x . $x^2 + bx + c = (x \quad)(x \quad)$

Step 2. Find two numbers m and n that:

- Multiply to c , $m \cdot n = c$
- Add to b , $m + n = b$

Step 3. Use m and n as the last terms of the factors. $(x + m)(x + n)$

Step 4. Check by multiplying the factors.

Example A: Factor $x^2 - 3x - 88$.

Solution: Write the factor as two binomials with first term x . $(x \quad)(x \quad)$

Find two number that multiply to -88 and add to -3

Factor of -88	Sum of factors
1, -88	$1 + (-88) = -87$
2, -44	$2 + (-44) = -42$
4, -22	$4 + (-22) = -18$
8, -11	$8 + (-11) = -3$

Use 8 and -11 as the last terms of the factors.

$(x + 8)(x - 11)$

$x^2 - 3x - 88 = (x + 8)(x - 11)$

Check: $(x + 8)(x - 11) = x^2 - 11x + 8x - 88 = x^2 - 3x - 88$

Example B: Factor $x^2 + 5x + 2$.

Solution: Write the factor as two binomials with first term x . $(x \quad)(x \quad)$

Find two number that multiply to 2 and add to 5

Factor of 2	Sum of factors
1, 2	$1 + 2 = 3$
-1, -2	$-1 + (-2) = -3$

Since none of the factors of 2 sum up to 5, hence this is an example of Prime factorization.

Another word this means that $x^2 + 5x + 2$ is not factorable.

Factor Trinomials of the Form $ax^2 + bx + c$ using the “ac” Method:

Step 1. Factor any GCF.

Step 2. Find the product ac .

Step 3. Find two numbers m and n that:

- Multiply to ac , $m \cdot n = a \cdot c$
- Add to b , $m + n = b$

Step 4. Split the middle term using m and n . Rewrite $ax^2 + bx + c$ as $ax^2 + mx + nx + c$

Step 5. Factor by grouping.

Step 6. Check by multiplying factors.

Example C: Factor $28x^2 - 58x - 30$.

Solution: Factor out any GCF first.

$$28x^2 - 58x - 30 = 2(14x^2 - 29x - 15)$$

Find the product ac : $ac = 14 \cdot (-15) = -210$

Find two numbers that multiply to -210 and sum up to -29

$$\begin{aligned}(-35) \cdot 6 &= -210 \\ -35 + 6 &= -29\end{aligned}$$

Split the middle term: $2(14x^2 - 29x - 15)$

$$2(14x^2 + 6x - 35x - 15)$$

Factor the trinomial by grouping:

$$\begin{aligned}2(2x(7x + 3) - 5(7x + 3)) \\ 2(7x + 3)(2x - 5)\end{aligned}$$

Check by multiplying all three factors.

$$\begin{aligned}2(7x + 3)(2x - 5) \\ 2(14x^2 - 35x + 6x - 15) \\ 2(14x^2 - 29x - 15) \\ 28x^2 - 58x - 30\end{aligned}$$

Factor Using Substitution: It is standard to use u for the substitution that allow us to make it fit the $ax^2 + bx + c$ form.

Example D: Factor by substitution: $x^4 - 2x^2 - 48$.

Solution: The variable part of the middle term is x^2 and its square, x^4 , is the variable part of the first term. If we let $u = x^2$, we can put our trinomial in the $ax^2 + bx + c$ form we need to factor it.

Let $u = x^2$ $u^2 - 2u - 48$

Factor the trinomial. Find two number that multiply to -48 and add to -2 .

$$\begin{aligned}(-8) \cdot 6 &= -48 \\ -8 + 6 &= -2\end{aligned}$$

Hence the factor of $u^2 - 2u - 48$ is $(u - 8)(u + 6)$

Replace u with x^2 : $(x^2 - 8)(x^2 + 6)$

Check: $(x^2 - 8)(x^2 + 6)$

$$x^4 + 6x^2 - 8x^2 - 48$$

$$x^4 - 2x^2 - 48$$

Example E: Factor by substitution: $5(x + 1)^2 + 4(x + 1) - 12$.

Solution: The binomial in the middle term, $(x + 1)$ is squared in the first term. If we let $u = x + 1$ and substitute, our trinomial will be in $ax^2 + bx + c$ form.

Let $u = x + 1$

$$5u^2 + 4u - 12$$

Factor the trinomial. Find two number that multiply to -60 and add to 4 .

$$-6 \cdot 10 = -60$$

$$-6 + 10 = 4$$

Split the middle term: $5u^2 + 4u - 12 = 5u^2 + 10u - 6u - 12$

Factor by grouping: $5u(u + 2) - 6(u + 2)$

Hence the factor is $(u + 2)(5u - 6)$

Replace u with $x + 1$: $((x + 1) + 2)(5(x + 1) - 6)$

Simplify: $(x + 3)(5x - 1)$

Worksheet 6.2: Factor Trinomials

Factor Trinomials of the Form $x^2 + bx + c$: In the following exercises, factor each trinomial.

1) $a^2 + 7a + 10$

4) $r^2 - 24r + 143$

7) $r^2 + 3rs - 180s^2$

2) $x^2 - 2x + 17$

5) $2p^2 - 28p + 98$

8) $-3x^3 + 6x^2 + 189x$

3) $y^2 + 13y - 14$

6) $x^2 + 5xy - 40y^2$

9) $5a^4 - 10a^3 - 120a^2$

Factor Trinomials of the Form $ax^2 + bx + c$ using the “ac” Method: In the following exercises, factor using the “ac” Method.

10) $2x^2 + 9x + 4$

13) $20a^2 - 67 + 56$

16) $-42y^2 - 36y + 6$

11) $5a^2 - 14a + 8$

14) $-3x^2 + 14x - 15$

17) $32x^2 - 48x + 18$

12) $10x^2 - x - 24$

15) $18r^2 - 33r + 14$

18) $-30p^2 + 16p + 24$

Factor Using Substitution: In the following exercises, factor using substitution.

19) $x^4 + 6x^2 - 27$

22) $(x - 1)^2 - 17(x - 1) + 72$

20) $x^4 + 8x^2 + 12$

23) $(x + 4)^2 + 19(x + 4) + 84$

21) $3x^4 + 14x^2 + 8$

24) $6(x + 2)^2 - 4(x + 2) - 2$

6.3 Factor Special Products

Learning Objectives:

In this section, you will:

- Factor perfect square trinomials
- Factor differences of squares
- Factor sums and differences of cubes

Perfect Square Trinomials Pattern: If a and b are real numbers

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Step 1. Does the trinomial fit the pattern?

- Check to see if the leading coefficient is a perfect square, a^2 .
- Check that the last term is a perfect square, b^2 .
- Check the middle term is it the product of $2ab$.

Step 2. Write the square of the binomial.

Step 3. Check by multiplying.

Example A: Factor the following trinomial: a) $4x^2 + 36x + 81$; b) $49x^2 - 168xy + 144y^2$

Solution:

a) Are the first and last terms perfect squares?

Yes: $(2x)^2$ $(9)^2$

Check the middle term.

$$2(2x)(9) = 36x \quad \text{Check!}$$

Write as the square of a binomial.

$$(2x + 9)^2$$

Check by multiplying:

$$(2x + 9)(2x + 9) = 4x^2 + 18x + 18x + 81 \\ 4x^2 + 36x + 81 \quad \text{Check!}$$

b) Are the first and last terms perfect squares?

Yes: $(7x)^2$ $(12y)^2$

Check the middle term.

$$2(7x)(12y) = 168xy \quad \text{Check!}$$

Write as the square of a binomial.

$$(7x - 12y)^2$$

Check by multiplying:

$$(7x - 12y)(7x - 12y) = 49x^2 - 84x - 84x + 144 \\ 49x^2 - 168xy + 144y^2 \quad \text{Check!}$$

Differences of Squares Pattern: If a and b are real numbers, $a^2 - b^2 = (a - b)(a + b)$.

Step 1. Does the binomial fit the pattern?

- Is this a difference?
- Are the first and last terms perfect squares?

Step 2. Write them as squares.

Step 3. Write the product of conjugates.

Step 4. Check by multiplying.

Example B: Factor the following: a) $200x^3 - 72x$; b) $x^2 + 4x + 4 - 9y^4$

Solution:

a) Is there a GCT? Yes, $8x$ factor it out:

$$8x(25x^2 - 9)$$

Is the binomial a difference of squares?

$$\text{Yes: } 8x((5x)^2 - (3)^2)$$

Factor as a product of conjugates.

$$8x(5x - 3)(5x + 3)$$

Check by multiplying:

$$\begin{aligned} &8x(5x - 3)(5x + 3) \\ &= 8x(25x^2 - 9) \\ &200x^3 - 72x \quad \text{Check!} \end{aligned}$$

b) Use the perfect square trinomial pattern.

$$\begin{aligned} &x^2 + 4x + 4 - 9y^4 \\ &(x + 2)^2 - 9y^4 \end{aligned}$$

Is this a difference of square? Yes:

$$(x + 2)^2 - (3y^2)^2$$

Factor as a product of conjugates.

$$((x + 2) - 3y^2)((x + 2) + 3y^2)$$

Rewrite the solution

$$(3y^2 + x + 2)(-3y^2 + x + 2)$$

Sum and Difference of Cubes Pattern:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Step 1. Does the binomial fit the sum or difference of cubes pattern?

- Is it a sum or difference?
- Are the first and last terms perfect cubes?
- Check the middle term is it the product of $2ab$.

Step 2. Write them as cubes.

Step 3. Use either the sum or difference of cubes pattern.

Step 4. Simplify inside the parentheses.

Step 3. Check by multiplying the factors.

Example C: Factor the following: a) $16x^3y^2 - 686y^5$; b) $(x + 1)^3 + y^3$

Solution:

a) Is there a GCT? Yes, $4y^2$ factor it out:

$$2y^2(8x^3 - 343y^3)$$

This binomial is a difference of two cubes.

$$2y^2((2x)^3 - (7y)^3)$$

Use the difference of cubes pattern:

$$\begin{aligned} &(a - b)(a^2 + ab + b^2) \\ &2y^2(2x - 7y)(4x^2 + 14xy + 49y^2) \end{aligned}$$

b) This binomial is a sum of two cubes.

$$(x + 1)^3 + (y)^3$$

Use the sum of cubes pattern:

$$\begin{aligned} &(a + b)(a^2 - ab + b^2) \\ &((x + 1) + y)((x + 1)^2 - (x + 1)y + (y)^2) \end{aligned}$$

Simplify.

$$(x + y + 1)(x^2 + 2x + 1 - xy - y + y^2)$$

Worksheet 6.3: Factor Special Products

Factor Perfect Square Trinomials: In the following exercises, factor completely using the perfect square trinomial pattern.

1) $9x^2 + 48x + 64$

4) $300a^3 - 660a^2b + 363ab^2$

2) $4x^2 - 28x + 49$

5) $16x^4 + 24x^2y + 9y^2$

3) $2p^4 + 36p^2q^6 + 162q^{12}$

6) $100x^2 - 40xy^3 + 4y^6$

Factor Difference of Squares: In the following exercises, factor completely using difference of squares pattern, if possible.

7) $4x^2 - \frac{1}{9}$

10) $64p^4 - \frac{1}{16}q^2$

13) $-72x^3 + 338y^4x$

8) $144x^8 - 1$

11) $12x^3 - 81x$

14) $x^2 + 2xy + y^2 - 25x^4$

9) $25x^2 + 4$

12) $3x^2y - 12y^3$

15) $a^2 - 8a + 16 - 36b^2$

Factor Sums and Difference of Cubes: In the following exercises, factor completely using sums and difference of cubes pattern, if possible.

16) $x^3 + 1000$

19) $250p^3 + 2$

22) $1024q + 2p^9q$

17) $64 - 27a^3$

20) $27p^3q + \frac{q}{64}$

23) $729 - (x + 1)^3$

18) $8y^6 - \frac{1}{27}$

21) $8x^3y^2 - 216y^2$

24) $(x - 2)^3 - 27y^3$

6.4 General Strategy for Factoring Polynomials

Learning Objectives:

In this section, you will:

- Recognize and use the appropriate method to factor a polynomial completely

Use a general strategy for factoring polynomials:

Step 1. Factor out Greatest Common Factor if possible.

Step 2. Determine if the polynomial is a binomial, trinomial or are there more than three terms?

- If it is a binomial:
 - Is it a sum?
Of squares? Sums of squares do not factor.
Of cubes? Use the sum of cubes pattern.
 - Is it difference?
Of squares? Factor as the product of conjugates.
Of cubes? Use the difference of cubes pattern.
- If it is a trinomial:
 - Is it of the form $x^2 + bx + c$? Undo FOIL.
 - Is it of the form $ax^2 + bx + c$?
If a and c are squares, check if it fits the perfect square trinomial pattern.
Otherwise use “ac” method.
- If it has more than three terms:
 - Use the grouping method

Step 3. Check.

Is it factored completely?

Do the factors multiply back to the original polynomial?

Example A: Factor completely: $4x^3 - 40x^2 - 156x$

Solution:

Is there a GCF? Yes, $4x$

Factor out the GCF. $4x(x^2 - 10x - 39)$

In the parentheses, is a trinomial with leading coefficient 1. Find two number that multiply to -39 and add to -10 .

$$-13 \cdot 3 = -39$$

$$-13 + 3 = -10$$

Undo FOIL: $4x(x - 13)(x + 3)$

Is the expression factored completely? Yes

Check your answer: $4x(x - 13)(x + 3) = 4x(x^2 - 10x - 39) = 4x^3 - 40x^2 - 156x$

Example B: Factor completely: $4x^2 - 34x + 60$

Solution:

Is there a GCF? Yes, 2

Factor out the GCF. $2(2x^2 - 17x + 30)$

In the parentheses, is a trinomial with leading coefficient 2. Use “ac” method to factor the trinomial. Find two number that multiply to 60 and add to -17 .

$$-12 \cdot (-5) = 60$$

$$-12 + (-5) = -17$$

Split the middle term: $2(2x^2 - 12x - 5x + 30)$

Factor by grouping: $2(2x(x - 6) - 5(x - 6))$

Factor out the binomial: $2(x - 6)(2x - 5)$

Is the expression factored completely? Yes

Check your answer: $2(x - 6)(2x - 5) = 2(2x^2 - 12x - 5x + 30) = 2(2x^2 - 17x + 30)$
 $= 4x^2 - 34x + 60$ check!

Example C: Factor completely: $200p^2q - \frac{2q}{81}$

Solution:

Is there a GCF? Yes, $2q$

Factor out the GCF. $2q(100p^2 - \frac{1}{81})$

In the parentheses, is a binomial with the difference of two squares. $2q \left((10p)^2 - \left(\frac{1}{9}\right)^2 \right)$

Write as a product of conjugates

$$2q \left(10p - \frac{1}{9} \right) \left(10p + \frac{1}{9} \right)$$

Is the expression factored completely? Yes

Check your answer:

$$2q \left(10p - \frac{1}{9} \right) \left(10p + \frac{1}{9} \right) = 2q \left(100p^2 - \frac{1}{81} \right) = 200p^2q - \frac{2q}{81} \quad \text{check!}$$

Example D:) Factor completely: $p^2q - \frac{2q}{81}$

Solution:

Is there a GCF? Yes, $2q$

Factor out the GCF. $2q(100p^2 - \frac{1}{81})$

In the parentheses, is a binomial with the difference of two squares. $2q \left((10p)^2 - \left(\frac{1}{9}\right)^2 \right)$

Write as a product of conjugates

$$2q \left(10p - \frac{1}{9} \right) \left(10p + \frac{1}{9} \right)$$

Is the expression factored completely? Yes

Check your answer:

$$2q \left(10p - \frac{1}{9} \right) \left(10p + \frac{1}{9} \right) = 2q \left(100p^2 - \frac{1}{81} \right) = 200p^2q - \frac{2q}{81} \quad \text{check!}$$

Worksheet 6.4: General Strategy for Factoring Polynomials

Recognize and Use the Appropriate Method to Factor a Polynomial Completely: In the following exercises, factor completely.

1) $3x^3 - 27x^2 + 81x$

2) $5a^5 - 25a^3$

3) $8r^3 - 72r^2 + 144r$

4) $24y^3 - 144y^2 + 216y$

5) $15x^5y^2 - 240xy^2$

6) $30rs - 24 - 30r + 24s$

7) $18uv - 9u + 24v - 12$

8) $36x^2 - 120xy + 100y^2$

9) $12p^5 + 12p^2q^3$

10) $6x^4y + 15xy^4$

11) $49a^2 + 70ab + 25b^2$

12) $24x^6 - 81y^3$

13) $5x^4 - 5000$

14) $45r^2s - 63rs + 18s$

15) $72x^2y - 90xy + 36y$

16) $25x^2 - 30xy + 9y^2 - 36$

17) $56p^2q - 72pq + 28q$

18) $36p^2 - 48pq + 16q^2 - 81$

19) $(2z + 3)^2 - 4(2z + 3) + 4$

20) $4x^4 + 17x^2 + 12$

21) $6r^4 + 26r^2 + 16$

22) $16(y + 2)^2 + 28(y + 2) + 49$

6.5 Polynomial Equations

Learning Objectives:

In this section, you will:

- Use the Zero Product Property
- Solve quadratic equations by factoring
- Solve equation with polynomial functions
- Solve applications modeled by polynomial equations

Polynomial Equation: A **polynomial equation** is an equation that contains a polynomial expression. The **degree of the polynomial equation** is the degree of the polynomial.

Quadratic Equation: An equation of the form $ax^2 + bx + c = 0$ is called a quadratic equation. a , b , and c are real numbers and $a \neq 0$.

Use the Zero Product Property: If $a \cdot b = 0$, then $a = 0$ or $b = 0$ or both.

Step 1. Set each factor equal to zero.

Step 2. Solve the linear equations.

Step 3. Check.

Example A: Solve $(8x + 3)(7x - 28) = 0$

Solution: Since the product equals zero, set each factor equal to zero.

$$8x + 3 = 0 \text{ or } 7x - 28 = 0$$

Solve each equation: $x = -\frac{3}{8}$ or $x = 4$

Check: when $x = -\frac{3}{8}$

$$\begin{aligned} \left(8\left(-\frac{3}{8}\right) + 3\right)\left(7\left(-\frac{3}{8}\right) - 28\right) &= 0 \\ (-3 + 3)\left(-\frac{21}{8} - 28\right) &= 0 \\ (0)\left(-\frac{21}{8} - \frac{224}{8}\right) &= 0 \end{aligned}$$

$$0 = 0 \text{ Check!}$$

Check: when $x = 4$

$$\begin{aligned} (8(4) + 3)(7(4) - 28) &= 0 \\ (32 + 3)(28 - 28) &= 0 \\ (35)(0) &= 0 \\ 0 &= 0 \text{ Check!} \end{aligned}$$

Solve Quadratic Equations by Factoring:

Step 1. Write the quadratic equation in standard form, $ax^2 + bx + c = 0$.

Step 2. Factor the quadratic expression.

Step 3. Use the Zero Product Property.

Step 4. Solve the linear equations.

Step 5. Check. Substitute each solution separately into the original equation.

Example B: Solve $28x^2 - 6 = 13x$

Solution: Write the quadratic equation in standard form. $28x^2 - 13x - 6 = 0$

Factor the quadratic expression using “ac” method.

Find two numbers that multiply to -168 and sum up to -13

$$(-21) \cdot 8 = -168$$

$$-21 + 8 = -13$$

Split the middle term: $28x^2 + 8x - 21x - 6 = 0$

Factor the trinomial by grouping:

$$\begin{aligned} 4x(7x + 2) - 3(7x + 2) &= 0 \\ (7x + 2)(4x - 3) &= 0 \end{aligned}$$

Use the Zero Product Property: $(7x + 2) = 0$ or $(4x - 3) = 0$

Solve the linear equations: $x = -\frac{2}{7}$ or $x = \frac{3}{4}$

Check: when $x = -\frac{2}{7}$

$$\begin{aligned} 28\left(-\frac{2}{7}\right)^2 - 6 &= 13\left(-\frac{2}{7}\right) \\ 28\left(\frac{4}{49}\right) - 6 &= -\frac{26}{7} \\ \frac{112}{49} - \frac{294}{49} &= -\frac{26}{7} \\ -\frac{182}{49} &= -\frac{26}{7} \text{ Check!} \end{aligned}$$

Check: when $x = \frac{3}{4}$

$$\begin{aligned} 28\left(\frac{3}{4}\right)^2 - 6 &= 13\left(\frac{3}{4}\right) \\ 28\left(\frac{9}{16}\right) - 6 &= \frac{39}{4} \\ \frac{63}{4} - \frac{24}{4} &= \frac{39}{4} \\ \frac{39}{4} &= \frac{39}{4} \text{ Check!} \end{aligned}$$

Example C: Solve $(2x - 5)(x + 2) = 7x$

Solution: Multiply the binomials. $2x^2 - x - 10 = 7x$

Write the quadratic equation in standard form. $2x^2 - 8x - 10 = 0$

Factor out GCF. $2(x^2 - 4x - 5) = 0$

Factor the trinomial. $2(x - 5)(x + 1) = 0$

Simplify by dividing 2 to both sides. $\frac{2(x-5)(x+1)=0}{2} \quad \frac{2}{2}$
 $(x - 5)(x + 1) = 0$

Use the Zero Product Property to set each factor to 0. $x - 5 = 0$ or $x + 1 = 0$

Solve each equation. $x = 5$ or $x = -1$

Solve Equations with Polynomial Functions

Example D: For the function $f(x) = x^2 - 2x - 15$, a) find x when $f(x) = 20$; b) find two points that lie on the graph of the function.

Solution:

a) Substitute 20 for $f(x)$. $20 = x^2 - 2x - 15$

Put the quadratic in standard form. $0 = x^2 - 2x - 35$

Factor the trinomial. $0 = (x - 7)(x + 5)$

Use the zero product property. $x - 7 = 0$ or $x + 5 = 0$

Solve. $x = 7$ or $x = -5$

Check: when $x = 7$

$$f(7) = (7)^2 - 2(7) - 15$$

$$f(7) = 49 - 14 - 15$$

$$f(7) = 20 \text{ Check!}$$

Check: when $x = -5$

$$f(-5) = (-5)^2 - 2(-5) - 15$$

$$f(-5) = 25 + 10 - 15$$

$$f(-5) = 20 \text{ Check!}$$

b) Since $f(7) = 20$ and $f(-5) = 20$, the points $(7, 20)$ and $(-5, 20)$ lie on the graph of the function.

Zero of a Function: For any function f , if $f(x) = 0$, then x is a **zero of the function**.

Example E: For the function $f(x) = 6x^2 + 17x - 14$, find a) the zeros of the function; b) any x -intercepts of the graph of the function; c) any y -intercepts of the graph of the function.

Solution:

a) To find the zeros of the function, set $f(x) = 0$.

$$0 = 6x^2 + 17x - 14$$

Factor the trinomial.

Use the zero product property.

Solve.

$$(3x - 2)(2x + 7) = 0$$

$$3x - 2 = 0 \text{ or } 2x + 7 = 0$$

$$x = \frac{2}{3} \text{ or } x = -\frac{7}{2}$$

b) An x -intercept occurs when $y = 0$. Since $f\left(\frac{2}{3}\right) = 0$ and $f\left(-\frac{7}{2}\right) = 0$, the points $\left(\frac{2}{3}, 0\right)$ and $\left(-\frac{7}{2}, 0\right)$ lie on the graph. These points are x -intercepts of the function.

c) A y -intercept occurs when $x = 0$. To find the y -intercepts we need to find $f(0)$.

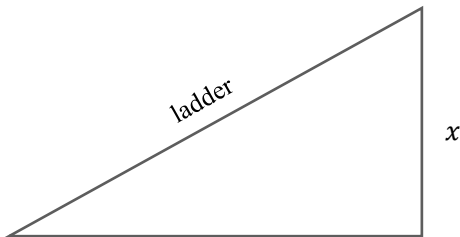
$$f(0) = 6(0)^2 + 17(0) - 14$$

$$\text{Simplify. } f(0) = -14$$

Since $f(0) = -14$, the point $(0, -14)$ lies on the graph. This point is the y -intercept of the function.

Solve Applications Modeled by Polynomial Equations:

Example F: A ladder leans against a wall. The length of the ladder is 20 feet. The distance from the base of the ladder to the wall is 4 feet more than the height at which the ladder touches the wall. Find the height at which the ladder touches the wall.



Solution: x be $x + 4$ in feet from the floor to the point where the ladder touches the wall.

The distance from the base of the ladder to the wall is $x + 4$ feet.

Since this is a right triangle, we can use the Pythagorean Theorem. $a^2 + b^2 = c^2$.

Substitute in the variables.

$$x^2 + (x + 4)^2 = 20^2$$

Solve the equation.

$$x^2 + (x^2 + 8x + 16) = 400$$

Simplify.

$$2x^2 + 8x + 16 = 400$$

Rewrite in Standard form.

$$2x^2 + 8x - 384 = 0$$

Factor out GCF.

$$2(x^2 + 4x - 192) = 0$$

Factor the trinomial.

$$2(x + 16)(x - 12) = 0$$

Use the Zero Product Property. $2 \neq 0$ or $x + 16 = 0$ or $x - 12 = 0$

Solve.

$$2 \neq 0 \text{ or } x = -16 \text{ or } x = 12$$

Since x is a side of the triangle, $x = -16$ does not make sense. Hence $x = 12$.

The height at which the ladder touches the wall is 12 feet.

Worksheet 6.5: Polynomial Equations

Use the Zero Product Property: In the following exercises, solve.

1) $(b - 3)(2b + 5) = 0$

3) $y(6y - 7)^2 = 0$

2) $(x + 2)(3x - 8)(5x + 6) = 0$

4) $4x(2x + 9)(3x - 5) = 0$

Solve Quadratic Equations by Factoring: In the following exercises, solve.

5) $x^2 = 10x - 16$

10) $(x + 1)(x - 2) = 10$

6) $-9x + 15 = -x^2$

11) $(3a - 2)(2a + 5) = 4a$

7) $2x^2 + 9x = 5$

12) $4m^2 - 8m = 12$

8) $64s^2 = 196$

13) $6n^2 - 11n = -4$

9) $4r^2 - 28r = -49$

14) $12x^3 - 18x^2 = -6x$

Solve Equations with Polynomial Functions: In the following exercise, solve.

15) For the function, $f(x) = x^2 - 7x + 10$, a) when $f(x) = 2$, b) use this information to find two points that lie on the graph of the function.

16) For the function, $f(x) = 4x^2 - 12x + 10$, a) when $f(x) = 1$, b) use this information to find two points that lie on the graph of the function.

In the following exercises, for each function, find a) the zeros of the function; b) the x -intercepts of the graph of the function; c) the y -intercepts of the graph of the function.

17) $f(x) = 2x^2 + 5x - 3$

19) $f(x) = 25x^2 + 20x + 4$

18) $f(x) = x^2 - 16$

20) $f(x) = -2x^2 + 3x - 1$

Solve Applications Modeled by Quadratic Equations: In the following exercises, solve.

21) A triangular garden bed has one leg that is 3 meters longer than the other leg. The hypotenuse of the triangle is 15 meters. Calculate the lengths of the legs.

22) A ladder leans against a building. The ladder is 25 feet long. The distance from the base of the ladder to the building is 5 feet less than the height at which the ladder touches the building. Find the height at which the ladder touches the building.

23) A rectangular room has an area of 150 square feet. The length is 5 feet more than twice the width. Find the dimensions of the room.

24) A student throws a ball from a rooftop that is 50 feet above a playground. The function $h(t) = -16t^2 + 20t + 50$ describes the height h of the ball above the playground as a function of time t in seconds. Determine:

a) The time(s) when the ball lands on the playground.

- b) When the ball is at a height of 50 feet.
- c) The height of the ball at $t = 1$ second.

7.1 Multiply and Divide Rational Expressions

Learning Objectives

By the end of this section, you will be able to:

- Determine the values for which a rational expression is undefined
- Simplify rational expressions
- Multiply and divide rational expressions
- Multiply and divide rational functions

A **rational expression** is an expression of the form $\frac{P}{Q}$ where P and Q are polynomials and $Q \neq 0$.

Examples: $\frac{13}{20}$ $\frac{5x^4}{7y}$ $\frac{2x+6}{x^2-9}$ $\frac{x^2-2x+1}{x-1}$

If the denominator of a rational expression is zero, the expression is **undefined**.

To find values that would make the rational expression undefined:

1. Set the denominator equal to 0.
2. Solve the equation.

Example A: Determine the value for which the rational expression is undefined.

a) $\frac{2x-7}{3x+5}$

b) $\frac{x+2}{x^2+5x+6}$

- a) Set the denominator equal to zero
Solve the linear equation by “getting x alone”

$$\begin{aligned} 3x + 5 &= 0 \\ 3x &= -5 \\ x &= -\frac{5}{3} \end{aligned}$$

- b) Set the denominator equal to zero
Solve the quadratic equation by factoring

$$\begin{aligned} x^2 + 5x + 6 &= 0 \\ (x + 2)(x + 3) &= 0 \\ x + 2 = 0 \text{ or } x + 3 &= 0 \\ x = -2 \text{ or } x &= -3 \end{aligned}$$

Simplifying Rational Expressions

To simplify a rational expression:

1. Factor the numerator and denominator completely
2. Simplify by dividing out common factors

NOTE: Be sure you only cancel out common FACTORS. Factors are pieces that are being MULTIPLIED. Never cancel out terms (pieces being added/subtracted).

Can cancel the x's: $\frac{3x}{5x}$

Cannot cancel the x's: $\frac{3+x}{5-x}$

Example B: Simplify $\frac{x^2+7x+12}{x^2+2x-8}$

$$\frac{x^2 + 7x + 12}{x^2 + 2x - 8}$$

$$\frac{(x+3)(x+4)}{(x-2)(x+4)}$$

$$\frac{(x+3)\cancel{(x+4)}}{(x-2)\cancel{(x+4)}}$$

$$\frac{x+3}{x-2}$$

Factor the numerator and denominator

Cancel the common factors

Note that we stop here and do not cancel the x's since they are terms

Example C: Simplify $\frac{1-x^2}{x^2-3x+2}$

$$\frac{1 - x^2}{x^2 - 3x + 2}$$

$$\frac{(1+x)(1-x)}{(x-2)(x-1)}$$

$$\frac{(1+x)\cancel{(1-x)}}{(x-2)\cancel{(x-1)}} (-1)$$

$$\frac{1+x}{x-2}$$

Factor the numerator and denominator

$(1-x)$ and $(x-1)$ are **opposites**

$$\frac{(1-x)}{(x-1)} = \frac{-x+1}{x-1} = \frac{-(x-1)}{x-1} = -1$$

When you cancel opposites, multiply by -1

Note that we stop here and do not cancel the x's since they are terms

Multiplying and Dividing Rational Expressions

To multiply or divide rational expressions:

1. If dividing, rewrite as multiplication ("keep, change, flip")

2. Factor the numerator and denominator completely
3. Multiply the numerators and multiply the denominators
4. Simplify by dividing out common factors

Example D: Multiply $\frac{x^2+7x+10}{4x} \cdot \frac{8x^2}{x^2-25}$

$$\frac{x^2+7x+10}{4x} \cdot \frac{8x^2}{x^2-25}$$

Factor

$$\frac{(x+5)(x+2)}{4x} \cdot \frac{8x^2}{(x+5)(x-5)}$$

Multiply numerators and denominators

$$\frac{8x^2(x+5)(x+2)}{4x(x+5)(x-5)}$$

Cancel common factors

$$\frac{\cancel{8x^2}(x+5)(x+2)}{\cancel{4x}(x+5)(x-5)}$$

Reduce $\frac{8x^2}{4x}$

$$\frac{2x(x+2)}{x-5}$$

Example E: Divide $\frac{x^2+8x+15}{x+5} \div \frac{x^2-16}{5x-20}$

$$\frac{x^2+8x+15}{x+5} \div \frac{x^2-16}{5x-20}$$

Rewrite as multiplication

$$\frac{x^2+8x+15}{x+5} \cdot \frac{5x-20}{x^2-16}$$

Factor

$$\frac{(x+5)(x+3)}{x+5} \cdot \frac{5(x-4)}{(x+4)(x-4)}$$

Multiply numerators and denominators

$$\frac{5(x+5)(x+3)(x-4)}{(x+5)(x+4)(x-4)}$$

Cancel common factors

$$\frac{5\cancel{(x+5)}(x+3)\cancel{(x-4)}}{\cancel{(x+5)}(x+4)\cancel{(x-4)}}$$

$$\frac{5(x+3)}{x+4}$$

Note that this division problem could have also been written as:

Simplify:
$$\frac{\frac{x^2+8x+15}{x+5}}{\frac{x^2-16}{5x-20}}$$

This is called a **complex fraction**. Since a fraction bar means division, a complex fraction is another way of writing division of two fractions.

Rational Functions

A **rational function** is a function of the form $R(x) = \frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$.

The **domain** of a rational function is all real numbers except for those that would cause zero in the denominator.

To find the domain of a rational function:

1. Set the denominator equal to 0.
2. Solve the equation.
3. The domain is all real numbers excluding the values found in step 2.

Example F: Find the domain of $R(x) = \frac{5x}{2x^2+x-3}$

$$2x^2 + x - 3 = 0$$

Set the denominator equal to zero

$$(2x+3)(x-1) = 0$$

Solve the equation (it's quadratic, so factor)

$$2x+3 = 0 \text{ or } x-1 = 0$$

$$x = -\frac{3}{2} \text{ or } x = 1$$

Write the domain: The domain is all real numbers where $x \neq -\frac{3}{2}$ and $x \neq 1$

Example G: Find $R(x) = \frac{f(x)}{g(x)}$ where $f(x) = \frac{2x}{x+3}$ and $g(x) = \frac{2x^2-4x}{x^2-9}$

$$R(x) = \frac{f(x)}{g(x)} = \frac{\frac{2x}{x+3}}{\frac{2x^2-4x}{x^2-9}}$$

Rewrite the complex fraction as a division problem

$$R(x) = \frac{2x}{x+3} \div \frac{2x^2-4x}{x^2-9}$$

Rewrite as multiplication

$$R(x) = \frac{2x}{x+3} \cdot \frac{x^2-9}{2x^2-4x}$$

Factor

$$R(x) = \frac{2x}{x+3} \cdot \frac{(x+3)(x-3)}{2x(x-2)}$$

Multiply

$$R(x) = \frac{2x(x+3)(x-3)}{2x(x+3)(x-2)}$$

Cancel common factors

$$R(x) = \frac{\cancel{2x}(x+3)(x-3)}{\cancel{2x}(x+3)(x-2)}$$

$$R(x) = \frac{x-3}{x-2}$$

Worksheet 7.1 Multiply and Divide Rational Expressions

1. Determine the value for which the rational expression is undefined: $\frac{4x^2}{x-9}$
2. Find the domain of $R(x) = \frac{2x+3}{x^2+x-12}$
3. Simplify $\frac{3x-9x^2}{9x^2-1}$
4. Simplify $\frac{2x^2-5x-3}{x^2-2x-3}$
5. Multiply $\frac{x^2-x-6}{5x^2-15x} \cdot \frac{10x^4-10x^3}{x^2+3x-4}$
6. Divide $\frac{3x}{5x^2+5x} \div \frac{9x^2}{15x^2+15x}$
7. Simplify $\frac{\frac{x^2+2x-3}{x^2+8x+12}}{\frac{x^2-x-12}{x^2-36}}$
8. Perform the indicated operations: $\frac{8x^2-6x}{x-4} \cdot \frac{x^2-16}{4x^2+x-3} \div \frac{2x+8}{5x+5}$
9. Find $R(x) = f(x) \cdot g(x)$ where $f(x) = \frac{4x}{x^2-16}$ and $g(x) = \frac{x^2+5x+4}{12x^2+12x}$
10. Find $R(x) = \frac{f(x)}{g(x)}$ where $f(x) = \frac{2x^4y^5}{xy^2}$ and $g(x) = \frac{10x^5y}{x^7y^9}$

7.2 Add and Subtract Rational Expressions

Learning Objectives

By the end of this section, you will be able to:

- Add and subtract rational expressions with a common denominator, opposite denominators, and unlike denominators
- Find the least common denominator of rational expressions
- Add and subtract rational functions

Recall that a common denominator is needed to add or subtract fractions.

How to Add or Subtract Rational Expressions

1. Determine if the expressions have a common denominator, opposite denominators, or unlike denominators
 - a. **Common** – go to step 2
 - b. **Opposite** – create common denominators by multiplying one fraction by $\frac{-1}{-1}$
 - c. **Unlike** – find the LCD by factoring each denominator and keeping one of each factor, then rewrite each expression with the LCD
2. Add or subtract by combining the numerators and keeping the denominators the same
3. Simplify, if possible, by factoring and cancelling common factors

Example A: Subtract: $\frac{2x^2+5x-7}{x+5} - \frac{x^2+3x+8}{x+5}$

Denominators are **common**, so go right to subtracting. Be sure to distribute the minus sign.

$$\frac{2x^2 + 5x - 7 - x^2 - 3x - 8}{x + 5}$$

Combine like terms in the numerator

$$\frac{x^2 + 2x - 15}{x + 5}$$

Simplify by factoring then cancelling

$$\frac{\cancel{(x+5)}(x-3)}{\cancel{x+5}} \\ x - 3$$

Example B: Subtract: $\frac{9x-4}{3x-2} - \frac{3x-4}{2-3x}$

$$\frac{9x-4}{3x-2} - \frac{3x-4}{2-3x} \cdot \frac{-1}{-1}$$

Denominators are **opposites**, so multiply a fraction by -1 (I'll choose the second fraction).

$$\frac{9x-4}{3x-2} - \frac{-3x+4}{3x-2}$$

Subtract. Be sure to distribute the minus sign.

$$\begin{array}{r} 9x-4+3x-4 \\ \hline 3x-2 \\ 12x-8 \\ \hline 3x-2 \\ 4(3x-2) \\ \hline 3x-2 \end{array}$$

Combine like terms in the numerator

Simplify by factoring then cancelling

4

Example C: Add: $\frac{2x}{x^2+2x-8} + \frac{4}{x^2+3x-10}$

Denominators are **unlike**, so start by finding the least common denominator (LCD).

Factor each denominator: $x^2 + 2x - 8 = (x+4)(x-2)$

$$x^2 + 3x - 10 = (x+5)(x-2)$$

Take one of each different factor, so the LCD is $(x+4)(x-2)(x+5)$

$$\frac{2x}{(x+4)(x-2)} + \frac{4}{(x+5)(x-2)}$$

Rewrite each expression with the LCD

$$\frac{2x(x+5)}{(x+4)(x-2)(x+5)} + \frac{4(x+4)}{(x+5)(x-2)(x+4)}$$

Simplify each numerator

$$\frac{2x^2+10x}{(x+4)(x-2)(x+5)} + \frac{4x+16}{(x+5)(x-2)(x+4)}$$

Add

$$\frac{2x^2+10x+4x+16}{(x+4)(x-2)(x+5)}$$

Combine like terms in the numerator

$$\frac{2x^2+14x+16}{(x+4)(x-2)(x+5)}$$

Simplify by factoring then cancelling

$$\frac{2(x^2+7x+8)}{(x+4)(x-2)(x+5)}$$

The numerator doesn't factor any further and there are no common factors to cancel, so this is the simplified answer.

Example D: Find $R(x) = f(x) - g(x)$ where $f(x) = \frac{3}{x+5}$ and $g(x) = \frac{4}{x-1}$

Denominators are **unlike**, so start by finding the least common denominator (LCD).

The denominators don't factor, so just take one of each different factor

The LCD is $(x + 5)(x - 1)$

$$\frac{3}{x+5} - \frac{4}{x-1}$$

Rewrite each expression with the LCD

$$\frac{3(x-1)}{(x+5)(x-1)} - \frac{4(x+5)}{(x-1)(x+5)}$$

Simplify each numerator

$$\frac{3x-3}{(x+5)(x-1)} - \frac{4x+20}{(x-1)(x+5)}$$

Subtract. Be sure to distribute the minus sign.

$$\frac{3x-3-4x-20}{(x+5)(x-1)}$$

Combine like terms in the numerator

$$\frac{-x-23}{(x+5)(x-1)}$$

$-x - 23$ cannot be factored, so this is simplified

Worksheet 7.2 Add and Subtract Rational Expressions

Perform the indicated operations.

1. $\frac{3x^2}{x+1} + \frac{x-2}{x+1}$

2. $\frac{6x+21}{x^2-3x-10} - \frac{3x+15}{x^2-3x+10}$

3. $\frac{x^2+5x+1}{x+3} - \frac{4x-5}{x+3} + \frac{7x+9}{x+3}$

4. $\frac{x^2+9x}{2x-1} - \frac{x^2-5}{1-2x}$

5. $\frac{1}{x^2-4} - \frac{1}{2-x}$

6. $\frac{3}{x+4} + \frac{2}{x-2}$

7. $\frac{x}{x^2+4x+3} - \frac{3}{x^2-4x-5}$

8. $\frac{x-1}{x+1} - \frac{x+1}{x-1} + \frac{x^2-5}{x^2-1}$

9. Find $R(x) = f(x) + g(x)$ when $f(x) = \frac{-3x-2}{x^2-2x-3}$ and $g(x) = \frac{2}{x+1}$

10. Find $R(x) = f(x) - g(x)$ when $f(x) = \frac{x}{x^2+4x+3}$ and $g(x) = \frac{3}{x^2-4x-5}$

7.2 Multiply and Divide Rational Expressions

1. $3x - 2$

2. $\frac{3}{x-5}$

3. $x + 5$

4. $x + 5$

5. $\frac{x+3}{(x+2)(x-2)}$

6. $\frac{5x+2}{(x+4)(x-2)}$

7. $\frac{x-9}{(x+3)(x-5)}$

8. $\frac{x-5}{x-1}$

9. $\frac{-x-8}{(x+1)(x-3)}$

10. $\frac{x-9}{(x+3)(x-5)}$

7.3 Simplify Complex Rational Expressions

Learning Objectives

By the end of this section, you will be able to:

- Simplify a complex rational expression by using the LCD

A **complex rational expression** is a rational expression in which the numerator and/or the denominator contains a rational expression.

Examples: $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{a^2} - \frac{1}{b^2}}$, $\frac{\frac{2}{x-1} + 2}{\frac{2}{x-1} - 2}$, $\frac{\frac{1}{x+2} + \frac{3}{x-4}}{\frac{5}{x^2 - 2x - 8}}$

Our goal is to *simplify* the expression. A complex rational expression is *simplified* when its numerator and denominator no longer contain any fractions.

Not simplified: $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{a^2} - \frac{1}{b^2}}$

Simplified: $\frac{ab}{b-a}$

How to simplify a complex rational expression by using the LCD.

- Find the LCD of all fractions in the complex rational expression.
 - TIP: Be sure to factor all the denominators first so you can find the LCD
- Multiply the numerator and denominator by the LCD.
 - TIP: Be sure to multiply the LCD to *each term* in the numerator and denominator
- Simplify the expression.

Example A: Simplify $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{a^2} - \frac{1}{b^2}}$

$$\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{a^2} - \frac{1}{b^2}}$$

The LCD is a^2b^2 . Multiply each term in the numerator and denominator by it.

$$\frac{(a^2b^2)\left(\frac{1}{a}\right) + (a^2b^2)\left(\frac{1}{b}\right)}{(a^2b^2)\left(\frac{1}{a^2}\right) - (a^2b^2)\left(\frac{1}{b^2}\right)}$$

$$\begin{array}{r}
 \frac{(\cancel{a^2}b^2)\left(\frac{1}{\cancel{a}}\right) + (\cancel{a^2}b^2)\left(\frac{1}{\cancel{b}}\right)}{(\cancel{a^2}b^2)\left(\frac{1}{\cancel{a^2}}\right) - (\cancel{a^2}b^2)\left(\frac{1}{\cancel{b^2}}\right)} \\
 \frac{ab^2 + a^2b}{b^2 - a^2} \quad \text{Simplify by factoring and cancelling.} \\
 \frac{ab(b+a)}{(b+a)(b-a)} \\
 \frac{ab}{b-a}
 \end{array}$$

Example B: Simplify $\frac{\frac{2}{x-1} + 2}{\frac{2}{x-1} - 2}$

$$\begin{array}{r}
 \frac{\frac{2}{x-1} + 2}{\frac{2}{x-1} - 2} \quad \text{The LCD is } x - 1. \text{ Multiply the numerator and denominator by it.} \\
 \frac{(x-1)\left(\frac{2}{x-1}\right) + (x-1)(2)}{(x-1)\left(\frac{2}{x-1}\right) - (x-1)(2)} \\
 \frac{\cancel{(x-1)}\left(\frac{2}{\cancel{x-1}}\right) + (x-1)(2)}{\cancel{(x-1)}\left(\frac{2}{\cancel{x-1}}\right) - (x-1)(2)} \quad \text{Note: Distribute 2 into } (x-1) \\
 \frac{2 + 2x - 2}{2 - 2x - 2} \quad \text{Combine like terms} \\
 \frac{2x}{-2x} \\
 -1 \quad \text{Reduce}
 \end{array}$$

Example C: Simplify $\frac{\frac{1}{x+2} + \frac{3}{x-4}}{\frac{5}{x^2 - 2x - 8}}$

$$\begin{array}{r}
 \frac{\frac{1}{x+2} + \frac{3}{x-4}}{\frac{5}{x^2 - 2x - 8}} \quad \text{Factor the trinomial to find the LCD: } x^2 - 2x - 8 = (x+2)(x-4) \\
 \frac{\frac{1}{x+2} + \frac{3}{x-4}}{\frac{5}{x^2 - 2x - 8}} \quad \text{The LCD is } (x+2)(x-4). \text{ Multiply the numerator and denominator by it.} \\
 \frac{(\cancel{x+2})(x-4)\left(\frac{1}{\cancel{x+2}}\right) + (\cancel{x+2})(x-4)\left(\frac{3}{\cancel{x-4}}\right)}{(\cancel{x+2})(x-4)\left(\frac{5}{(\cancel{x+2})(x-4)}\right)} \\
 \frac{x-4+3x+6}{5} \quad \text{Combine like terms}
 \end{array}$$

$$\frac{4x+2}{5}$$

Worksheet 7.3 Simplify Complex Rational Expressions

Simplify.

$$1. \frac{\frac{1-\frac{1}{y}}{x}}{\frac{1+\frac{1}{y}}{x}}$$

$$2. \frac{\frac{x-5}{5}}{\frac{1+\frac{1}{x}}{5}}$$

$$3. \frac{\frac{1}{x+2}}{\frac{5}{x^2-4}}$$

$$4. \frac{\frac{3}{x+1} - \frac{3}{x-1}}{\frac{5}{x^2-1}}$$

$$5. \frac{\frac{x}{x-1} + \frac{3}{x+5}}{\frac{5}{x+5} - \frac{x}{x-1}}$$

$$6. \frac{\frac{3}{x^2-3x-40}}{\frac{5}{x+5} - \frac{2}{x-8}}$$

$$7. \frac{y + \frac{1}{y^2-9}}{\frac{1}{y+3}}$$

$$8. \frac{4 + \frac{4}{a-5}}{\frac{a}{4} + \frac{1}{a-5}}$$

7.4 Solve Rational Equations

Learning Objectives

By the end of this section, you will be able to:

- Solve rational equations
- Use rational functions

Rational Equations

How to solve rational equations.

Steps

1. Note any value of the variable that would make any denominator zero.
2. Find the LCD of *all* denominators in the equation.
3. Clear the fractions by multiplying both sides of the equation by the LCD.
TIP: Be sure to multiply the LCD to *each term* in the equation
4. Solve the resulting equation.
5. Check – If any values found in Step 1 are algebraic solutions, discard them. (These are called **extraneous solutions**).

Example A: Solve $1 - \frac{7}{x} = \frac{-12}{x^2}$

Step 1: Note that if $x = 0$, then both denominators would equal zero.

So, $x \neq 0$

Step 2: Find the LCD of all denominators.

LCD = x^2

Step 3: Multiply all terms of the equation by the LCD

$$x^2(1) - x^2\left(\frac{7}{x}\right) = x^2\left(\frac{-12}{x^2}\right)$$

$$x^2(1) - \cancel{x^2}\left(\frac{7}{\cancel{x}}\right) = \cancel{x^2}\left(\frac{-12}{\cancel{x^2}}\right)$$

$$x^2 - 7x = -12$$

Step 4: Solve. Notice it's quadratic, so put it in standard form and factor to solve.

$$x^2 - 7x + 12 = 0$$

$$(x - 3)(x - 4) = 0$$

$$x - 3 = 0 \text{ or } x - 4 = 0$$

$$x = 3 \text{ or } x = 4$$

Step 5: Check. We did not get $x = 0$ as an algebraic solution, so there is nothing to discard.

$$x = 3 \text{ or } x = 4$$

Example B: Solve $\frac{x}{x+5} = \frac{50}{x^2-25} + 6$

When any denominators can be factored, do that first:

$$\frac{x}{x+5} = \frac{50}{(x+5)(x-5)} + 6$$

Step 1: Note that if $x = -5$ or $x = 5$, then denominators would equal zero.

$$\text{So, } x \neq -5, x \neq 5$$

Step 2: Find the LCD of all denominators.

$$\text{LCD} = (x+5)(x-5)$$

Step 3: Multiply all terms of the equation by the LCD

$$\cancel{(x+5)}(x-5) \left(\frac{x}{\cancel{x+5}} \right) = \cancel{(x+5)}(x-5) \left(\frac{50}{\cancel{(x+5)}(x-5)} \right) + (x+5)(x-5)(6)$$

$$x(x-5) = 50 + 6(x+5)(x-5)$$

Step 4: Solve. First combine like terms. Then notice the equation is quadratic.

$$x(x-5) = 50 + 6(x^2 - 25)$$

$$x^2 - 5x = 50 + 6x^2 - 150$$

$$0 = 5x^2 + 5x - 100$$

$$0 = 5(x^2 + x - 20)$$

$$0 = 5(x-4)(x+5)$$

$$x-4 = 0 \text{ or } x+5 = 0$$

$$x = 4 \text{ or } x = -5$$

Step 5: Check. $x = -5$ is an extraneous solution, so discard it.

$$x = 4$$

Example C: Solve $\frac{3}{x+2} - \frac{7}{x+3} = \frac{7}{x^2+5x+6}$

When any denominators can be factored, do that first:

$$\frac{3}{x+2} - \frac{7}{x+3} = \frac{7}{(x+2)(x+3)}$$

Step 1: Note that if $x = -2$ or $x = -3$, then denominators would equal zero.

$$\text{So, } x \neq -2, x \neq -3$$

Step 2: Find the LCD of all denominators.

$$\text{LCD} = (x+2)(x+3)$$

Step 3: Multiply all terms of the equation by the LCD

$$\cancel{(x+2)}(x+3) \left(\frac{3}{\cancel{x+2}} \right) - (x+2)\cancel{(x+3)} \left(\frac{7}{\cancel{x+3}} \right) = (x+2)(x+3) \left(\frac{7}{\cancel{(x+2)}(x+3)} \right)$$

$$3(x+3) - 7(x+2) = 7$$

Step 4: Solve. First combine like terms. Then notice the equation is linear.

$$3x + 9 - 7x - 14 = 7$$

$$-4x - 5 = 7$$

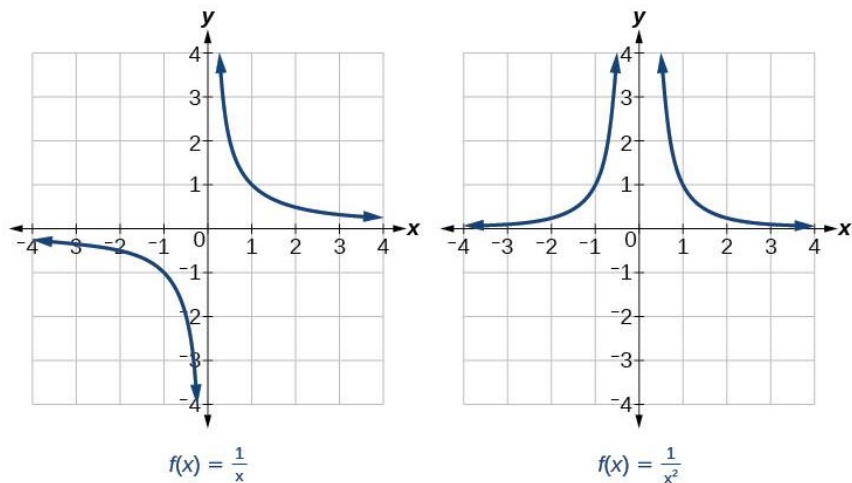
$$-4x = 12$$

$$x = -3$$

Step 5: Check. $x = -3$ is an extraneous solution, so discard it, *no solution*

Rational Functions

Graphs of basic rational functions:



Domain of a rational function: All real numbers except those that make the rational expression undefined.

- Find the domain by setting the denominator equal to zero and solving.
- The domain is all real numbers except those found.

Example D: For the rational function $f(x) = \frac{x+1}{x^2-2x-3}$, (a) find the domain of the function, (b) solve $f(x) = 1$, and (c) find the points on the graph at this function value.

a) Find the domain of the function by setting the denominator equal to zero and solving.

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3 \quad \text{or} \quad x = -1$$

Domain: *all real numbers except $x \neq 3, x \neq -1$*

b) Solve $f(x) = 1$ using the same steps as above

$$\frac{x+1}{x^2-2x-3} = 1$$

$$\frac{x+1}{(x-3)(x+1)} = 1$$

$$\cancel{(x-3)}(x+1) \left(\frac{x+1}{\cancel{(x-3)}(x+1)} \right) = (x-3)(x+1)(1)$$

$$x + 1 = x^2 - 2x - 3$$

$$0 = x^2 - 3x - 4$$

$$0 = (x - 4)(x + 1)$$

$$x = 4 \quad \text{or} \quad x = -1$$

$$x = 4$$

But $x = -1$ is not in the domain of the function

- c) We know from part (b) that $f(x) = 1$ and $x = 4$. Remember $f(x)$ is the same as y .
So $y = 1$ and $x = 4$ gives us the point $(4,1)$.

Worksheet 7.4 Solve Rational Equations

In numbers 1 – 7, solve the equation.

1. $\frac{1}{x} + \frac{2}{3} = \frac{3}{5}$

2. $1 = \frac{2}{x^2} - \frac{1}{x}$

3. $\frac{x}{x+4} = 3 - \frac{4}{x+4}$

4. $x + \frac{6}{x-3} = \frac{2x}{x-3}$

5. $\frac{x}{x-1} - \frac{1}{x-2} = \frac{11}{x^2-3x+2}$

6. $\frac{3}{x+4} + \frac{1}{x-4} = \frac{x+2}{x^2-16}$

7. $\frac{5}{x^2+4x+3} + \frac{2}{x^2+x-6} = \frac{3}{x^2-x-2}$

8. For the rational function $f(x) = \frac{x-1}{x^2-6x+5}$, (a) find the domain of the function, (b) solve $f(x) = 4$, and (c) find the points on the graph at this function value.

8.1 Simplify Expressions with Roots

Learning Objectives:

In this section, you will:

- Simplify expressions with roots
- Estimate and approximate roots
- Simplify variable expressions with roots

Notation		
	$\sqrt[n]{a}$	
$n = \text{index,}$	$\sqrt{} = \text{radical,}$	$a = \text{radicand}$

Point: The index tells us what root is it. If it is a square root, we do not put the index of 2, i.e

$$\sqrt[2]{a} = \sqrt{a}$$

Point: Since every positive number have two square roots, a positive value and a negative value, we call the positive value the principle square root

Nth Root of a Number	
If $b^n = a$, then b is the nth root of a	
The principle nth root of a is $\sqrt[n]{a}$	

Recall that two negatives multiplied together is a positive, and the same holds for other even higher powers. Because of this, we cannot take the even root of a negative number.

Example A: Simplify $\sqrt{16}$

$$\sqrt{16} = 4$$

$$\text{Since } 4^2 = 16$$

Example B: Simplify: $\sqrt[3]{-27}$

$$\sqrt[3]{-27} = -3$$

$$\text{Since } (-3)^3 = -27$$

Since we cannot take an even root of a negative number, we need to ensure that variables in even roots that simplify out are positive. Here is how we handle that.

For any integer $n \geq$

When the index n is odd

$$\sqrt[n]{a^n} = a$$

When the index n is even

$$\sqrt[n]{a^n} = |a|$$

Example C: Simplify $\sqrt[4]{x^4}$

$$\sqrt[4]{x^4} = |x|$$

Since the index is even, $\sqrt[n]{a^n} = |a|$

Example D: Simplify $\sqrt{x^{10}}$

$$\sqrt{x^{10}} = \sqrt{(x^5)^2}$$

Rewrite since $(x^5)^2 = x^{10}$

$$|x^5|$$

Since the index is even, we use the absolute value

Point: If the power on the x was even after simplifying, we would not need to use the absolute value since an even number of negatives must be positive.

If we want to estimate what two integers a root lies between, we take perfect powers of numbers based on the index until we find the two its between

Example E: Estimate between what two integer values the root is $\sqrt{53}$

Number	Square root
36	6
49	7
64	8

$$49 < 53 < 64$$

Locate 53 between two perfect squares

$$7 < \sqrt{53} < 8$$

$\sqrt{53}$ is between their square roots

Worksheet 8.1: Simplify Expressions with Roots

Simplify:

1) $\sqrt{25}$

2) $-\sqrt{36}$

3) $\sqrt[3]{64}$

4) $\sqrt[4]{16}$

5) $\sqrt{-36}$

6) $\sqrt{x^2}$

7) $\sqrt[3]{r^3}$

8) $\sqrt[5]{a^5b^{10}}$

9) $\sqrt[6]{729z^6m^{18}}$

10) $\sqrt{400a^4}$

11) $-\sqrt[4]{256a^8b^{20}}$

12) $\sqrt[3]{125s^{10}t^5}$

13) $\sqrt[3]{-1000}$

14) $\sqrt{(-4)^2}$

15) $\sqrt[5]{(-4)^5}$

16) $\sqrt{1}$

17) $\sqrt{100x^{56}}$

18) $\sqrt{196a^2b^8c^{18}}$

Estimate between what two integer values the root is:

19) $\sqrt{45}$

20) $\sqrt[3]{100}$

21) $\sqrt[4]{18}$

22) $\sqrt{103}$

23) $\sqrt[3]{-157}$

8.2 Simplify Radical Expressions

Learning Objectives:

In this section, you will:

- Use the product property to simplify radical expressions
- Use the quotient property to simplify radical expressions

Point: Simplifying a radical or root expression means to remove all perfect squares, cubes, etc. from the problem depending on the index of the root

We have two different methods for taking roots:

For numbers

1. Find the prime factorization of the number
2. Look for sets of a number that match the root
3. Every set means one number come outside the root and the others are removed
4. Any number that is not part of a set stays under the root.
5. Everything outside the root is multiplied together, and everything inside the root is multiplied together

For variables

1. Divide the power of the variable by the root
2. The whole number portion of the solution is the new power of the variable outside the radical, where the remainder is the new power of the variable inside the radical

This method removes the need for seeing the perfect square in a square root, or the perfect cube in a cube root and so on.

Point: Everything outside the radical is multiplied together, and everything in the radical is multiplied together

Example A: Simplify $\sqrt{200}$

$$\sqrt{200} = \sqrt{2 \cdot 2 \cdot 2 \cdot 5 \cdot 5}$$

Find the prime factorization

$$\sqrt{200} = 2 \cdot 5\sqrt{2} = 10\sqrt{2}$$

A square root wants pairs of numbers. We see a pair of 2's and a pair of 5's, so one of each goes outside the radical. That leaves one 2 left that stays in the radical

Example B: Simplify $\sqrt{48x^2}$

$$\sqrt{48x^2} = \sqrt{2 \cdot 2 \cdot 2 \cdot 2 \cdot 3} \cdot \sqrt{x^2}$$

Find the prime factorization, handling numbers and variables separately

$$\sqrt{48} = 2 \cdot 2\sqrt{3}$$

There are two pairs of twos, so take one two for each pair out of the radical

$$\sqrt{x^2} = |x|$$

Power of two divided by two is one, so one x goes outside the radical

$$\sqrt{48x^2} = 4|x|\sqrt{3}$$

Combine numbers and variables together

Example C: Simplify: $\sqrt[3]{56x^7}$

$$\sqrt[3]{56x^8} = \sqrt[3]{2 \cdot 2 \cdot 2 \cdot 7} \cdot \sqrt[3]{x^7}$$

Find the prime factorization handling numbers and variables separately

$$\sqrt[3]{56} = 2\sqrt[3]{7}$$

There is a 2's that match the index so one of the sets comes outside the radical while the 7 stays inside

$$\sqrt[3]{x^7} = x^2\sqrt[3]{x}$$

Power of 7 divided by 3 is 2 remainder 1 so 2 powers go outside the root and one power stays under the root

$$\sqrt[3]{56x^8} = 2x^2\sqrt[3]{x}$$

Combine numbers and variables

Example D: Simplify $\sqrt{\frac{54x^5y^3}{xy^2}}$

$$\sqrt{\frac{54x^4y^3}{xy^2}} = \sqrt{18x^4y}$$

Simplify the fraction

$$\sqrt{54x^4y} = \sqrt{2 \cdot 3 \cdot 3 \cdot 3} \cdot \sqrt{x^4} \cdot \sqrt{y}$$

Find the prime factorization, handling numbers and variable separately

$$\sqrt{54} = 3\sqrt{6}$$

There are a pair of 3's, so we write one in front of the radical. That leaves a 2 and another 3 under the radical

$$\sqrt{x^4} = x^2$$

Power of 4 divided by 4 is 2 remainder 0, so 2 powers go outside the root and none stay underneath. Note that y stays under as well since its power is smaller than the root

$$\sqrt{\frac{54x^5y^3}{xy^2}} = 3x^2\sqrt{6y}$$

Combine numbers and variables

Example E: Simplify $\frac{\sqrt{81}}{\sqrt{3}}$

$$\sqrt{\frac{81}{3}}$$

Combine under one radical

$$\sqrt{27}$$

Simplify the fraction

$$\sqrt{27} = \sqrt{3 \times 3 \times 3} = 3\sqrt{3}$$

There are a pair of 3's, so we write one in front of the radical. That leaves another 3 under the radical

Worksheet 8.2: Simplify Radical Expressions

Simplify:

1) $\sqrt{24}$

2) $\sqrt{12}$

3) $\sqrt[3]{250}$

4) $\sqrt[4]{32}$

5) $\sqrt[3]{243}$

6) $\sqrt{x^3}$

7) $\sqrt[3]{x^5 r^3}$

8) $\sqrt[5]{a^7 b^{11}}$

9) $\sqrt[6]{c^6 z^{17}}$

10) $\sqrt{600a^5}$

11) $\sqrt[4]{1296n^2m^9}$

12) $\sqrt[3]{147s^{12}t^8}$

13) $\sqrt[3]{-80}$

14) $\sqrt{-4}$

15) $\sqrt[5]{-256}$

16) $7 + \sqrt{60}$

17) $\frac{4+\sqrt{50}}{2}$

18) $\frac{20-\sqrt[3]{750}}{5}$

19) $\sqrt{\frac{45}{500}}$

20) $\sqrt[3]{\frac{40x^4}{5x}}$

21) $\sqrt[4]{\frac{r^{18}t^{11}}{t^{19}}}$

22) $\frac{\sqrt{120q^7}}{\sqrt{10q^2}}$

23) $\frac{\sqrt[3]{-32}}{\sqrt[3]{2}}$

8.3 Simplify Rational Exponents

Learning Objectives:

In this section, you will:

- Simplify expressions with $a^{\frac{1}{n}}$.
- Simplify expressions with $a^{\frac{m}{n}}$.

Recall: A rational number is any number that can be written as a fraction.

When in the form of an exponent, a rational number's numerator tells us the power on the variable, and a rational number's denominator tells us the root on the variable

$$a^{\frac{1}{n}} = \sqrt[n]{a} \quad \text{and} \quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

Example A: Write as a radical expression: $x^{\frac{1}{2}}$

$$x^{\frac{1}{2}} = \sqrt{x}$$

The denominator of 2 means square root.

Example B: Write as a radical expression: $y^{\frac{2}{3}}$

$$y^{\frac{2}{3}} = \sqrt[3]{y^2}$$

The denominator of 3 means cube root. The numerator of 2 means to raise the y inside the radical to the second power.

Example C: Write with a rational exponent $7\sqrt[4]{2x^3y}$

$$7\sqrt[4]{2x^3y} = 7 \cdot (2y)^{\frac{1}{4}} \cdot x^{\frac{3}{4}}$$

Note the 7 is outside the radical, so its power will not change. The rest is the fourth root, so all their powers will have 4 in the denominator.

Point: We can simplify rational exponents by working with them as radicals and using our exponent rules. Generally, we will take the root first then apply the power to the term.

Example D: Simplify $27^{\frac{2}{3}}$

$$27^{\frac{2}{3}} = (\sqrt[3]{27})^2$$

The 3 in the denominator means cube root. The 2 in the numerator means to square the root

$$(\sqrt[3]{27})^2 = (3)^2 = 9$$

Simplify the root and raise it to the power

Example E: Simplify $625^{-\frac{3}{4}}$

$$625^{-\frac{3}{4}} = \frac{1}{625^{\frac{3}{4}}}$$

Rewrite with a positive exponent

$$\frac{1}{625^{\frac{3}{4}}} = \frac{1}{(\sqrt[4]{625})^3}$$

The 4 in the denominator means fourth root. The 3 in the numerator means to cube the root

$$\frac{1}{(\sqrt[4]{625})^3} = \frac{1}{(5)^3} = \frac{1}{125}$$

Simplify the root and raise it to the power

Example F: Simplify $\left(\frac{8x^{\frac{1}{2}}y^{-\frac{3}{4}}}{z^6}\right)^{\frac{4}{3}}$

$$\left(\frac{8x^{\frac{1}{2}}y^{-\frac{3}{4}}}{z^6}\right)^{\frac{4}{3}} = \left(\frac{8x^{\frac{1}{2}}}{z^6y^{\frac{3}{4}}}\right)^{\frac{4}{3}}$$

Rewrite with a positive exponent

$$\frac{(8)^{\frac{4}{3}}(x^{\frac{1}{2}})^{\frac{4}{3}}}{(z^6)^{\frac{4}{3}}(y^{\frac{3}{4}})^{\frac{4}{3}}}$$

Use the exponent rule $(ab)^n = a^n b^n$

$$\frac{16x^{\frac{2}{3}}}{z^8y}$$

Multiply the exponents and simplify

Worksheet 8.3: Simplify Rational Exponents

Write as a radical expression:

1) $z^{\frac{2}{3}}$

2) $r^{-\frac{1}{2}}$

3) $2x^{\frac{5}{7}}$

Write with a rational exponent:

4) $\sqrt{x^5}$

5) $\sqrt[3]{y^2}$

6) $\sqrt{\left(\frac{2x^3}{3y}\right)^3}$

Simplify

7) $16^{\frac{3}{4}}$

8) $-64^{\frac{1}{2}}$

9) $(-81)^{\frac{1}{2}}$

10) $(25)^{-\frac{1}{2}}$

11) $625^{\frac{1}{4}}$

12) $(-64)^{\frac{1}{3}}$

13) $81^{\frac{3}{2}}$

14) $-16^{-\frac{3}{4}}$

15) $(-125)^{\frac{2}{3}}$

Simplify using the laws of exponents, assuming all variables are positive

16) $a^{\frac{1}{3}} \cdot a^{\frac{2}{3}}$

17) $(k^3)^{\frac{6}{5}}$

18) $25^{\frac{3}{2}} \cdot 25^{\frac{5}{2}}$

19) $\left(32r^{\frac{2}{7}}\right)^{\frac{3}{5}}$

20) $\left(a^{\frac{5}{3}}b^{\frac{1}{2}}\right)^{\frac{6}{5}}$

21) $\frac{z^{\frac{1}{3}} \cdot z^{-\frac{5}{6}}}{z^{-\frac{2}{5}}}$

8.4 Add, Subtract, and Multiply Radical Expressions

Learning Objectives:

In this section, you will:

- Add and subtract radical expressions
- Multiply radical expressions
- Use polynomial multiplication to multiply radical expressions

Like Radicals: Radical expressions with the same index and same radicand.

It helps to view radicals like variables. Variables have to match exactly, including powers, in order to consider them like terms so we can do things like add and subtract them. It is the same for radicals; if they do not match exactly in every way, they are not like terms. Thinking of the square root of 2 as an x so that it can only be added to other square roots of 2 is helpful. Unlike variables though, we must simplify the radical first in to find like radicals

Point: Simplify then do the operation(add, subtract, multiply, rationalize, etc.)

Example A: Simplify: $\sqrt{7} - 8\sqrt{7} + 3\sqrt{7z}$

$$1\sqrt{7} - 8\sqrt{7} = -7\sqrt{7}$$

The first two radicals are like radicals, so they are combined. Remember that if there is no coefficient than it is a 1 for the coefficient.

$$-7\sqrt{7} + 3\sqrt{7z}$$

The third term is not a like radical, so it cannot be combined.

Example B: Simplify: $3\sqrt{20} + 7\sqrt{125}$

$$\sqrt{20} = 2\sqrt{5} \text{ and } \sqrt{125} = 5\sqrt{5}$$

Simplify each radical expression.

$$3(2\sqrt{5}) + 7(5\sqrt{5}) = 6\sqrt{5} + 35\sqrt{5}$$

Multiply the numbers outside of the radical together

$$41\sqrt{5}$$

Since the radicals are like radicals, they can be added together.

For Multiplication, everything outside the radical goes together and everything inside the radical goes together. Use the following rule to help:

For any real numbers a and b and for any integer $n \geq 2$

$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$$

Example C: Multiply and simplify: $(2\sqrt[3]{54x^4})(5\sqrt[3]{16x})$

$$2 \cdot 5\sqrt[3]{54x^4 \cdot 16x}$$

Multiply the coefficients together and the radicands together.

$$2 \cdot 5\sqrt[3]{3^3 \cdot 2^5 \cdot x^5}$$

Factor the values in the radicand to simplify it.

$$10(3 \cdot 2)x^3\sqrt[3]{2^2x^2} = 60x^3\sqrt[3]{4x^2}$$

Simplify the radical, making sure to multiply everything outside the radical together and everything in the radical together.

Point: Since we view radicals the same as variables, that means we multiply them with distribution the same way we have for just variables.

Example D: Multiply and Simplify: $\sqrt{7}(3 - \sqrt{7})$

$$3\sqrt{7} - \sqrt{7^2}$$

Distribute out the term in front of the parentheses

$$3\sqrt{7} - 7$$

Simplify the radical

Example E: Multiply and Simplify: $(\sqrt{2} - 3\sqrt{5})(\sqrt{10} + \sqrt{3})$

$$\sqrt{2} \cdot \sqrt{10} + \sqrt{2} \cdot \sqrt{3} - 3\sqrt{5} \cdot \sqrt{10} - 3\sqrt{5} \cdot \sqrt{3}$$

Distribute

$$2\sqrt{5} + \sqrt{6} - 3(5\sqrt{2}) - 3\sqrt{15}$$

Simplify the radicals

$$2\sqrt{5} + \sqrt{6} - 15\sqrt{2} - 3\sqrt{15}$$

Since there are no like terms, we multiply the $3 \cdot 5$ *together* for the final answer

Example F: Multiply and Simplify: $(2\sqrt{5} - 3\sqrt{2})^2$

$$(2\sqrt{5})^2 - 2(2\sqrt{5})(3\sqrt{2}) + (3\sqrt{2})^2$$

Use $(a \pm b)^2 = a^2 \pm 2ab + b^2$

$$4 \cdot 5 - 12\sqrt{10} + 9 \cdot 2$$

Multiply each term, remembering $(\sqrt{n})^2 = n$ if $n \geq 0$

$$38 - 12\sqrt{10}$$

Simplify

Worksheet 8.4: Add, Subtract, and Multiply Radical Expressions

Simplify:

1) $2\sqrt{x} + 3\sqrt{x}$

2) $\sqrt[3]{2n} - 4\sqrt[3]{2n}$

3) $3\sqrt{7} - 4\sqrt{7}$

4) $6\sqrt{5a} - 11\sqrt{5a} + 2\sqrt{5a}$

5) $\sqrt{48} + \sqrt{75}$

6) $2\sqrt[3]{135} - \frac{1}{2}\sqrt[3]{40}$

7) $\sqrt{16x^3} + x\sqrt{25x}$

8) $\sqrt{a^5b^7} - \sqrt{ab}$

9) $\sqrt{2} + \sqrt[3]{2}$

Multiply and simplify:

10) $(-4\sqrt{5})(3\sqrt{10})$

11) $(7\sqrt[4]{27})(3\sqrt[4]{9})$

12) $(6\sqrt{6xy^3})(2\sqrt{12xy})$

13) $3\sqrt{7}(2 - 5\sqrt{5})$

14) $-\sqrt{14}(\sqrt{21} - 4\sqrt{7})$

15) $(\sqrt{2} - \sqrt{3})(\sqrt{5} + \sqrt{7})$

16) $(9 - \sqrt[3]{2x})(1 - \sqrt[3]{7x})$

17) $(2 + \sqrt{5})^2$

18) $(\sqrt[4]{3x} - 2\sqrt[4]{27})^2$

19) $(\sqrt{10} + \sqrt{7})(\sqrt{10} - \sqrt{7})$

20) $(\sqrt[3]{3} + 2)(\sqrt[3]{3} - 2)$

21) $\sqrt{3}(1 - 9\sqrt{2})^2$

8.5 Divide Radical Expressions

Learning Objectives: 10.5

Divide radical expressions

- Rationalize a one-term denominator
- Rationalize a two-term denominator

Remember we can combine radicals or separate them using the quotient property

If $\sqrt[n]{a}$ and $\sqrt[n]{b}$ are real numbers, $b \neq 0$ and for any integer $n \geq 2$ then,

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}} \text{ and } \frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$$

Point: We want to put the radicands together so we can simplify them before we take the root.

For a radical expression to be considered simplified it requires:

1. No factors in the radicand have perfect powers of the index
2. No fractions in the radicand
3. No radicals in the denominator of the fraction

We need to remove all perfect powers, and be sure the radical only exists in the numerator. To remove the radical in the denominator, we can rationalize the denominator.

Point: When rationalizing the denominator, we have two methods depending on the number of terms in the denominator:

- For one term, we look at the factored form of the radicand in the denominator and multiply it by whatever term we need to make every term in the denominator have the power of the index

For two terms, we use the difference of perfect squares for square roots or difference/sum of perfect cubes for cube roots

Example A: Simplify: $\frac{\sqrt{125}}{\sqrt{20}}$

$$\sqrt{\frac{125}{20}}$$

Rewrite using the quotient property

$$\sqrt{\frac{5 \cdot 25}{5 \cdot 4}}$$

Remove common factors

$$\frac{\sqrt{25}}{\sqrt{4}} = \frac{5}{2}$$

Simplify the radical

Example B: *Simplify:* $\frac{\sqrt{1240x^3y^6}}{\sqrt{10xy}}$

$$\sqrt{\frac{1240x^3y^6}{10xy}}$$

Rewrite using the quotient property

$$\sqrt{\frac{124 \cdot \cancel{10} \cdot x^2 \cdot \cancel{x} \cdot y^5 \cdot \cancel{y}}{\cancel{10}xy}}$$

Remove common factors

$$\sqrt{124x^2y^5} = 2xy^2\sqrt{31y}$$

Simplify the radical

Example C: *Rationalize the fraction:* $\frac{\sqrt[3]{7}}{\sqrt[3]{12}}$

$$\frac{\sqrt[3]{7}}{\sqrt[3]{2^2 \cdot 3}}$$

Factor the denominator inside the radical

$$\frac{\sqrt[3]{7}}{\sqrt[3]{2^2 \cdot 3}} \cdot \frac{\sqrt[3]{2 \cdot 3^2}}{\sqrt[3]{2 \cdot 3^2}}$$

Multiply the fraction by however many copies of each factor to make it match the index

$$\frac{\sqrt[3]{126}}{\sqrt[3]{2^3 \cdot 3^3}} = \frac{\sqrt[3]{126}}{6}$$

Simplify the radical

Example D: *Rationalize the fraction:* $\frac{1}{\sqrt{2}-3}$

$$\frac{1}{\sqrt{2}-3} \cdot \frac{\sqrt{2}+3}{\sqrt{2}+3}$$

Multiply by the perfect square pair

$$\frac{\sqrt{2}+3}{(\sqrt{2})^2-3^2}$$

$$\frac{\sqrt{2}+3}{2-9} = \frac{\sqrt{2}+3}{-7}$$

Use the difference of perfect squares:

$$(a + b)(a - b) = a^2 - b^2$$

Simplify the radical

Worksheet 8.5: Divide Radical Expressions

Divide and rationalize:

$$1) \frac{\sqrt{50}}{\sqrt{8}}$$

$$2) \frac{\sqrt{125}}{\sqrt{15}}$$

$$3) \frac{3\sqrt{6}}{7\sqrt{8}}$$

$$4) \frac{2\sqrt{3}}{2\sqrt{12}}$$

$$5) \frac{5\sqrt{2x^7}}{\sqrt{x^3}}$$

$$6) \frac{\sqrt[3]{54}}{\sqrt[3]{3}}$$

$$7) \frac{\sqrt[4]{320x^{10}}}{\sqrt[4]{5x^3}}$$

$$8) \frac{2}{\sqrt{5}}$$

$$9) \frac{1}{\sqrt[3]{54}}$$

$$10) \frac{xy}{\sqrt{x}+\sqrt{y}}$$

$$11) \frac{3+\sqrt{5}}{\sqrt{10}-\sqrt{15}}$$

$$12) \frac{t-\sqrt{3}}{t+\sqrt{3}}$$

$$13) \frac{\sqrt[3]{-625a^5b}}{\sqrt[3]{5ab^2}}$$

$$14) \sqrt{\frac{72}{54}}$$

$$15) \sqrt[3]{\frac{-160}{2x}}$$

8.6 Solve Radical Equations

Learning Objectives:

In this section, you will:

- Solve radical equations
- Solve radical equations with two radicals
- Use radicals in applications

Radical Equation

An equation in which a variable is in the radicand of a radical expression is called a radical equation

Solving any type of equation is based on two overall steps:

1. Isolate the radical
2. Remove the radical

When we are solving radical equations, the goal is the root. If we can isolate the root, we can raise it to the power of the index to remove the radical. The method follows this idea:

1. Isolate a radical in the equation
2. Raise both sides to the power of the index of the isolated radical
3. Repeat steps 1 and 2 until no more radicals remain
4. Solve the resulting equation
5. Check the solutions(s)

Point: We must check the solutions to see if we have any extraneous solutions. Extraneous solutions are solutions that solve the problem we removed the radicals from but do not work in the original radical equation. Remember that since we are taking powers of the problem, we are altering it. We can be sure we do not lose any solutions when we use this method, but we may find solutions that do not work.

Point: Everything outside the radical is multiplied together, and everything in the radical is multiplied together

Example A: Solve: $\sqrt{x+3} + 2 = 5$

$$\sqrt{x+3} = 3$$

Isolate the radical

$$(\sqrt{x+3})^2 = 3^2$$

Raise both sides to the power of the index

$$x + 3 = 9$$

Simplify

$$x = 6$$

Solve

$$\sqrt{6+3} + 2 = 5$$

Check

Example B: Solve $\sqrt{x+1} - 1 = x$

$$\sqrt{x+1} = x + 1$$

Isolate the radical

$$(\sqrt{x+1})^2 = (x+1)^2$$

Raise both sides to the power of the index

$$x + 1 = x^2 + 2x + 1$$

Simplify

$$x = 0, -1$$

Solve

$$\text{For } x = 0, \sqrt{0+1} - 1 = 0$$

Check

$$\text{For } x = -1, \sqrt{-1+1} - 1 = -1$$

Example C: Solve: $2\sqrt[3]{4-p} - 1 = 7$

$$2\sqrt[3]{4-p} = 8$$

$$\sqrt[3]{4-p} = 4$$

Isolate the radical

$$(\sqrt[3]{4-p})^3 = (4)^3$$

Raise both sides to the power of the index

$$4 - p = 64$$

Simplify

$$p = -60$$

Solve

$$2\sqrt[3]{4 - (-60)} - 1 = 7$$

Check

Example D: Solve $\sqrt{2x-1} = \sqrt{x-4}$

$$(\sqrt{2x-1})^2 = (\sqrt{x-4})^2$$

$$2x - 1 = x - 4$$

$$x = -3$$

$$\sqrt{2(-3)-1} \neq \sqrt{-3-4}$$

Since the radicals are isolated, we raise each side to the power of the index

Simplify

Solve

Check, since we cannot take an even root of a negative number, there is no solution

Example E: Solve: $\sqrt{2x+10} - \sqrt{x+6} = 1$

$$\sqrt{2x+10} = \sqrt{x+6} + 1$$

Isolate a radical

$$(\sqrt{2x+10})^2 = (\sqrt{x+6} + 1)^2$$

Raise each side to the power of the index

$$\begin{aligned} 2x + 10 &= x + 6 + 2\sqrt{x+6} + 1 \\ 2x + 10 &= x + 7 + 2\sqrt{x+6} \end{aligned}$$

Simplify

$$x + 3 = 2\sqrt{x+6}$$

Isolate the radical, in this case, choosing to keep the coefficient to avoid fractions

$$(x+3)^2 = (2\sqrt{x+6})^2$$

Raise both sides to the power of the index

$$x^2 + 6x + 9 = 4(x+6)$$

Simplify

$$\begin{aligned} x^2 + 6x + 9 &= 4x + 24 \\ x^2 + 2x - 15 &= 0 \\ x &= 3, -5 \end{aligned}$$

Solve

$$\begin{aligned} \text{For } x = 3 \quad \sqrt{2(3)+10} - \sqrt{3+6} &= 1 \\ \text{For } x = -5 \quad \sqrt{2(-5)+10} - \sqrt{-5+6} &\neq 1 \end{aligned}$$

Check. Since 3 works but -5 does not, the only solution is $x = 3$

Worksheet 8.6: Solve Radical Equations

Solve:

1) $\sqrt{2x-1} = 3$

2) $\sqrt{x} = -1$

3) $\sqrt{3x+4} = 5$

4) $\sqrt[4]{x-1} = 2$

5) $\sqrt{1-z} + 5 = 9$

6) $2\sqrt{4x-3} + 3 = 13$

7) $\sqrt{x-3} = x-5$

8) $2\sqrt{2x+5} - x = 4$

9) $\sqrt{4-11x} - x + 2 = 0$

10) $\sqrt{2x-1} + 2 = x$

11) $\sqrt[3]{4x^2+7} - 2 = 0$

12) $\sqrt{3x^2+10x} - 5 = 0$

13) $7 + \sqrt{b-3} = 2$

14) $\sqrt[3]{2x+3} + 5 = 2$

15) $\sqrt{x} + 1 = \sqrt{4x+12}$

16) $\sqrt[3]{4x-8} = 2\sqrt[3]{x-4}$

17) $\sqrt{x-8} = 4\sqrt{x-3}$

18) $\sqrt{3x-5} + \sqrt{x-1} = 2$

Use radicals to solve the word problem. Round as needed to the hundredth decimal place.

19) John drops a stone from a 150 foot tower into a pond below. Use the formula to $t = \frac{\sqrt{h}}{4}$ determine when it will hit the water.

20) Alice bought enough dirt to cover a square garden bed that's 80 square feet. Use the formula $s = \sqrt{A}$ to find the length of each side.

21) Qutin found that the formula relating revenue and price is as follows: $R = \sqrt{\frac{p}{2}}$ What revenue will a price of \$12 earn ?

8.7 Use Radicals in Functions

Learning Objectives

In this section, you will:

- Evaluate a radical function
- Find the domain of a radical function
- Graph radical functions

Radical Function

A function that is defined by a radical expression of any index

Point: We evaluate a radical function the same way we do every function, we plug in the value/expression in for the variable in the function. When doing this, we do need to consider the domain

If the index is even, the radicand must be greater than or equal to zero, if the index is odd, there is no restriction on the radicand.

This means our natural domain has three restrictions

1. Cannot divide by zero
2. Even roots must be nonnegative
3. Logarithms must be positive inside

To deal with even roots, we set the radicand ≥ 0 and solve

Example A: For the following function: $f(x) = \sqrt{3x + 1}$

Evaluate: $f(5)$

$$f(5) = \sqrt{3(5) + 1} \quad \text{Substitute 5 for } x$$

$$f(5) = \sqrt{16} \quad \text{Simplify}$$

$$f(5) = 4 \quad \text{Take the square root}$$

Evaluate: $f(-1)$

$$f(-1) = \sqrt{3(-1) + 1}$$

Substitute -1 for x

$$f(-1) = \sqrt{-2}$$

Simplify

No solution

Since we cannot take the square root of a negative, there is no solution

Example B: Evaluate: $h(x) = \sqrt[3]{4 - x}$ for $h(12)$

$$h(12) = \sqrt[3]{4 - 12}$$

Substitute 12 for x

$$h(12) = \sqrt[3]{-8}$$

Simplify

$$h(12) = -2$$

Take the cube root

Example C: Find the domain of the radical function: $f(x) = \sqrt{3x - 6}$

$$3x - 6 \geq 0$$

Since we have an even index, we set the radicand ≥ 0

$$x \geq 2$$

Solve

$$[2, \infty)$$

Write in interval notation

Example D: Find the domain of the radical function: $h(x) = \sqrt{\frac{2}{x-7}}$

$$x - 7 > 0$$

Since it is an even index, the radicand cannot be negative. Since there is a denominator, the denominator cannot be zero. We combine these rules.

$$x > 7$$

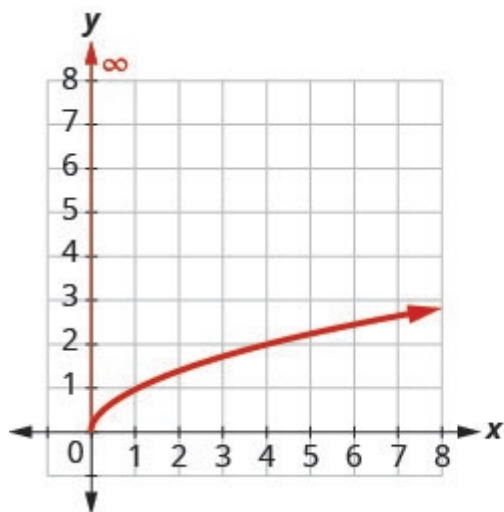
Solve

$$(7, \infty)$$

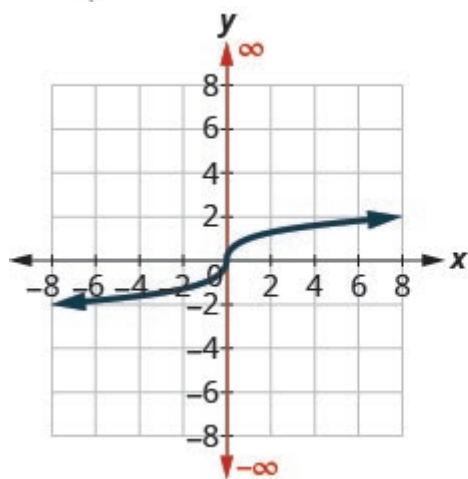
Write in interval notation

Graphing Radical Functions

To graph a radical function, we can evaluate perfect index multiples to get useful and simple points to graph. Below are the graphs for $\sqrt{x}$ and $\sqrt[3]{x}$ that illustrate this point.



x	$f(x) = \sqrt{x}$	$(x, f(x))$
0	0	(0, 0)
1	1	(1, 1)
4	2	(4, 2)
9	3	(9, 3)



x	$f(x) = \sqrt[3]{x}$	$(x, f(x))$
-8	-2	(-8, -2)
-1	-1	(-1, -1)
0	0	(0, 0)
1	1	(1, 1)
8	2	(8, 2)

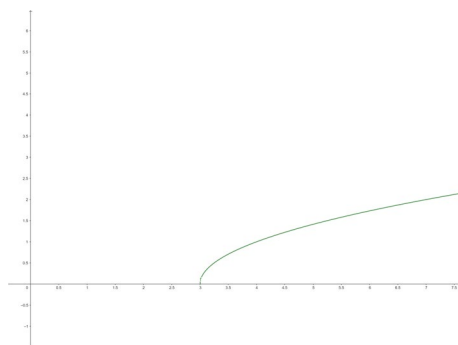
If we want to graph a more complicated radical function, we ensure the radicand is a perfect multiple, not the variable. This means if we are graphing $\sqrt{x+2}$, then $x+2$ must be a perfect square.

Point: Since even indexed roots cannot be negative, solve the radicand equal to 0 as a starting point for your points.

Example E: Graph $f(x) = \sqrt{x-3}$

To graph this, we use x values that will result in perfect squares. Once we have them, we plot those points and connect them smoothly.

x	$f(x) = \sqrt{x - 3}$	$(x, f(x))$
3	0	(3,0)
4	1	(4,1)
7	2	(7,2)
12	3	(12,3)
28	5	(28,5)



Worksheet 8.7: Use Radicals as Functions

Evaluate the function: $f(x) = \sqrt{2x - 1}$

1) $f(5)$

2) $f(0)$

3) $f\left(\frac{1}{2}\right)$

Evaluate the function: $f(x) = \sqrt[3]{3x - 4}$

4) $f(6)$

5) $f(1)$

6) $f\left(\frac{35}{6}\right)$

Evaluate the function: $h(x) = \sqrt{x^2 + 2}$

7) $h(5)$

8) $h(-8)$

9) $h(2)$

Find the domain and write in interval notation

10) $f(x) = \sqrt{2x - 1}$

11) $h(x) = \sqrt[3]{3x^2 - 2}$

12) $g(x) = \sqrt{\frac{3}{5x-1}}$

13) $f(x) = \sqrt{\frac{x-1}{x+1}}$

14) $g(x) = \sqrt{4 - 9x}$

15) $h(x) = \sqrt[3]{\frac{1}{x-16}}$

Graph the following:

16) $f(x) = \sqrt[3]{x + 1}$

17) $f(x) = -\sqrt{x} + 1$

18) $f(x) = \sqrt{2x - 1}$

8.8 Use the Complex Number System

Learning Objectives

In this section, you will:

- Evaluate the square root of a negative number
- Add and subtract complex numbers
- Multiply complex numbers
- Divide complex numbers
- Simplify powers of i

Many problems require mathematical terms we do not have naturally, so they are created. One such thing is the imaginary unit i .

Imaginary Unit

The imaginary unit i is the number whose square is -1

$$i^2 = -1 \text{ or } i = \sqrt{-1}$$

We can now take the even roots of negative numbers, as we just get the root with an i .

Example A: Write in terms of i and simplify $\sqrt{-36}$

$$\sqrt{-36} = \sqrt{36}i$$

Use the definition of i

$$\sqrt{36}i = 6i$$

Simplify the radical

We now expand our ideas of numbers to include imaginary numbers. This combination is what we call complex numbers

Complex Numbers

A number in form $a + bi$ where a and b are real numbers

If $a=0$, then we say it is a pure imaginary number. Otherwise, the combination is a complex number, with a real component a and an imaginary component b .

Point: We treat imaginary numbers the same way we do variables from a rules perspective, except we know what it is at all points and we know what powers of i are. This means to add, we need like terms.

Example B: Add: $\sqrt{-125} + \sqrt{-20}$

$$\sqrt{-125} = 5i\sqrt{5} \text{ and } \sqrt{-20} = 2i\sqrt{5}$$

Simplify each square root

$$5i\sqrt{5} + 2i\sqrt{5} = 7i\sqrt{5}$$

Since the two terms are both $i\sqrt{5}$, we combine them as like terms

Simplify Powers of i

Since we know what i represents, we can work with multiple powers of it. Importantly, we can establish a pattern.

i
$i^2 = \sqrt{-1} \cdot \sqrt{-1} = -1$
$i^3 = i \cdot i^2 = -i$
$i^4 = i^2 \cdot i^2 = -1 \cdot -1 = 1$
$i^5 = i^4 \cdot i = i$
$i^6 = i^4 \cdot i^2 = -1$

What we should notice is every four powers is a 1, so the powers of i cycle in a pattern of four.

To simplify powers of i , just divide the power by 4 and note the remainder. That remainder gives us the equivalent power of i to use to solve.

Example C: Simplify: i^{38}

$$\frac{38}{4} = 9R2$$

Divide the power by 4

$$i^2 = -1$$

Raise i to the remainder power and simplify

Multiply and Divide Complex Numbers

As stated, we will treat imaginary numbers the same way we treat variables, except that we know how to simplify powers of i . This means we distribute and collect like terms the same way we have for all of algebra.

Example D: Multiply and simplify: $(3 - 2i)(7 + 4i)$

$$21 + 12i - 14i - 8i^2$$

Distribute out the product

$$-8i^2 = -8 \cdot -1 = 8$$

Recall that $i^2 = -1$ and simplify

$$21 + 12i - 14i + 8 = 29 - 2i$$

Simplify

Point: Rationalizing a fraction is very similar to dividing with complex numbers. Much like how we did not want roots in the denominator, we also do not want imaginary numbers in the denominator. Do to this, we will use the complex conjugate pair

Complex Conjugate Pair

The complex conjugate pair is of the form $a + bi, a - bi$

This works since $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$

Example E: Simplify $\frac{3+2i}{4+3i}$

$$\frac{3+2i}{4+3i} \cdot \frac{4-3i}{4-3i}$$

Multiply by the complex conjugate of the denominator to the numerator and denominator

$$\frac{12-9i+8i-6i^2}{16+9}$$

Distribute

$$\frac{18-i}{25}$$

Simplify

Worksheet 8.8: Use the Complex Number System

Simplify and write in terms of i :

1) $\sqrt{-25}$

2) $-\sqrt{-36}$

3) $\sqrt{-150}$

Add or Subtract the complex numbers:

4) $\sqrt{-50} + 3\sqrt{-8}$

5) $9\sqrt{-12} - 7\sqrt{-27}$

6) $(4 + 5i) + (6 - 7i)$

7) $(6 - 9i) - (-3 - 8i)$

8) $(2 + 6i) - i$

9) $(7 - \sqrt{-200}) + (3 + 4\sqrt{-32})$

Multiply or divide:

10) $(7 + 8i)(2 - i)$

11) $(3 + \sqrt{-72})(4 - \sqrt{-27})$

12) $(2 - 3i)(2 + 3i)$

13) $(5 + 4i)^2$

14) $\frac{-3+2i}{6i}$

15) $\frac{2+7i}{4-2i}$

16) $\frac{3+5i}{3-5i}$

17) $\frac{-10-\sqrt{-18}}{3+2\sqrt{-98}}$

Simplify:

18) i^9

19) i^{47}

20) i^{213}

21) i^{-18}

9.1 Solve Quadratic Equations Using the Square Root Property

Learning Objectives

By the end of this section, you will be able to:

- Solve quadratic equations of the form $ax^2 = k$ using the Square Root Property
- Solve quadratic equations of the form $a(x + h)^2 = k$ using the Square Root Property

SQUARE ROOT PROPERTY

$$\text{If } x^2 = k, \text{ then } x = \pm\sqrt{k} \text{ or } \quad \text{If } x^2 = k, \text{ then } x = \sqrt{k} \text{ or } x = -\sqrt{k}$$

HOW TO: Solve a quadratic equation using the square root property.

STEPS

1. Isolate the quadratic term and make its coefficient 1.
2. Use the square root property.
3. Simplify the radical.

Example A: Solve $2x^2 - 8 = 0$

$$2x^2 - 8 = 0$$

First get x^2 alone.

$$2x^2 = 8$$

$$x^2 = 4$$

Apply the square root property.

$$x = \pm\sqrt{4}$$

Simplify the radical.

$$x = \pm 2$$

$$\boxed{x = 2, x = -2}$$

Example B: Solve $x^2 - 18 = 0$

$$x^2 - 18 = 0$$

First get x^2 alone.

$$x^2 = 18$$

Apply the square root property.

$$x = \pm\sqrt{18}$$

$$x = \pm 3\sqrt{2}$$

$$\boxed{x = 3\sqrt{2}, x = -3\sqrt{2}}$$

Simplify the radical.

Example C: Solve $x^2 + 50 = 0$

$$2x^2 = -100$$

$$x^2 = -50$$

$$x = \pm\sqrt{-50}$$

$$x = \pm 5i\sqrt{2}$$

$$\boxed{x = 5i\sqrt{2}, x = -5i\sqrt{2}}$$

First get x^2 alone.

Apply the square root property.

Simplify the radical.

Example D: Solve $3x^2 + 5 = 30$

$$3x^2 + 5 = 30$$

$$3x^2 = 25$$

$$x^2 = \frac{25}{3}$$

$$x = \pm\sqrt{\frac{25}{3}}$$

$$x = \pm \frac{\sqrt{25}}{\sqrt{3}}$$

$$x = \pm \frac{5}{\sqrt{3}}$$

$$x = \pm \frac{5}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$x = \pm \frac{5\sqrt{3}}{3}$$

$$\boxed{x = \frac{5\sqrt{3}}{3}, x = -\frac{5\sqrt{3}}{3}}$$

First get x^2 alone.

Apply the square root property.

Simplify the radical.

Rationalize the denominator.

Example E: Solve $(x - 3)^2 = 16$

Note the different look of this equation. We can use the square root property to solve this as well. Just treat the parentheses like you treat the x when you apply the property:

$$x^2 = 16$$

$$x = \pm\sqrt{16}$$

$$(x - 3)^2 = 16$$

$$x - 3 = \pm\sqrt{16}$$

$$(x - 3)^2 = 16$$

x^2 is already alone. Apply the square root property

$$x - 3 = \pm\sqrt{16}$$

Simplify the radical.

$$x - 3 = \pm 4$$

Get x alone by adding 3 to both sides.

$$x = 3 \pm 4$$

Obtain each solution.

$$x = 3 + 4 \quad \text{or} \quad x = 3 - 4$$

$$\boxed{x = 7, \quad x = -1}$$

Rationalize the denominator.

Example F: Solve $2(x + 4)^2 + 3 = 57$

$$2(x + 4)^2 + 3 = 57$$

First get $(x + 4)^2$ alone.

$$2(x + 4)^2 = 54$$

$$(x + 4)^2 = 27$$

Apply the square root property.

$$x + 4 = \pm\sqrt{27}$$

Simplify the radical.

$$x + 4 = \pm 3\sqrt{3}$$

Get x alone.

$$x = -4 \pm 3\sqrt{3}$$

$$\boxed{x = -4 + 3\sqrt{3}, \quad x = -4 - 3\sqrt{3}}$$

Worksheet 9.1 Solve Quadratic Equations Using the Square Root Property

Solve each equation.

1. $x^2 - 36 = 0$

2. $x^2 = 75$

3. $x^2 + 20 = 4$

4. $3x^2 + 50 = 14$

5. $5x^2 + 2 = 3$

6. $\frac{3}{2}x^2 - 7 = 20$

7. $(x + 1)^2 = 100$

8. $5(x - 4)^2 = 100$

9. $\left(x + \frac{1}{5}\right)^2 = \frac{3}{25}$

10. $(2x + 3)^2 + 1 = -7$

9.2 Solve Quadratic Equations by Completing the Square

Learning Objectives

By the end of this section, you will be able to:

- Complete the square of a binomial expression
- Solve quadratic equations of the form $x^2 + bx + c = 0$ by completing the square
- Solve quadratic equations of the form $ax^2 + bx + c = 0$ by completing the square

Completing the Square of a Binomial Expression

Our goal is to turn the given binomial into a perfect square trinomial and then factor it. We will then want to factor the perfect square trinomial. This will allow us to use the completing the square method to solve quadratic equations, which we'll see later.

HOW TO: Complete a square of $x^2 + bx$.

STEPS

1. Identify b , the coefficient of x .
2. Find $\left(\frac{1}{2}b\right)^2$, the number to complete the square.
3. Add the $\left(\frac{1}{2}b\right)^2$ to $x^2 + bx$.
4. Factor the perfect square trinomial, writing it as a binomial squared.
5. Recall:
 - a. $a^2 + 2ab + b^2 = (a + b)^2$
 - b. $a^2 - 2ab + b^2 = (a - b)^2$

Example A: Complete the square of $x^2 + 10x$ to make a perfect square trinomial. Then write the result as a binomial squared.

Step 1: $b = 10$

Step 2: Find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot 10\right)^2 = (5)^2 = 25$

Step 3: Add the number above to your binomial: $x^2 + 10x + 25$

Step 4: Factor to $\boxed{(x + 5)^2}$

Example B: Complete the square of $x^2 - \frac{1}{3}x$ to make a perfect square trinomial. Then write the result as a binomial squared.

Step 1: $b = -\frac{1}{3}$

Step 2: Find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot -\frac{1}{3}\right)^2 = \left(-\frac{1}{6}\right)^2 = \frac{1}{36}$

Step 3: Add the number above to your binomial: $x^2 - \frac{1}{3}x + \frac{1}{36}$

Step 4: Factor to $\boxed{\left(x - \frac{1}{6}\right)^2}$

Solving Quadratic Equations by Completing the Square

HOW TO: Solve a quadratic equation of the form $ax^2 + bx + c = 0$ by completing the square.

STEPS

1. If $a \neq 1$, divide the equation by a to make the coefficient of the x^2 term 1.
2. Isolate the variable terms on one side of the equation and the constant terms on the other.
3. Find $\left(\frac{1}{2}b\right)^2$ and add it to both sides of the equation.
4. Factor the perfect square trinomial and write it as a binomial squared. On the right side of the equation, add the terms.
5. Use the square root property.
6. Simplify the radical and then solve the two resulting equations.

Example C: Solve by completing the square: $x^2 - 4x = 12$

Step 1: $a = 1$ so no division is necessary

Step 2: The variables are on the left and the constant is on the right of the equation, so no rearranging is necessary

Step 3: Find $\left(\frac{1}{2}b\right)^2$:

$$\left(\frac{1}{2} \cdot -4\right)^2 = (-2)^2 = 4$$

Add 4 to both sides of the equation:

$$x^2 - 4x + 4 = 12 + 4$$

Step 4: Factor and simplify:

$$(x - 2)^2 = 16$$

Step 5: Use the square root property:

$$x - 2 = \pm\sqrt{16}$$

Step 6: Simplify the radical and solve:

$$x - 2 = \pm 4$$

$$x - 2 = 4, \quad x - 2 = -4$$

$$\boxed{x = 6, \quad x = -2}$$

Example D: Solve by completing the square: $x^2 + 2x - 17 = 0$

$$x^2 + 2x - 17 = 0$$

$a = 1$ but the equation needs to be rearranged

$$x^2 + 2x = 17$$

Now find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot 2\right)^2 = (1)^2 = 1$ and add to both sides

$$x^2 + 2x + 1 = 17 + 1$$

Factor and simplify

$$(x + 1)^2 = 18$$

Use the square root property and solve

$$x + 1 = \pm\sqrt{18}$$

$$x + 1 = \pm 3\sqrt{2}$$

$$x = -1 \pm 3\sqrt{2}$$

$$\boxed{x = -1 + 3\sqrt{2}, \quad x = -1 - 3\sqrt{2}}$$

Example E: Solve by completing the square: $x^2 + 7x + 15 = 0$

$$x^2 + 7x + 15 = 0$$

$a = 1$ but the equation needs to be rearranged

$$x^2 + 7x = -15$$

Now find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot 7\right)^2 = \left(\frac{7}{2}\right)^2 = \frac{49}{4}$ and add to both sides

$$x^2 + 7x + \frac{49}{4} = -15 + \frac{49}{4}$$

Factor and simplify

$$\left(x + \frac{7}{2}\right)^2 = -\frac{11}{4}$$

Use the square root property and solve

$$x + \frac{7}{2} = \pm \sqrt{-\frac{11}{4}}$$

$$x + \frac{7}{2} = \pm \frac{\sqrt{-11}}{\sqrt{4}}$$

$$x + \frac{7}{2} = \pm \frac{i\sqrt{11}}{2}$$

$$x = -\frac{7}{2} \pm \frac{i\sqrt{11}}{2}$$

$$\boxed{x = -\frac{7}{2} + \frac{i\sqrt{11}}{2}, \quad x = -\frac{7}{2} - \frac{i\sqrt{11}}{2}}$$

Example F: Solve by completing the square: $4x^2 + 20x - 3 = 0$

$$4x^2 + 20x - 3 = 0$$

$a \neq 1$ so divide the whole equation by 4 (the coefficient of x^2)

$$\frac{4x^2}{4} + \frac{20x}{4} - \frac{3}{4} = \frac{0}{4}$$

$$x^2 + 5x - \frac{3}{4} = 0$$

Rearrange terms: add $\frac{3}{4}$ to both sides

$$x^2 + 5x = \frac{3}{4}$$

Now find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot 5\right)^2 = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$ and add to both sides

$$x^2 + 5x + \frac{25}{4} = \frac{3}{4} + \frac{25}{4}$$

Factor and simplify

$$\left(x + \frac{5}{2}\right)^2 = \frac{28}{4}$$

$$\left(x + \frac{5}{2}\right)^2 = 7$$

Use the square root property and solve

$$x + \frac{5}{2} = \pm \sqrt{7}$$

$$x = -\frac{5}{2} \pm \sqrt{7}$$

$$\boxed{x = -\frac{5}{2} + \sqrt{7}, \quad x = -\frac{5}{2} - \sqrt{7}}$$

Example G: Solve by completing the square: $3x^2 - 2x = 21$

$$3x^2 - 2x = 21$$

$a \neq 1$ so divide the whole equation by 3 (the coefficient of x^2)

$$\frac{3x^2}{3} - \frac{2x}{3} = \frac{21}{3}$$

$$x^2 - \frac{2}{3}x = \frac{21}{3}$$

Now find $\left(\frac{1}{2}b\right)^2 = \left(\frac{1}{2} \cdot -\frac{2}{3}\right)^2 = \left(-\frac{2}{6}\right)^2 = \left(-\frac{1}{3}\right)^2 = \frac{1}{9}$
and add

$$x^2 - \frac{2}{3}x + \frac{1}{9} = \frac{21}{3} + \frac{1}{9}$$

Factor and simplify

$$\left(x - \frac{1}{3}\right)^2 = \frac{64}{9}$$

Use the square root property and solve

$$x - \frac{1}{3} = \pm \sqrt{\frac{64}{9}}$$

$$x - \frac{1}{3} = \pm \frac{8}{3}$$

$$x = \frac{1}{3} \pm \frac{8}{3}$$

$$x = \frac{1}{3} + \frac{8}{3}, \quad x = \frac{1}{3} - \frac{8}{3}$$

$$x = \frac{9}{3}, \quad x = \frac{-7}{3}$$

$$\boxed{x = 3, \quad x = \frac{-7}{3}}$$

Worksheet 9.2 Solve Quadratic Equations by Completing the Square

Solve each equation by completing the square.

11. $x^2 - 2x = 8$

12. $x^2 + 6x = -2$

13. $x^2 - 10x + 12 = 0$

14. $x^2 - 4x - 14 = 0$

15. $x^2 + 2x = -20$

16. $x^2 + 3x = -\frac{1}{4}$

17. $4x^2 - 28x + 24 = 0$

18. $3x^2 - 36x = -120$

19. $9x^2 - 12x = 9$

20. $3x^2 - 8x = -4$

9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 1)

Learning Objectives

By the end of this section, you will be able to:

- Solve quadratic equations using the quadratic formula

QUADRATIC FORMULA

The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$, are given by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To use the quadratic formula, note the values of a , b , and c in the given equation, and plug them into the formula. Then, simplify the formula. (Be sure the equation is in standard form first!)

$$ax^2 + bx + c = 0 \qquad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example A: Solve using the quadratic formula: $2x^2 + 11x + 12 = 0$

First, note that the equation is in standard form.

Identify the values of a , b , and c and plug them in:

$$2x^2 + 11x + 12 = 0 \qquad x = \frac{-11 \pm \sqrt{11^2 - 4(2)(12)}}{2(2)}$$

Simplify the formula:

$$x = \frac{-11 \pm \sqrt{121 - 96}}{4}$$

$$x = \frac{-11 \pm \sqrt{25}}{4}$$

Simplify the square root

$$x = \frac{-11 \pm 5}{4}$$

Split into two equations

$$x = \frac{-11 + 5}{4}$$

$$x = \frac{-11 - 5}{4}$$

$$x = \frac{-6}{4}$$

$$x = \frac{-16}{4}$$

$$\boxed{x = -\frac{3}{2} \quad x = -4}$$

Example B: Solve using the quadratic formula: $x^2 - 6x = 2$

First, note that the equation is NOT in standard form. Rewrite it: $x^2 - 6x - 2 = 0$
Identify the values of a , b , and c and plug them in:

$$1x^2 - 6x - 2 = 0$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(-2)}}{2(1)}$$

Simplify the formula:

$$x = \frac{6 \pm \sqrt{36 + 8}}{2}$$

$$x = \frac{6 \pm \sqrt{44}}{2}$$

Simplify the radical

$$x = \frac{6 \pm 2\sqrt{11}}{2}$$

Simplify by factoring and canceling

$$x = \frac{2(3 \pm 1\sqrt{11})}{2}$$

$$x = 3 \pm 1\sqrt{11}$$

$$\boxed{x = 3 + \sqrt{11} \quad x = 3 - \sqrt{11}}$$

Example C: Solve using the quadratic formula: $3x^2 - x + 2 = 0$

First, note that the equation is in standard form.
Identify the values of a , b , and c and plug them in:

$$3x^2 - 1x + 2 = 0$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(3)(2)}}{2(3)}$$

Simplify the formula:

$$x = \frac{1 \pm \sqrt{1-24}}{6}$$

$$x = \frac{1 \pm \sqrt{-23}}{6}$$

$$x = \frac{1 \pm i\sqrt{23}}{6}$$

$$x = \frac{1}{6} \pm \frac{\sqrt{23}}{6}i$$

Simplify the radical

Write answers in standard form

$$x = \frac{1}{6} + \frac{\sqrt{23}}{6}i \quad x = \frac{1}{6} - \frac{\sqrt{23}}{6}i$$

Worksheet 9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 1)

Solve each equation by using the quadratic formula.

21. $x^2 + 5x - 14 = 0$

22. $3x^2 - x = 2$

23. $x^2 + 8x - 2 = 0$

24. $x^2 - 4x = 14$

25. $2x^2 + 10x + 11 = 0$

26. $2x^2 - x + 4 = 0$

27. $5x^2 + 2x + 4 = 4$

28. $x^2 + 8x + 16 = 0$

9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 2)

Learning Objectives

By the end of this section, you will be able to:

- Use the discriminant to predict the number and type of solutions of a quadratic equation
- Identify the most appropriate method to use to solve a quadratic equation

The Discriminant

DISCRIMINANT

In the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, the quantity $b^2 - 4ac$ is called the discriminant.

The discriminant can help us determine the number and type of solutions a quadratic equation has without actually solving the equation.

Sign of the Discriminant	Number and Type of Solutions	Examples		
		Quadratic Equation	Discriminant $b^2 - 4ac$	Solutions
positive	2 real solutions	$2x^2 + 11x + 12 = 0$	$11^2 - 4(2)(12) = 25$	$x = -\frac{3}{2}$ $x = -4$
zero	1 real solution	$x^2 + 2x + 1 = 0$	$2^2 - 4(1)(1) = 0$	$x = -1$
negative	2 complex solutions	$x^2 - 8x + 20 = 0$	$(-8)^2 - 4(1)(20) = -16$	$x = 4 + 2i$ $x = 4 - 2i$

Example A: Determine the number of solutions to the quadratic equation $4x^2 - 5x + 6 = 0$.

Find the discriminant:

$$\begin{aligned}b^2 - 4ac &= (-5)^2 - 4(4)(6) \\&= 25 - 96 \\&= -71\end{aligned}$$

Since the discriminant is **negative**, there are *two complex solutions*.

Example B: Determine the number of solutions to the quadratic equation $3x^2 - 5x - 12 = 0$.

Find the discriminant:

$$\begin{aligned}b^2 - 4ac &= (-5)^2 - 4(3)(-12) \\&= 25 + 144 \\&= 169\end{aligned}$$

Since the discriminant is **positive**, there are *two real solutions*.

Example C: Determine the number of solutions to the quadratic equation $x^2 - 8x + 16 = 0$.

Find the discriminant:

$$\begin{aligned}b^2 - 4ac &= (-8)^2 - 4(1)(16) \\&= 64 - 64 \\&= 0\end{aligned}$$

Since the discriminant is **zero**, there is *one real solution*.

Summary of Methods to Solve Quadratic Equations

We learned four methods to solve quadratic equations:

1. Factoring
2. Square Root Property
3. Completing the Square
4. Quadratic Formula

How can you decide which method to use?

HOW TO: Identify the most appropriate method to solve a quadratic equation.

STEPS	REASON
1. Try factoring first.	If the quadratic factors easily, this method is very quick.
2. Try the square root property next.	If the equation fits the form $ax^2 = k$ or $a(x - h)^2 = k$, it can be easily solved with this property.
3. Use the quadratic formula .	Any other equation is best solved with the formula. (NOTE: Any quadratic equation can be solved with the quadratic formula, but the above two methods may be easier.)

NOTE: The **completing the square** method is not recommended since many people find it cumbersome and prefer not to use it.

Example D: Identify the most appropriate method to solve $x^2 - 7x + 10 = 0$.

Since the quadratic can be easily factored, the most appropriate method is **factoring**.

Example E: Identify the most appropriate method to solve $5x^2 = 20$.

Since the equation is in the form $ax^2 = k$, the most appropriate method is the **square root property**.

Example F: Identify the most appropriate method to solve $x^2 + x + 1 = 0$.

Since the quadratic cannot be factored, the most appropriate method is the **quadratic formula**.

Worksheet 9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 2)

In numbers 1-6, determine the number and type of solutions to the quadratic equation.

29. $2x^2 + 3x - 2 = 0$

30. $x^2 - 4x + 4 = 0$

31. $3x^2 + 2x + 5 = 0$

32. $x^2 + 7x = -10$

33. $x^2 - 3x + 6 = 0$

34. $x^2 - 6x = -9$

In numbers 7-12, identify the most appropriate method to solve the quadratic equation.

35. $x^2 - 5x + 6 = 0$

36. $x^2 = 16$

37. $(x + 4)^2 = 25$

38. $5x^2 - 3x + 1 = 0$

39. $x^2 + 2x = -1$

40. $x^2 + 2x = -3$

9.6 Graph Quadratic Equations Using Properties

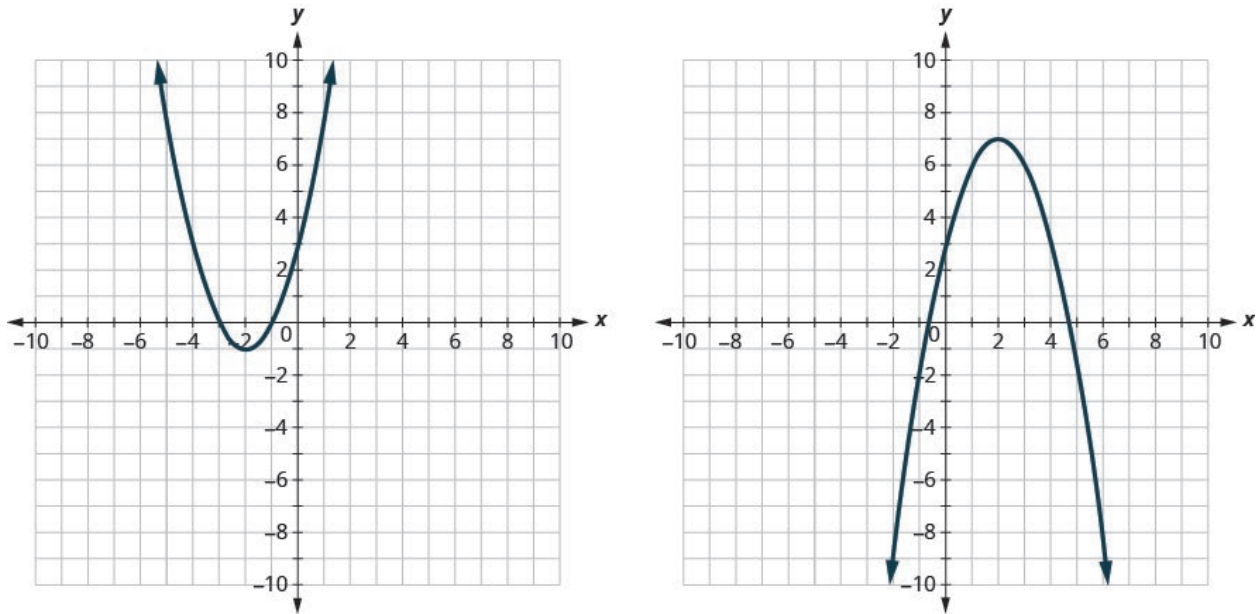
Learning Objectives

By the end of this section, you will be able to:

- Graph quadratic equations using properties
- Solve maximum and minimum applications

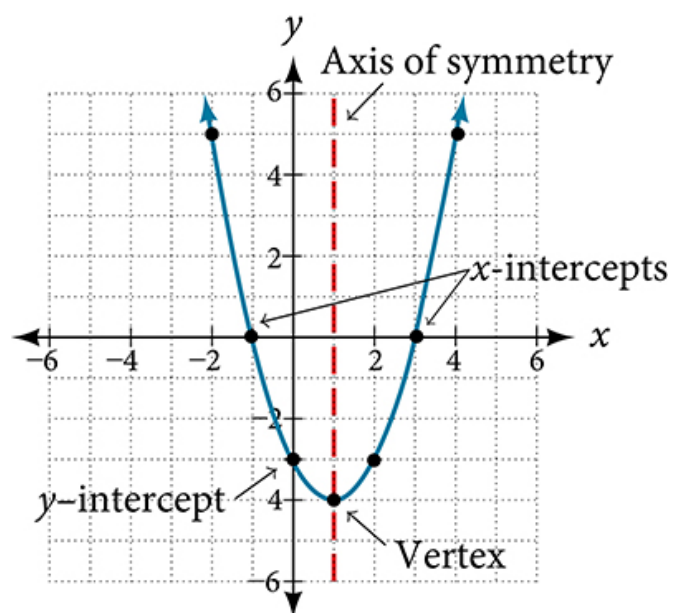
Graphing Quadratic Equations/Functions

The graph of a quadratic equation is called a **parabola**. Parabolas are U-shaped graphs, as seen in the image below.



We will use five features of parabolas to help us graph them:

1. Does the parabola open upward or downward?
2. Axis of symmetry – ‘fold line’ that shows the line of symmetry of the graph
3. Vertex – the point that is the lowest (parabola opens up) or highest (parabola opens down)
4. y-intercept – the point where the parabola crosses the y-axis
5. x-intercept(s) – the point(s) where the parabola crosses the x-axis



HOW TO: Graph a quadratic function using properties

For a quadratic function in the form $ax^2 + bx + c = 0$, use the following steps.

STEPS	METHOD
1. Determine whether the parabola opens upward or downward.	Look at the value of a . If... • $a > 0$, the parabola opens upward  • $a < 0$, the parabola opens downward 
2. Find the equation of the axis of symmetry.	The axis of symmetry is the vertical line given by the formula: $x = -\frac{b}{2a}$
3. Find the vertex.	The vertex is a point. Find its x - and y -coordinates: • x-coordinate is $-\frac{b}{2a}$ • y-coordinate is found by plugging this x-value into the quadratic equation
4. Find the y -intercept. Then find the point symmetric to the y -intercept across the axis of symmetry.	To find the y -intercept: Let $x = 0$ and solve for $f(x)$.
5. Find the x -intercepts.	Let $f(x) = 0$ and solve for x . <i>TIP: Use factoring, the square root property, or the quadratic formula.</i>
6. Graph the parabola	Plot the points you found above and connect with a smooth curve.

Example A: Graph the function by using its properties: $f(x) = x^2 + 2x - 3$

First, note that $a = 1$, $b = 2$, and $c = -3$

1. Since a is positive, the parabola **opens upward**.

2. Axis of symmetry: Find $x = -\frac{b}{2a}$

$$x = -\frac{2}{2(1)} = -1 \quad \text{The axis of symmetry is } x = -1$$

3. Vertex: Find the coordinates of x (same as axis of symmetry) and y (plug x into $f(x)$).

$$x = -1$$

$$y = (-1)^2 + 2(-1) - 3 = -4 \quad \text{The vertex is } (-1, -4)$$

4. y -intercept: Find $f(0)$

$$f(0) = 0^2 + 2(0) - 3 = -3 \quad \text{The } y\text{-intercept is } (0, -3)$$

Point symmetric to the y -intercept: Since the y -int ($x = 0$) is 1 unit to the right of the axis of symmetry ($x = -1$), find the point that is 1 unit to the left of the axis of symmetry ($x = -2$)

The point symmetric to the y-intercept is $(-2, -3)$

5. x -intercept: Solve $f(x) = 0$. For this equation, we will use factoring.

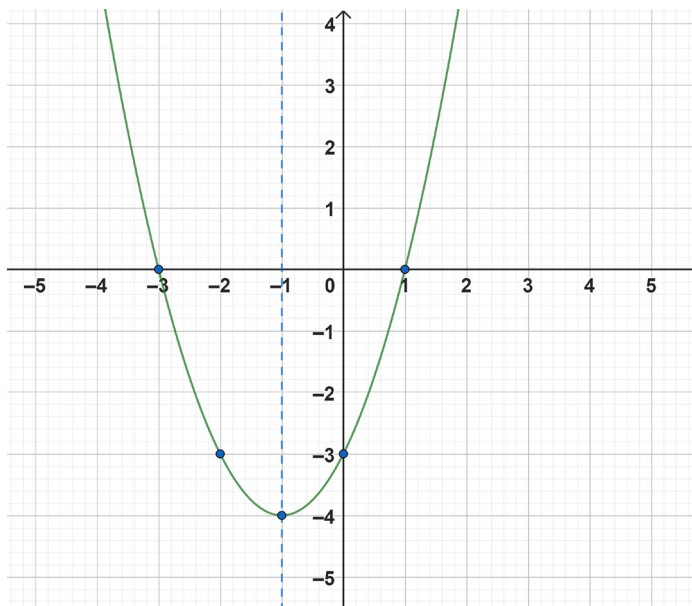
$$x^2 + 2x - 3 = 0$$

$$(x - 1)(x + 3) = 0$$

$$x = 1 \text{ or } x = -3$$

The x -intercepts are $(1, 0)$ and $(-3, 0)$

6. Graph the parabola



Example B: Graph the function by using its properties: $f(x) = -2x^2 - 8x - 8$

First, note that $a = -2$, $b = -8$, and $c = -8$

1. Since a is negative, the parabola **opens downward**.

2. Axis of symmetry: Find $x = -\frac{b}{2a}$

$$x = -\frac{(-8)}{2(-2)} = -2$$

The axis of symmetry is $x = -2$

3. Vertex: Find the coordinates of x (same as axis of symmetry) and y (plug x into $f(x)$).

$$x = -2$$

$$y = -2(-2)^2 - 8(-2) - 8 = 0 \quad \text{The vertex is } (-2, 0)$$

4. y -intercept: Find $f(0)$

$$f(0) = -2(0)^2 - 8(0) - 8 = -8$$

The y -intercept is $(0, -8)$

Point symmetric to the y -intercept: Since the y -int ($x = 0$) is 2 units to the right of the axis of symmetry ($x = -2$), find the point that is 2 units to the left of the axis of symmetry ($x = -4$)

The point symmetric to the y -intercept is $(-4, -8)$

5. x -intercept: Solve $f(x) = 0$. For this equation, we will use factoring.

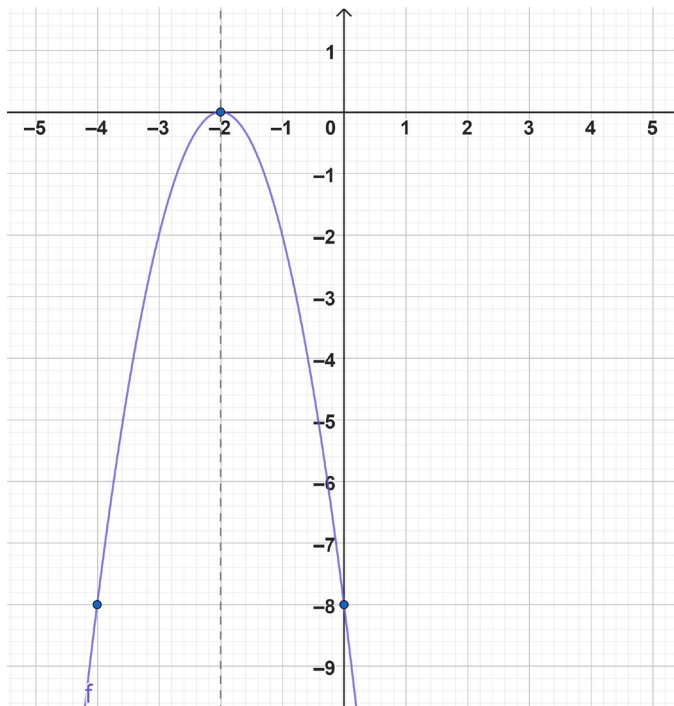
$$-2x^2 - 8x - 8 = 0$$

$$-2(x^2 + 4x + 4) = 0$$

$$\begin{aligned} -2(x+2)(x+2) &= 0 \\ x &= -2 \end{aligned}$$

The x-intercept is $(-2, 0)$

6. Graph the parabola



Example C: Graph the function by using its properties: $f(x) = x^2 + 2x + 4$

First, note that $a = 1$, $b = 2$, and $c = 4$

1. Since a is positive, the parabola **opens upward**.

2. Axis of symmetry: Find $x = -\frac{b}{2a}$

$$x = -\frac{2}{2(1)} = -1$$

The axis of symmetry is $x = -1$

3. Vertex: Find the coordinates of x (same as axis of symmetry) and y (plug x into $f(x)$).

$$x = -1$$

$$y = (-1)^2 + 2(-1) + 4 = 3$$

The vertex is $(-1, 3)$

4. y-intercept: Find $f(0)$

$$f(0) = (0)^2 + 2(0) + 4 = 4$$

The y-intercept is $(0, 4)$

Point symmetric to the y-intercept: Since the y-int ($x = 0$) is 1 unit to the right of the axis of symmetry ($x = -1$), find the point that is 1 unit to the left of the axis of symmetry ($x = -2$)

The point symmetric to the y-intercept is $(-2, 4)$

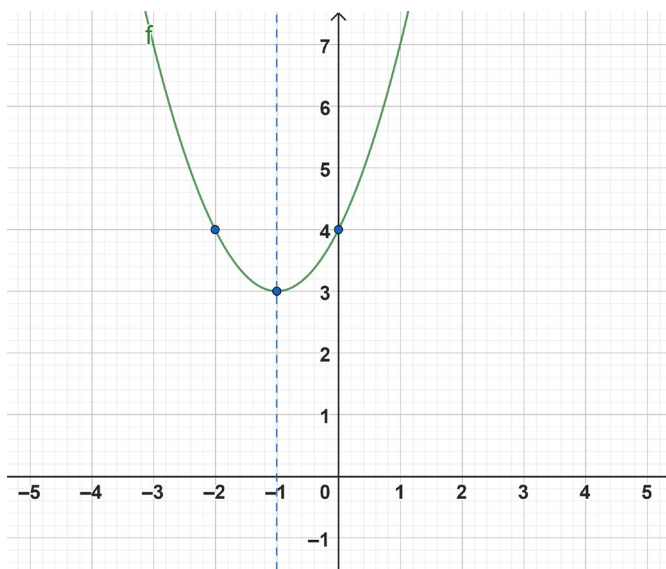
5. x-intercept: Solve $f(x) = 0$. This equation cannot be factored so we will use the quadratic formula.

$$x^2 + 2x + 4 = 0$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(4)}}{2(1)} = \frac{-2 \pm \sqrt{-12}}{2}$$

Since the discriminant is negative, there is no real solution and so **no x-intercept**.

6. Graph the parabola



Example D: Graph the function by using its properties: $f(x) = -3x^2 + 6x + 2$

First, note that $a = -3$, $b = 6$, and $c = 2$

1. Since a is negative, the parabola **opens downward**.

2. Axis of symmetry: Find $x = -\frac{b}{2a}$

$$x = -\frac{6}{2(-3)} = 1$$

The axis of symmetry is $x = 1$

3. Vertex: Find the coordinates of x (same as axis of symmetry) and y (plug x into $f(x)$).

$$x = 1$$

$$y = -3(1)^2 + 6(1) + 2 = 5$$

The vertex is $(1, 5)$

4. y -intercept: Find $f(0)$

$$f(0) = -3(0)^2 + 6(0) + 2 = 2 \quad \textbf{The y-intercept is } (0, 2)$$

Point symmetric to the y -intercept: Since the y -int ($x = 0$) is 1 unit to the left of the axis of symmetry ($x = 1$), find the point that is 1 unit to the right of the axis of symmetry ($x = 2$)

The point symmetric to the y -intercept is $(2, 2)$

5. x -intercept: Solve $f(x) = 0$. This equation cannot be factored so we will use the quadratic formula.

$$-3x^2 + 6x + 2 = 0$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(-3)(2)}}{2(-3)} = \frac{-6 \pm \sqrt{60}}{-6} = \frac{-6 \pm 2\sqrt{15}}{-6} = \frac{-3 \pm \sqrt{15}}{-3}$$

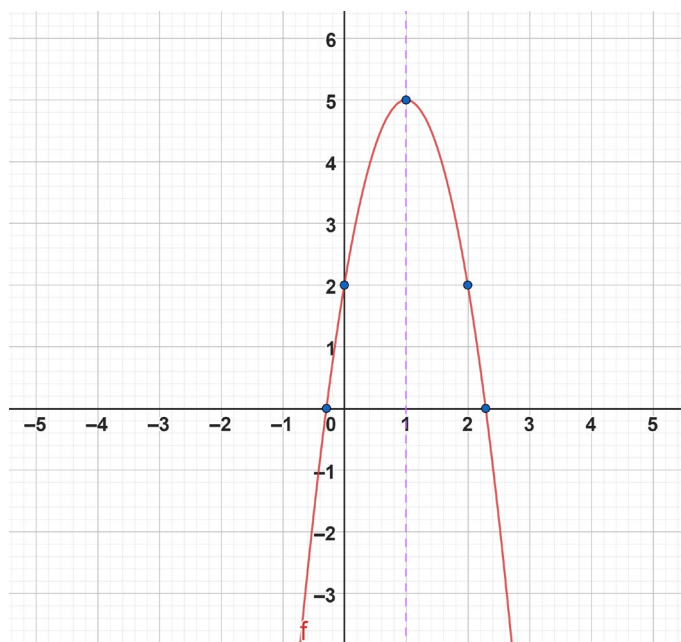
$$x = \frac{-3+\sqrt{15}}{-3}, \quad x = \frac{-3-\sqrt{15}}{-3}$$

Approximate the values:

$$x \approx -0.29, \quad x \approx 2.29$$

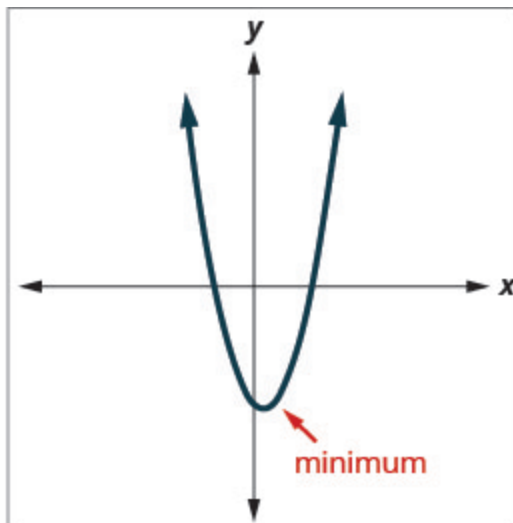
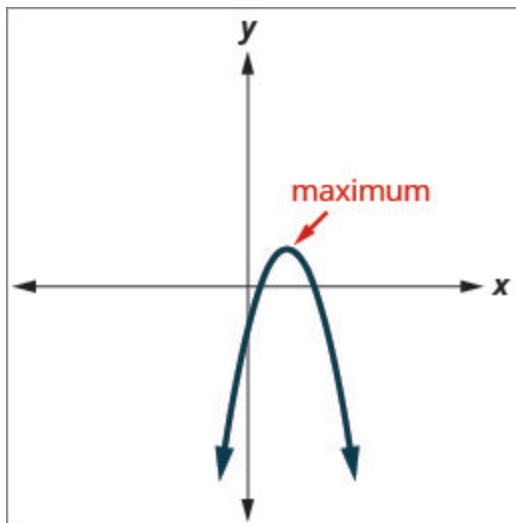
The approximate values of the x-intercepts are $(-0.29, 0)$ and $(2.29, 0)$

6. Graph the parabola



Maximum and Minimum Applications

Since the **vertex** is the highest or lowest point of the parabola, it gives us the maximum or minimum value of a quadratic function, as seen in the image below.



Example D: Find the maximum or minimum value of the quadratic function $f(x) = x^2 + 6x + 8$.

First, note that a is positive so this parabola opens upward and therefore has a **minimum**.
Find the vertex:

$$x = -\frac{b}{2a} = -\frac{6}{2(1)} = -3$$

$$y = (-3)^2 + 6(-3) + 8 = -1$$

The minimum value is **-1** and it occurs when $x = -3$.

Example E:

If a toy rocket is launched vertically upward from ground level with an initial velocity of 128 feet per second, then its height h after t seconds is given by the equation $h(t) = -16t^2 + 128t$. How long will it take the rocket to reach its maximum height? What is the maximum height?

First, note that a is negative so this parabola opens downward and has a **maximum**.

Also note that $a = -16$, $b = 128$, and $c = 0$.

Find the vertex:

$$x: t = -\frac{b}{2a} = -\frac{128}{2(-16)} = 4$$

$$y: h = -16(4)^2 + 128(4) = 256$$

The maximum value is 256 and it occurs when $t = 4$.

After 4 seconds, the toy rocket will reach its maximum height of 256 feet.

Worksheet 9.6 Graph Quadratic Equations Using Properties

1. Graph the function by using its properties: $f(x) = x^2 - 4x + 3$
2. Graph the function by using its properties: $f(x) = -x^2 + 4x - 4$
3. Graph the function by using its properties: $f(x) = 3x^2 + 6x + 3$
4. Graph the function by using its properties: $f(x) = -2x^2 + 12x - 10$
5. Graph the function by using its properties: $f(x) = -2x^2 + 4x + 1$
6. Find the maximum or minimum value of the quadratic function $f(x) = x^2 + 8x - 5$.
7. Find the maximum or minimum value of the quadratic function $f(x) = -5x^2 + 30x + 9$.
8. While on vacation Bobby jumped off a cliff into the ocean. His height h , in feet, after t seconds is given by $h(t) = -16t^2 + 16t + 480$. How long did it take Bobby to reach his maximum height? What was his maximum height?
9. A clothing retailer determines that the function $R(x) = -x^2 + 30x$ can be used to find their revenue R when the selling price of a t-shirt is x dollars. Find the selling price that would give the retailer the maximum revenue, and then find the amount of the maximum revenue.

10.2 Evaluate and Graph Exponential Functions

Learning Objectives:

In this section, you will:

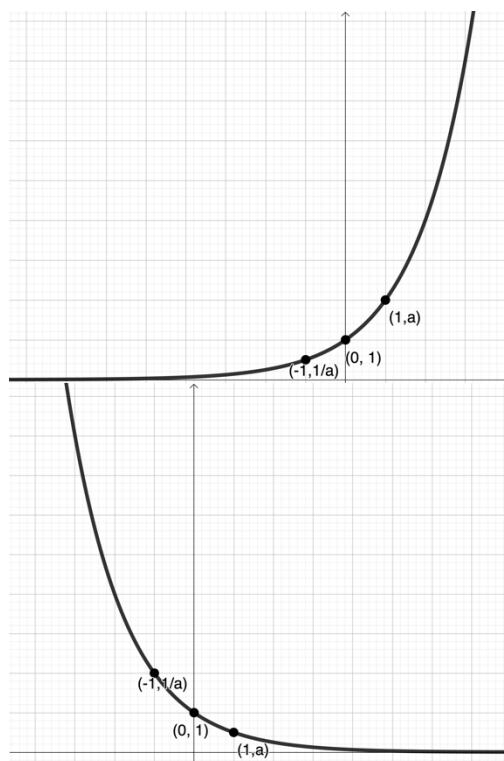
- Graph exponential functions

Exponential Function: An exponential function, where $a > 0$ and $a \neq 1$, is a function of the form $f(x) = a^x$

Properties of the Graph of $f(x) = a^x$:

When $a > 1$	
Domain	$(-\infty, \infty)$
Range	$(0, \infty)$
x -intercept	none
y -intercept	$(0, 1)$
Contains	$(1, a), (-1, \frac{1}{a})$
Asymptote	x -axis, the line $y = 0$
Basic shape	increasing

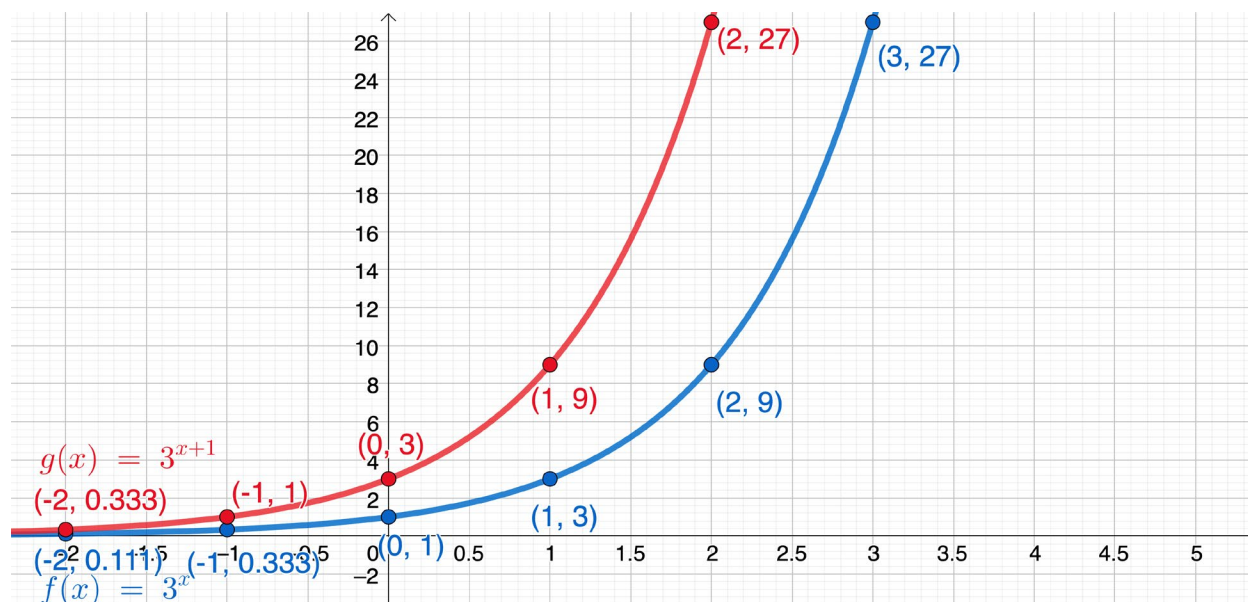
When $0 < a < 1$	
Domain	$(-\infty, \infty)$
Range	$(0, \infty)$
x -intercept	none
y -intercept	$(0, 1)$
Contains	$(1, a), (-1, \frac{1}{a})$
Asymptote	x -axis, the line $y = 0$
Basic shape	decreasing



Example A: On the same coordinate system graph $f(x) = 3^x$, and $g(x) = 3^{x+3}$

Solution:

x	$f(x) = 3^x$	$(x, f(x))$		x	$g(x) = 3^{x+1}$	$(x, f(x))$
-2	$3^{-2} = \frac{1}{9}$	$(-2, \frac{1}{9})$		-2	$3^{-2+1} = \frac{1}{3}$	$(-2, \frac{1}{3})$
-1	$3^{-1} = \frac{1}{3}$	$(-1, \frac{1}{3})$		-1	$3^0 = 1$	$(-1, 1)$
0	$3^0 = 1$	$(0, 1)$		0	$3^1 = 3$	$(0, 3)$
1	$3^1 = 3$	$(1, 3)$		1	$3^2 = 9$	$(1, 9)$
2	$3^2 = 9$	$(2, 9)$		2	$3^3 = 27$	$(2, 27)$



Natural Base e : The number e is defined as the value of $(1 + \frac{1}{n})^n$, as n increases without bound. We say, as n approaches infinity, $e \approx 2.718281827 \dots$

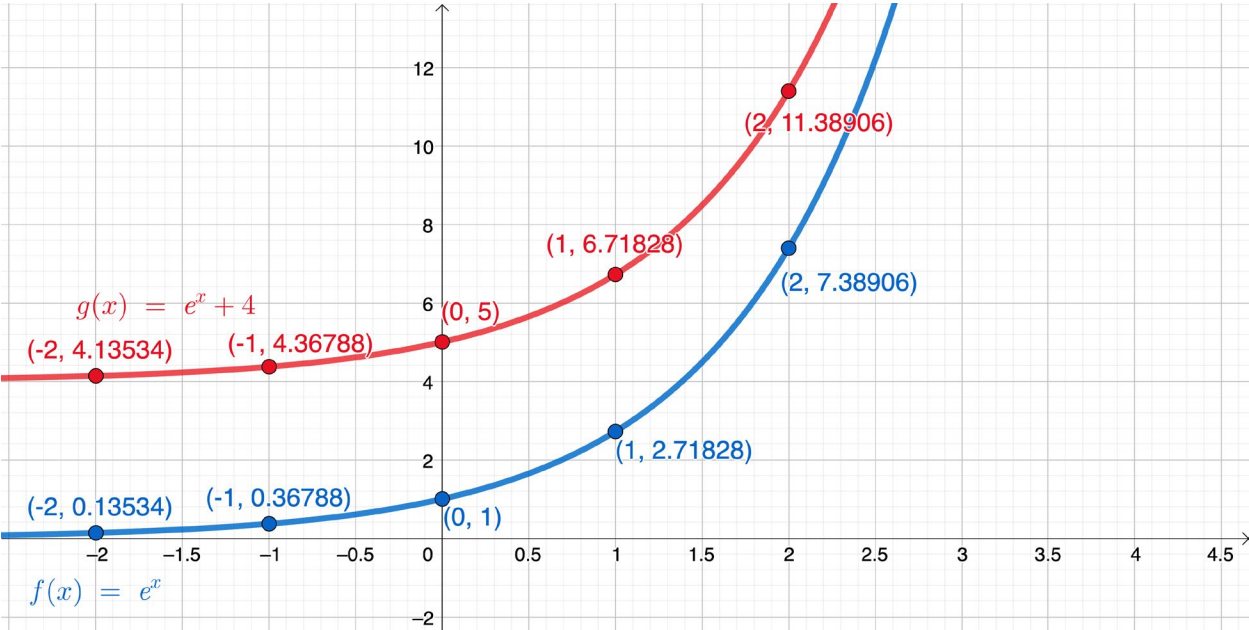
Natural Exponential Function: The natural exponential function is an exponential function whose base is e . $f(x) = e^x$ the domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.

Example B: On the same coordinate system graph $f(x) = e^x$, and $g(x) = e^x + 4$

Solution:

x	$f(x) = e^x$	$(x, f(x))$		x	$g(x) = e^x + 4$	$(x, f(x))$
-2	$e^{-2} = \frac{1}{e^2}$	$(-2, \frac{1}{e^2})$		-2	$e^{-2} = \frac{1}{e^2} + 4$	$(-2, \frac{1}{e^2} + 4)$
-1	$e^{-1} = \frac{1}{e}$	$(-1, \frac{1}{e})$		-1	$e^{-1} = \frac{1}{e} + 4$	$(-1, \frac{1}{e} + 4)$
0	$e^0 = 1$	$(0, 1)$		0	$e^0 + 4 = 5$	$(0, 5)$

1	$e^1 = e$	$(1, e)$		1	$e^1 + 4$	$(1, e^1 + 4)$
2	$e^2 = e^2$	$(2, e^2)$		2	$e^2 + 4$	$(2, e^2 + 4)$



Worksheet 10.2: Evaluate and Graph Exponential Functions

Graph Exponential Functions: In the following exercises, graph each exponential function.

1) $f(x) = 4^x$

4) $f(x) = \frac{1}{5}^x$

2) $f(x) = 5^x$

5) $f(x) = 2.4^x$

3) $f(x) = \frac{1}{4}^x$

6) $f(x) = 0.7^x$

In the following exercises, graph each function in the same coordinate system.

7) $f(x) = 2^x$, and $g(x) = 2^x - 4$

9) $f(x) = e^x$, and $g(x) = e^{x-1} + 3$

8) $f(x) = 4^x$, and $g(x) = 4^{x+4}$

10) $f(x) = \frac{1}{2}^x$, and $g(x) = \frac{1}{2}^{x-2}$

In the following exercises, graph each exponential function.

11) $f(x) = 3^{x-2} - 3$

13) $f(x) = e^x + 5$

12) $f(x) = \frac{1}{4}^{x+4} + 2$

14) $f(x) = -2^{x+1} + 2$

10.3 Evaluate and Graph Logarithmic Functions

Learning Objectives:

In this section, you will:

- Convert between exponential and logarithmic form
- Evaluate logarithmic functions
- Graph logarithmic functions

Convert Between Exponential and Logarithmic Form:

The function $f(x) = \log_a x$ is the **logarithmic function** with base a , where $a > 0$, $x > 0$, and $a \neq 1$.

$y = \log_a x$ is equivalent to $x = a^y$

Example A: Convert to logarithmic form: a) $3^4 = 81$, b) $6^{\frac{1}{3}} = \sqrt[3]{6}$, c) $\left(\frac{1}{3}\right)^5 = \frac{1}{243}$.

Solution:

a) $3^4 = 81$ $y = \log_a x$ $4 = \log_3 81$ If $3^4 = 81$, then $4 = \log_3 81$	b) $6^{\frac{1}{3}} = \sqrt[3]{6}$ $y = \log_a x$ $\frac{1}{3} = \log_6 \sqrt[3]{6}$ If $6^{\frac{1}{3}} = \sqrt[3]{6}$, then $\frac{1}{3} = \log_6 \sqrt[3]{6}$	c) $\frac{1}{3}^5 = \frac{1}{243}$ $y = \log_a x$ $5 = \log_{\frac{1}{3}} \frac{1}{243}$ If $\left(\frac{1}{3}\right)^5 = \frac{1}{243}$, then $5 = \log_{\frac{1}{3}} \frac{1}{243}$
--	--	---

Example B: Convert to exponential form: a) $3 = \log_2 8$, b) $0 = \log_7 1$, c) $-4 = \log_{10} \frac{1}{10000}$.

Solution:

a) $3 = \log_2 8$ $x = a^y$ $8 = 2^3$ If $3 = \log_2 8$, then $8 = 2^3$	b) $0 = \log_7 1$ $x = a^y$ $1 = 7^0$ If $0 = \log_7 1$, then $1 = 7^0$	c) $-4 = \log_{10} \frac{1}{10000}$ $x = a^y$ $\frac{1}{10000} = 10^{-4}$ If $-4 = \log_{10} \frac{1}{10000}$, then $\frac{1}{10000} = 10^{-4}$
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Evaluate Logarithmic Functions:

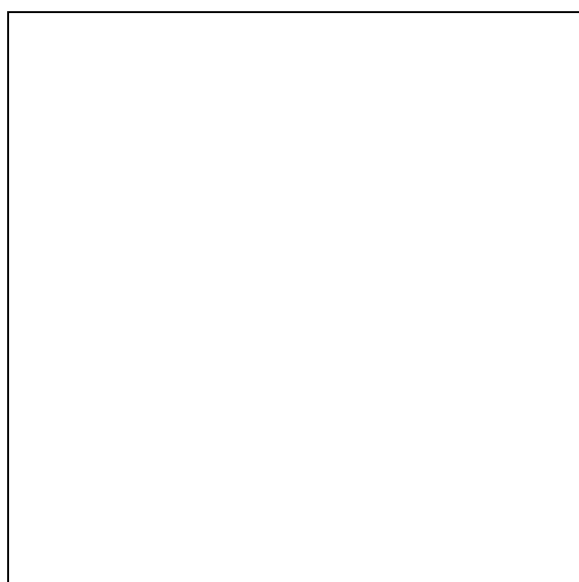
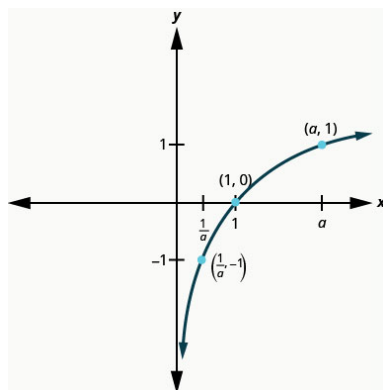
Example C: Find the exact value of each logarithm without using a calculator: a) $\log_6 216$,
b) $\log_{64} 4$, c) $\log_4 \frac{1}{16}$.

Solution:

- a) Set the expression equal to x : $\log_6 216 = x$
Change to exponential form: $6^x = 216$
Rewrite 216 as 6^3 : $6^x = 6^3$
With the same base the exponents must be equal: $x = 3$ therefore, $\log_6 216 = 3$.
- b) Set the expression equal to x : $\log_{64} 4 = x$
Change to exponential form: $64^x = 4$
Since $\sqrt[3]{64} = 4$: $64^{\frac{1}{3}} = 4$
This means that: $x = \frac{1}{3}$ therefore, $\log_{64} 4 = \frac{1}{3}$.
- c) Set the expression equal to x : $\log_4 \frac{1}{16} = x$
Change to exponential form: $4^x = \frac{1}{16}$
Since $4^{-2} = \frac{1}{16}$:
This means that: $x = -2$ therefore, $\log_4 \frac{1}{16} = -2$.

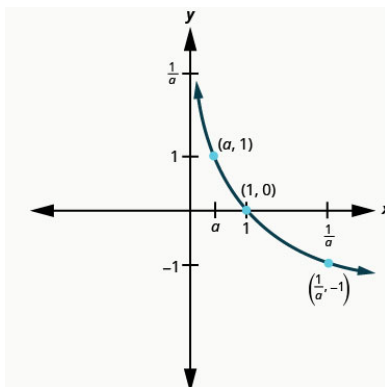
Graph Logarithmic Functions:

Properties of the Graph of $y = \log_a x$:



When $a > 1$	
Domain	$(0, \infty)$
Range	$(-\infty, \infty)$
x -intercept	$(1, 0)$
y -intercept	none
Contains	$(a, 1), (\frac{1}{a}, -1)$
Asymptote	y -axis
Basic shape	increasing

When $0 < a < 1$	
Domain	$(0, \infty)$
Range	$(-\infty, \infty)$
x-intercept	$(1, 0)$
y-intercept	none
Contains	$(a, 1), (\frac{1}{a}, -1)$
Asymptote	y-axis
Basic shape	decreasing

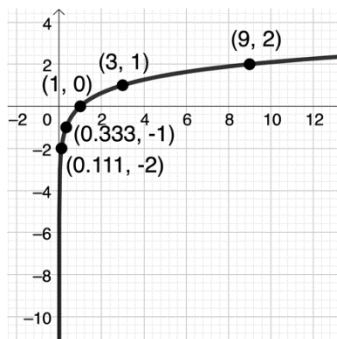


Example D) Graph $\log_3 x$.

Solution:

To graph the function, we will first rewrite the logarithmic equation, $y = \log_3 x$, in exponential form, $3^y = x$. we will use point plotting to graph the function. It will be easier to start with values of y and then get x .

y	$3^y = x$	(x, y)
-2	$3^{-2} = \frac{1}{9}$	$(\frac{1}{9}, -2)$
-1	$3^{-1} = \frac{1}{3}$	$(\frac{1}{3}, -1)$
0	$3^0 = 1$	$(1, 0)$
1	$3^1 = 3$	$(3, 1)$
2	$3^2 = 9$	$(9, 2)$

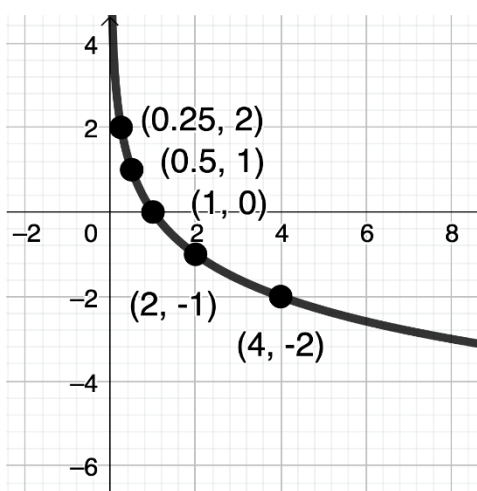


Example E) Graph $\log_{\frac{1}{2}} x$.

Solution:

To graph the function, we will first rewrite the logarithmic equation, $y = \log_{\frac{1}{2}} x$, in exponential form, $\frac{1}{2}^y = x$. we will use point plotting to graph the function. It will be easier to start with values of y and then get x .

y	$\frac{1^y}{2} = x$	(x, y)
-2	$\frac{1^{-2}}{2} = 4$	$(4, -2)$
-1	$\frac{1^{-1}}{2} = 2$	$(2, -1)$
0	$\frac{1^0}{2} = 1$	$(1, 0)$
1	$\frac{1^1}{2} = \frac{1}{2}$	$(\frac{1}{2}, 1)$
2	$\frac{1^2}{2} = \frac{1}{4}$	$(\frac{1}{4}, 2)$



Worksheet 10.3: Evaluate and Graph Logarithmic Functions

Convert Between Exponential and Logarithmic Form: In the following exercises, convert from exponential to logarithmic form.

- 1) $5^3 = 125$
- 2) $2^{-5} = \frac{1}{32}$
- 3) $19^x = \sqrt[3]{19}$

- 4) $x^{\frac{1}{3}} = \sqrt[3]{10}$
- 5) $(\frac{1}{6})^3 = \frac{1}{216}$
- 6) $e^4 = x$

In the following exercises, convert from logarithmic equation to exponential form.

- 7) $4 = \log_4 256$
- 8) $7 = \log_x 128$
- 9) $0 = \log_9 1$

- 10) $2 = \log_5 25$
- 11) $-5 = \log_{10} \frac{1}{100,000}$
- 12) $7 = \log_e x$

Graph Logarithmic Functions: In the following exercises, graph each logarithmic function.

- 13) $y = \log_5 x$
- 14) $y = \log_{3.5} x$

- 15) $y = \log_{\frac{1}{4}} x$
- 16) $y = \log_{0.7} x$

Answer Key

Chapter 2

2.1 Use a General Strategy to Solve Linear Equations

1. $x = -2$
2. $x = -3$
3. $x = 32$
4. $y = -1$
5. $y = 4$
6. $n = 11/5$
7. $x = 25/12$
8. no solution
9. all real numbers
10. $x = \frac{1}{4}$
11. $x = -14$
12. $y = -17/6$
13. $x = -7$
14. $n = 21$

2.3 Solve a Formula for a Specific Variable

1. $h = \frac{2A}{b}$
2. $b = 180 - a - c$
3. $W = \frac{P-2L}{2}$
4. $h = \frac{V}{\pi r^2}$
5. $r = \frac{A-P}{Pt}$
6. $y = \frac{5-6x}{2}$

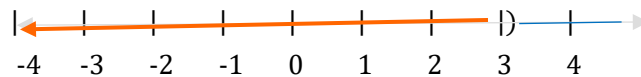
7. $y = \frac{-8-x}{-3}$

2.4 Solve Mixture and Uniform Motion Applications

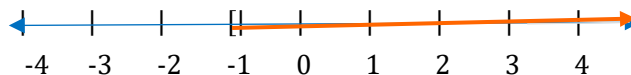
1. 55 quarters and 170 nickels
2. Eight 68-cent stamps and thirty 73-cent stamps
3. 40 children's tickets and 80 adult's tickets
4. Dylan's rate is 6 mph and Cassie's rate is 8 mph
5. 2 hours
6. Tom's speed is 80 mph and Ryan's speed is 70 mph

2.5 Solve Linear Inequalities

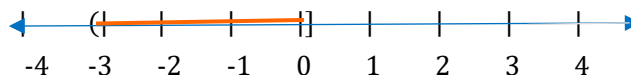
1.



2.



3.



3. $\left(\frac{5}{2}, \infty\right)$

4. $(-\infty, 2]$

5. $(-3, \infty)$

6. $\left(-\infty, -\frac{3}{2}\right]$

7. $\left(\frac{2}{7}, \infty\right)$

8. $(6, \infty)$
9. $(\infty, 2]$
10. $(-\infty, 5)$
11. $(2, \infty)$
12. $(-\infty, \frac{1}{3})$
13. $(-\infty, 4)$
14. $[\frac{3}{13}, \infty)$
15. $7 + x > 12, (5, \infty)$
16. $5v \leq v - 6, (-\infty, -\frac{3}{2}]$
17. $x + 15 \geq -15, [-30, \infty)$
18. 27 miles
19. 20 pairs
20. 182 minutes

2.6 Solve Compound Inequalities

1. None
2. $[-2, 2)$
3. $(-2, 5)$
4. $[-1, 4)$
5. $(-\infty, \frac{3}{2})$
6. $[1, 7]$
7. $[-6, 0)$
8. $(-5, -\frac{2}{3})$
9. $(-\infty, 7)$
10. $(-\infty, 0] \cup (1, \infty)$
11. $(-\infty, -1) \cup (3, \infty)$
12. $(-\infty, -3]$
13. $(-\infty, -11) \cup (-4, \infty)$
14. $(-\infty, 9] \cup [10, \infty)$
15. $6 + x > 10$ or $6 + x < 4, (-\infty, -2) \cup (4, \infty)$
16. $4 \geq x - 9 \geq 7, [13, 16]$

17. $0 < 5x + 3 < 18, \left(-\frac{3}{5}, 3\right)$

18. \$2.40 to \$4.80

2.7 Solve Absolute Value Inequalities

1. 5,-5

2. -5,2

3. -7,9

4. None

5. 5,-5

6. $-\frac{29}{15}, \frac{7}{3}$

7. $\frac{11}{2}, \frac{1}{4}$

8. $-1, -\frac{1}{3}$

9. $-\frac{1}{5}, -\frac{2}{3}$

10. $[-7, 7]$

11. $(-7, 1)$

12. $\left(-\infty, -\frac{78}{7}\right) \cup (10, \infty)$

13. $(-\infty, -7) \cup (7, \infty)$

14. $[-4, 20]$

15. $\left[-\frac{1}{2}, 4\right]$

16. $\left(-\frac{1}{2}, 1\right)$

17. $(-\infty, \infty)$

18. $(-\infty, \infty)$

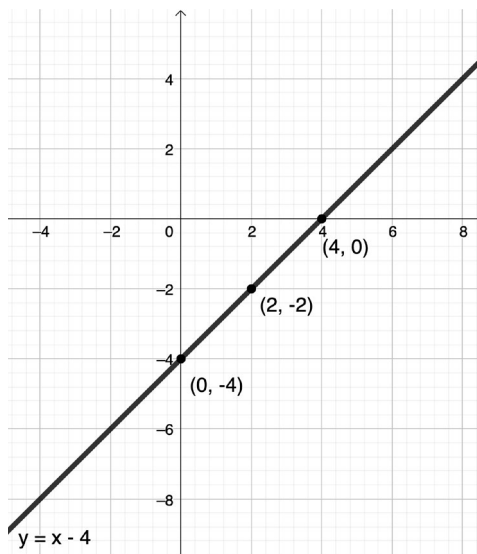
19. $[800, 1200]$

20. $[47.5, 52.5]$

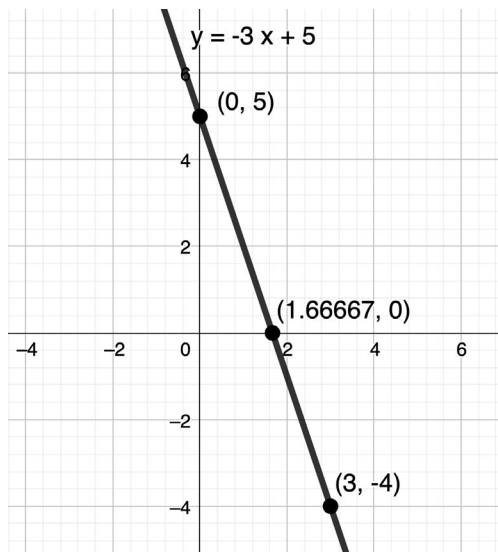
Chapter 3

3.1 Graph Linear Equations in Two Variables

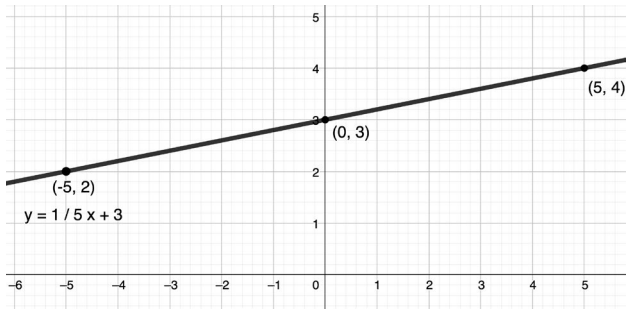
1.



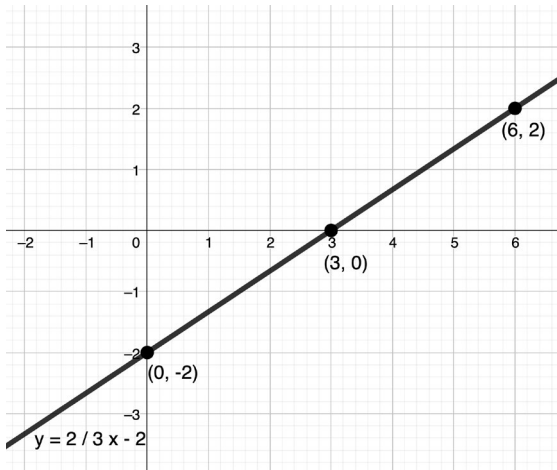
2.



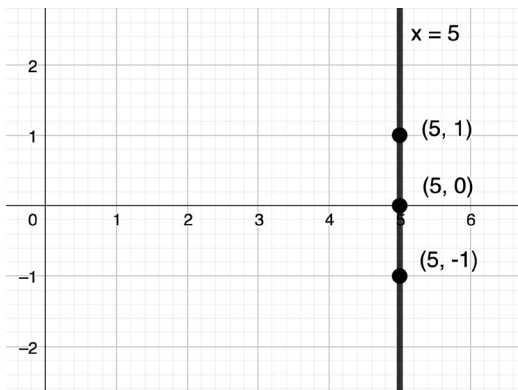
3.



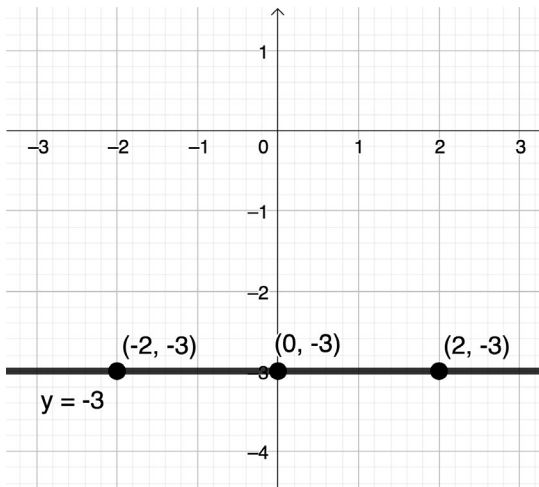
4.



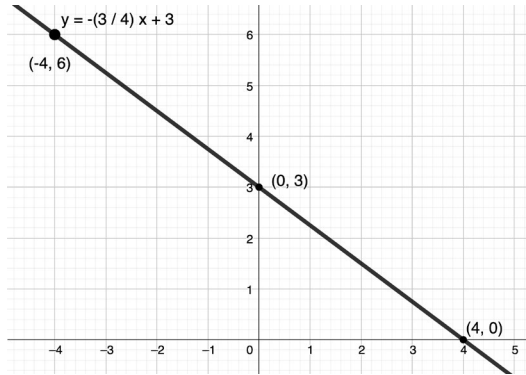
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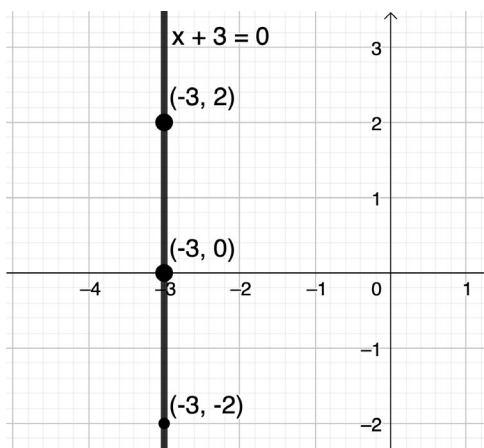
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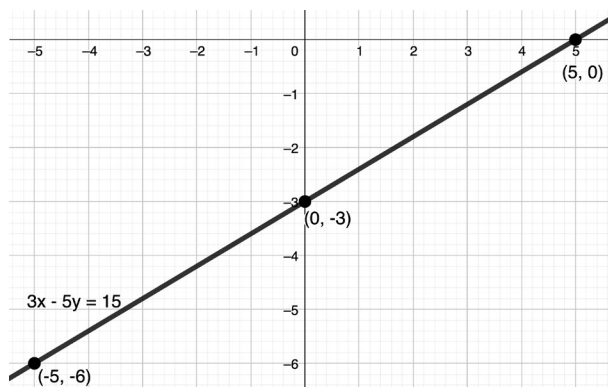
7.



8.



9.



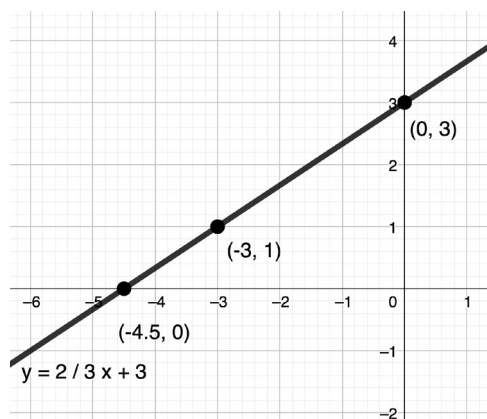
10. x -intercept: $(-3, 0)$ and y -intercept: $(0, 2)$

11. x -intercept: $(3, 0)$ and y -intercept: $(0, 4)$

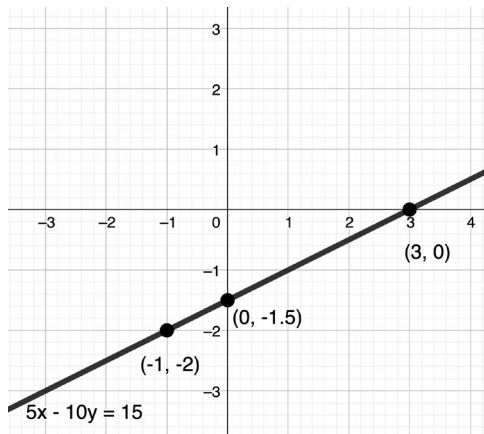
12. x -intercept: $(\frac{25}{2}, 0)$ and y -intercept: $(0, -5)$

13. x -intercept: $(-12, 0)$ and y -intercept: $(0, -9)$

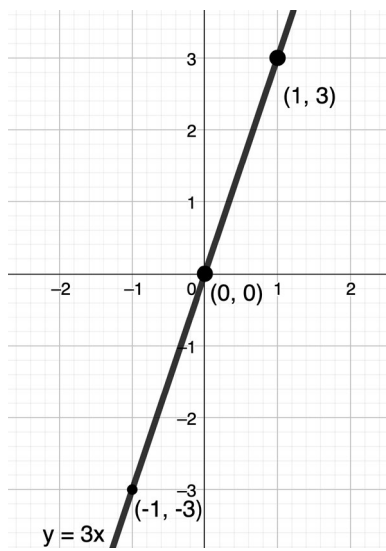
14.



15.



16.



3.2 Slope of a Line

1. $m = -\frac{4}{5}$

2. $m = \frac{4}{7}$

3. $m = \frac{3}{2}$

4. $m = -1$

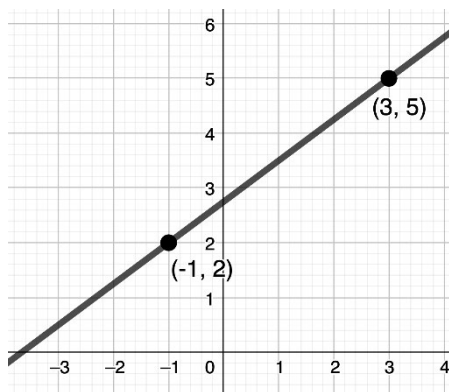
5. $m = -\frac{5}{6}$

6. $m = \text{undefined}$

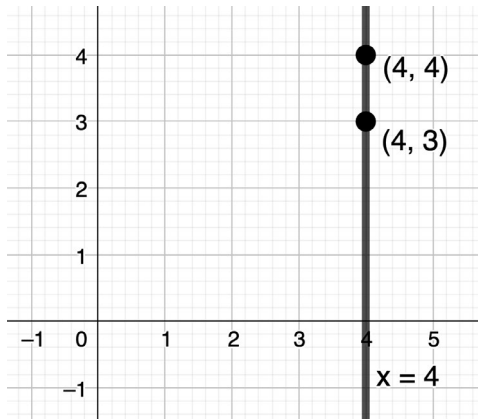
7. $m = -2$

8. $m = 0$

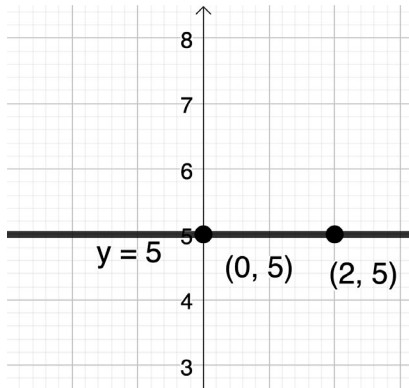
9.



10.



11.

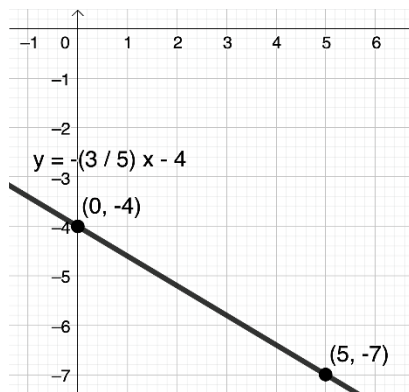


12. $m = 0$; y-intercept: $(0, 4)$

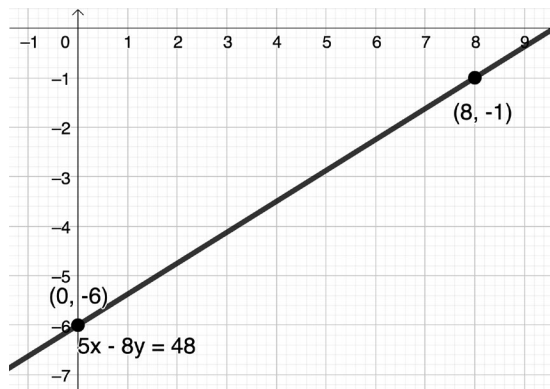
13. $m = -\frac{8}{7}$; y-intercept: $(0, 2)$

14. $m = 2$; y-intercept: $(0, -8)$

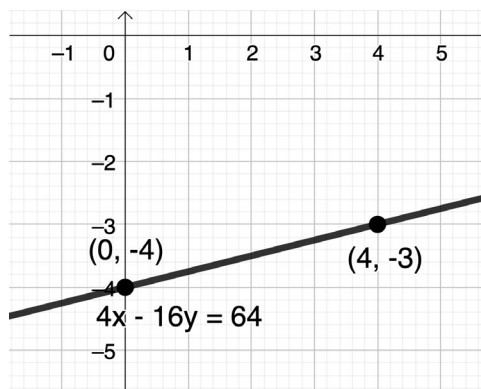
15.



16.



17.



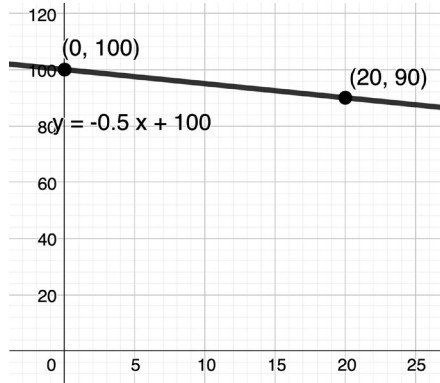
18. There is only one variable, x , the graph is a vertical line crossing the x -axis at -7 .

19. Since this equation is in $y = mx + b$ form, it will be easiest to graph this line by using the slope and y -intercepts.

20. This equation is the form $Ax + By = C$. The easiest way to graph it will be to find the intercepts and one more point.

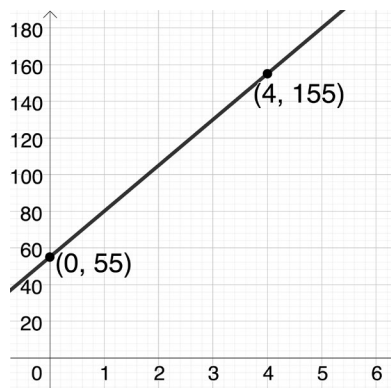
21.

- a) The slope of the temperature change is -0.5 . This represents the decrease in temperature per year.
- b) The initial temperature when $x = 0$, $T = 100$ degrees Celsius.
- c) To find the temperature after 8 years, $T = 96$ degrees Celsius.
- d)



22.

- a) The slope of the rental cost per day is \$25.
- b) The initial rental cost for renting the car for one day is \$80.
- c) The rental cost for renting the car for four days is \$155.
- d)



23. Per

24. Perpendicular

25. Parallel

26. Perpendicular

27. Neither

28. Parallel

29. Perpendicular

3.3 Find the Equation of a Line

1. $y = -\frac{3}{7}x - 3$

2. $y = 3x + 5$

3. $y = 6x - 6$

4. $y = -\frac{4}{7}x + 5$

5. $y = 3x - 18$

6. $y = -\frac{6}{7}x + 15$

7. $y = 9$

8. $y = \frac{4}{5}x + \frac{28}{5}$

9. $y = \frac{1}{2}x - 1$

10. $y = -\frac{2}{3}x + 3$

11. $x = 5$

12. $y = 2$

13. $y = -\frac{3}{4}x$

14. $y = \frac{5}{6}x + \frac{11}{6}$

15. $y = -\frac{3}{2}x + 5$

16. $y = \frac{1}{2}x + 2$

17. $y = \frac{3}{4}x + \frac{17}{2}$

18. $y = -\frac{5}{2}x + 4$

19. $y = -\frac{1}{2}x + 7$

20. $y = \frac{1}{3}x + \frac{13}{3}$

21. $y = -\frac{1}{4}x - \frac{5}{4}$

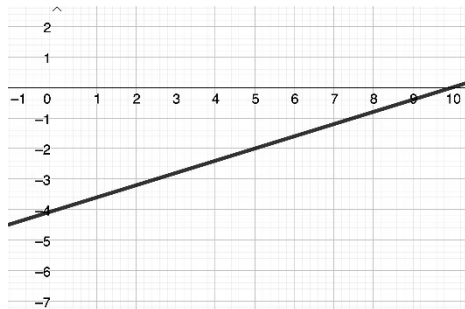
22. $y = -\frac{2}{5}x - \frac{13}{5}$

3.5 Relations and Functions

1. a) The domain of the relation is $\{4, 5, 7, 9, 11\}$
2. The range of the relation is $\{-4, 0, 8, 10\}$
3. a) The domain of the relation is $\{-5, -2, 0, 3\}$
4. The range of the relation is $\{-2, 1, 6, 8, 9\}$
5. a) The relation is not a function.
6. The domain of the relation is $\{-6, -3, 0, 4\}$
7. The range of the relation is $\{-2, -1, 0, 3, 5\}$
8. a) The relation is a function.
9. The domain of the relation is $\{-8, -5, 4, 6, 8\}$
10. The range of the relation is $\{0, 4, 5, 7\}$
11. Function
12. Not a Function
13. Function
14. Function
15. Not a Function
16. Function
17. a) $f(-3) = 13$, b) $f(s) = -2s + 7$, c) $f(s - 1) = -2s + 9$
18. a) $f(-3) = -5$, b) $f(s) = \frac{2}{3}s - 3$, c) $f(s - 1) = \frac{2}{3}s - \frac{11}{3}$
19. a) $f(-3) = 13$, b) $f(s) = s^2 - 2s - 2$, c) $f(s - 1) = s^2 - 4s + 1$
20. a) $f(-3) = -46$, b) $f(s) = -3s^2 + 5s - 4$, c) $f(s - 1) = -3s^2 + 11s - 12$
21. a) The independent variable is x , which represents the number of books sold. The dependent variable is $R(x)$, which represents the total monthly revenue.
22. $R(0) = 500$. This means that if no books are sold, the bookstore still earns \$500 in monthly revenue.
23. $R(100) = 1700$. This means that if the bookstore sells 100 books in a month, the total monthly revenue will be \$1700.
24. a) The independent variable is x , which represents the number of cups of coffee sold. The dependent variable is $P(x)$, which represents the total monthly profit.
25. $R(0) = 2500$. This means that if the coffee shop sells no cups of coffee, it incurs a loss of \$2500 for the month.
26. $P(3000) = 8000$. This means that if the coffee shop sells 3000 cups of coffee in a month, it will make a profit of \$8000.

3.6 Graphs of Functions

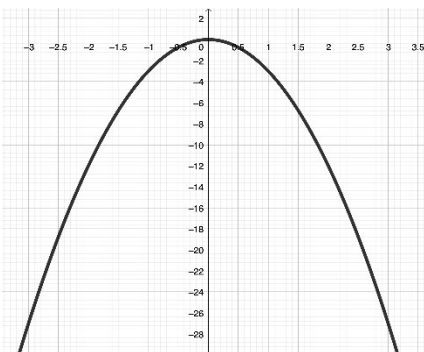
1. Not a Function
2. Function
3. Function
4. Not a Function
- 5.



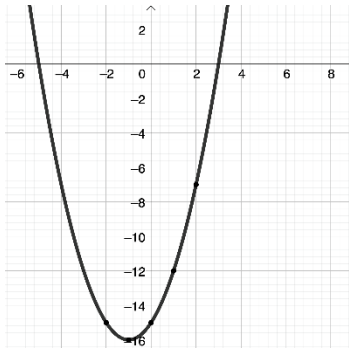
6.



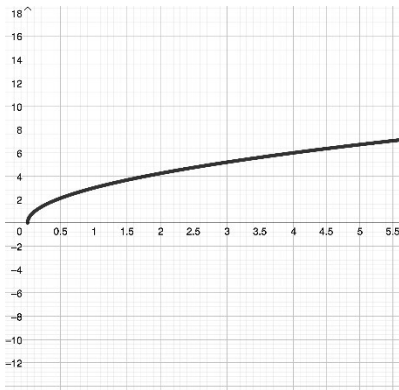
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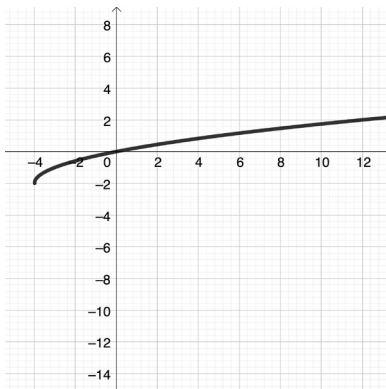
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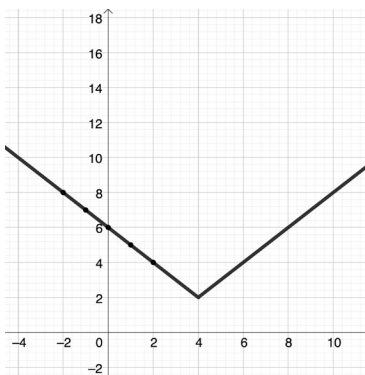
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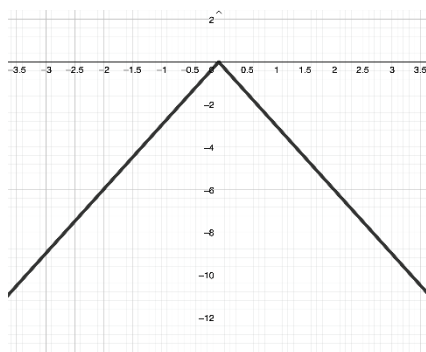


10.

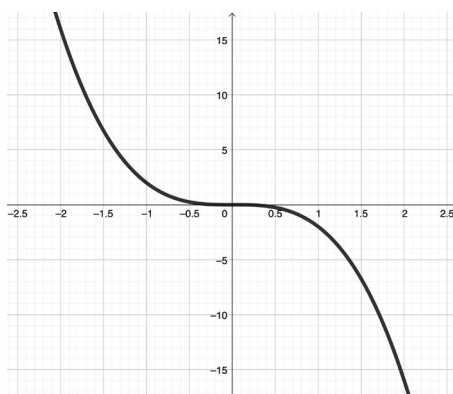


11.

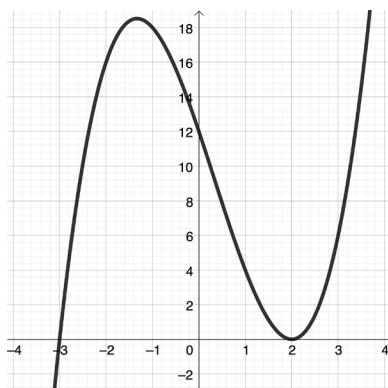




12.



13.



14.

15. When $x = -1$, the function crosses the y -axis at -9 . So, $f(-1) = -9$.

16. When $f(x) = -10$, the function crosses the x -axis at $x = 0$, and $x = 6$. So, $f(0) = -10$ and $f(6) = -10$.

17. The x -intercepts occur when $y = 0$. There are no x -intercepts occur when $f(x) = 0$. Resulted in no x -intercepts.

18. The y -intercept occur when $x = 0$. So the x -intercepts occur when $f(0)$. The y -intercept is $(0, -10)$.
19. This function has a value for all values of x . therefore, the domain in interval notation is $[-2, 8]$.
20. This function values, or y -values go for all values of x . therefore, the range in interval notation is $[-11, -6]$.
21. When $x = -4$, the function crosses the y -axis at 64. So, $f(-4) = 64$.
22. When $x = 2.36$, the function crosses the y -axis at 41.49. So, $f(2.36) = 41.49$.
23. The x -intercepts occur when $y = 0$. So the x -intercepts occur when $f(x) = 0$. The x -intercepts are $(-3, 0)$, $(0, 0)$, and $(4, 0)$.
24. The y -intercept occur when $x = 0$. So the x -intercepts occur when $f(0)$. The y -intercept is $(0,0)$.
25. This function has a value for all values of x . therefore, the domain in interval notation is $(-\infty, \infty)$.
26. This function values, or y -values go for all values of x . therefore, the range in interval notation is $(-\infty, \infty)$.

Chapter 4

4.1 Solve Systems of Linear Equations with Two Variables

1. Yes
2. No
3. No
4. $(-3, -4)$
5. $(3, -2)$
6. No Soln
7. $(3, 2)$
8. $(-1, 2)$
9. $x, 2x+2$
10. $(3, -4)$
11. 194, 172
12. $(4, -2)$

4.2 Solve Applications with Systems of Equations

1. 8 and 4
2. 10 and 30
3. 3 for Panucci, 2 for Cygnoid
4. 36 Water, 28 Soda
5. 18 Width, 22 Length
6. 12mph for the Boat, 9mph for the River
7. Steve drives 7 hours 20 minutes, Josh drives 3 hours 40 minutes
8. 180 feet not parallel to the river, and 190 feet parallel to the river

Chapter 5

5.2 Properties of Exponents and Scientific Notation

1. $8x^9$

2. $\frac{1}{y^3}$

3. $\frac{a^5}{b^4}$

4. -1

5. $\frac{125}{8}$

6. $\frac{5y^6}{x^3}$

7. $\frac{m^{38}}{n^{11}}$

8. $\frac{81x^{24}z^{36}}{16y^8}$

9. $\frac{1}{t^9}$

10. $\frac{256x^{20}z^{26}}{9}$

Chapter 6

6.1 Greatest Common Factor and Factor by Grouping

1. $22x^2$

2. $13m^3$

3. $24p^2q^2$

4. $15r^8s^6$

5. $8(7p - 3)$

6. $4a(a^2 + 2ab + 6b^2)$

7. $3mn^2(2mn + 3m^2 + 9n^2)$

8. $7p^2q(5p^2q^2 - 13q + 11p)$

9. $(2x - 3)(5x - 4)$

10. $(3r + 2s)(8r^2 - 9r)$

11. $(x + 1)(x^2 + 1)$

12. $(x + 4)(x - 1)$

13. $(x - 3y)(4x + 3)$

14. $5(x - 2y)(x + 3)$

15. $(6p - 1)(7p - 3q)$

16. $(m^2 - 2)(4 + n)$

17. $(4a - 5)(5a - 4)$

18. $(7p^2 - 11q)(8q - 5)$

6.2 Factor Trinomials

1. $(a + 5)(a + 2)$
2. Prime
3. $(y + 14)(y - 1)$
4. $(r - 11)(r - 13)$
5. $2(p - 7)(p - 7)$
6. $(x - 5y)(x + 8y)$
7. $(r - 12s)(r + 15s)$
8. $-3x(x + 7)(x - 9)$
9. $5a(a + 4)(a - 6)$
10. $(x + 4)(2x + 1)$
11. $(a - 2)(5a - 4)$
12. $(2x + 3)(5x - 8)$
13. $(4a - 7)(5a - 8)$
14. $(-x + 3)(3x - 5)$
15. $(6r - 7)(3r - 2)$
16. $-6(y + 1)(7y - 1)$
17. $2(4x - 3)(4x - 3)$
18. $-2(3p + 2)(5p - 6)$
19. $(x^2 - 3)(x^2 + 9)$
20. $(x^2 + 2)(x^2 + 6)$
21. $(x^2 + 4)(3x^2 + 2)$
22. $(x - 9)(x - 10)$
23. $(x + 11)(x + 16)$
24. $2(x + 1)(3x + 7)$

6.3 Factor Special Products

1. $(3x + 8)^2$
2. $(2x - 7)^2$
3. $2(p^2 + 9q^6)^2$
4. $3a(10a - 11b)^2$
5. $(4x^2 + 3y)^2$
6. $4(5x - y^3)^2$
7. $\left(2x - \frac{1}{3}\right)\left(2x + \frac{1}{3}\right)$
8. $(12x^4 - 1)(12x^4 + 1)$
9. Not a difference of two square, not factorable
10. $\left(8p^2 - \frac{1}{4}q\right)\left(8p^2 + \frac{1}{4}q\right)$
11. $3x(4x^2 - 27)$
12. $3y(x - 2y)(x + 2y)$
13. $-2x(6x - 13y^2)(6x + 13y^2)$
14. $(-5x^2 + x + y)(5x^2 + x + y)$
15. $(a - 6b - 4)(a + 6b - 4)$
16. $(x + 10)(x^2 - 10x + 100)$
17. $(4 - 3a)(16 + 12a + 9a^2)$
18. $\left(2y^2 - \frac{1}{3}\right)\left(4y^4 + \frac{2}{3}y^2 + \frac{1}{9}\right)$
19. $2(5p + 1)(25p^2 - 5p + 1)$
20. $q\left(3p + \frac{1}{4}\right)\left(9p^2 - \frac{3}{4}p + \frac{1}{16}\right)$
21. $8y^2(x - 3)(x^2 + 3x + 9)$
22. $2q(8 + p^3)(64 - 3p^3 + p^6)$
23. $(8 - x)(x^2 + 11x + 91)$
24. $(x - 2 - 3y)(x^2 - 4x + 4 + 3xy - 6y + 9y^2)$

6.4 General Strategy for Factoring Polynomials

1. $3x(x^2 - 9x + 27)$

2. $5a^3(a^2 - 5)$

3. $8r(r - 3)(r - 6)$

4. $24y(y - 3)^2$

5. $15xy^2(x^2 + 4)(x - 2)(x + 2)$

6. $6(s - 1)(5r + 4)$

7. $(9u + 12)(2v - 1)$

8. $2(3x - 5y)^2$

9. $12p^2(p + q)(p^2 - pq + q^2)$

10. $3xy(2x^3 + 5y^3)$

11. $(7a + 5b)^2$

12. $3(2x^2 - 3y)(4x^4 + 6x^2y + 9y^2)$

13. $(x^2 - 10)(x^2 + 10)$

14. $9s(5r - 2)(r - 1)$

15. $18y(4x^2 - 5x + 2)$

16. $(5x - 3y - 6)(5x - 3y + 6)$

17. $28q(2p - 1)(p - 1)$

18. $(6p - 4q - 9)(6p - 4q + 9)$

19. $(2z + 1)^2$

20. $(2x^2 + 3)(2x^2 + 4)$

21. $(3r^2 + 5)(2r^2 + 3)$

22. $(2y + 13)^2$

6.5 Polynomial Equations

1. $b = 3$ and $b = -\frac{5}{2}$
2. $x = -2$, $x = \frac{8}{3}$, and $x = -\frac{6}{5}$
3. $y = 0$ and $y = \frac{7}{6}$
4. $x = 0$, $x = -\frac{9}{2}$, and $x = \frac{5}{3}$
5. $x = 2$, and $x = 8$
6. $x = 3$, and $x = 5$
7. $x = -5$, and $x = \frac{1}{2}$
8. $s = -\frac{7}{4}$, and $s = \frac{7}{4}$
9. $r = \frac{7}{2}$
10. $x = 4$, and $x = -3$
11. $a = -2$, and $a = \frac{5}{6}$
12. $m = -1$, and $m = 3$
13. $n = \frac{4}{3}$, and $n = \frac{1}{2}$
14. $x = 0$, $x = \frac{1}{2}$, and $x = 1$
15.
 - a. $x = 1$, $x = 8$
 - b. $(1, 2)$ and $(8, 2)$
16.
 - a. $x = \frac{3}{2}$
 - b. $(\frac{3}{2}, 1)$
17.
 - a. The zeros of the function $f(x)$ are $x = \frac{1}{2}$ and $x = -3$
 - b. x -intercepts are $(\frac{1}{2}, 0)$ and $(-3, 0)$
- c. y -intercept is $(0, -3)$
18.
 - a. The zeros of the function $f(x)$ are $x = -4$ and $x = 4$
 - b. x -intercepts are $(-4, 0)$ and $(4, 0)$
 - c. y -intercept is $(0, -16)$
19.
 - a. The zeros of the function $f(x)$ are $x = -\frac{2}{5}$
 - b. x -intercepts are $(-\frac{2}{5}, 0)$
 - c. y -intercept is $(0, 4)$
20.
 - a. The zeros of the function $f(x)$ are $x = 1$ and $x = \frac{1}{2}$
 - b. x -intercepts are $(1, 0)$ and $(\frac{1}{2}, 0)$
 - c. y -intercept is $(0, -1)$
21. The lengths of the legs of the triangular garden bed are 9 meters and 12 meters.
22. The height at which the ladder touches the building is 20 feet.
23. The width of the room is 7.5 feet, and the length is 20 feet.
24.
 - a. The ball lands on the playground at $t = 2.5$ seconds
 - b. The ball is at a height of 50 feet at $t = 1.25$ seconds
 - c. At $t = 1$ second the ball is at a height of 54 feet.

Chapter 7

7.1 Multiply and Divide Rational Expressions

1. $x = 9$

2. $x \neq -4$ and $x \neq 3$

3. $-\frac{3x}{3x+1}$

4. $\frac{2x+1}{x+1}$

5. $\frac{2x^2(x+2)}{x+4}$

6. $\frac{1}{x}$

7. $\frac{(x-1)(x-6)}{(x+2)(x-4)}$

8. $5x$

9. $\frac{1}{3(x-4)}$

10. $\frac{x^5y^{11}}{5}$

7.2 Add and Subtract Rational Expressions

1. $\frac{3x-2}{x^2-4}$
2. $\frac{3x-5}{x^2-4}$
3. $\frac{x+5}{x^2-4}$
4. $\frac{x+5}{x^2-4}$
5. $\frac{x+3(x+2)(x-2)}{x^2-4}$
6. $\frac{5x+2(x+4)(x-2)}{x^2-4}$
7. $\frac{x-9(x+3)(x-5)}{x^2-4}$
8. $\frac{x-5x-1}{x^2-4}$
9. $\frac{-x-8(x+1)(x-3)}{x^2-4}$
10. $\frac{x-9(x+3)(x-5)}{x^2-4}$

7.3 Simplify Complex Rational Expressions

1. $\frac{y-x}{y+x}$

2. $x - 5$

3. $\frac{x-2}{5}$

4. $-\frac{6}{5}$

5. $\frac{x^2+8x-3}{-x^2-5}$

6. $\frac{3}{3x-50}$

7. $\frac{y^3-9y+1}{y-3}$

8. $\frac{16}{b-1}$

7.4 Solve Rational Equations

1. $x = -15$
2. $x = -2$ or $x = 1$
3. No solution
4. $x = 2$
5. $x = 5$ or $x = -2$
6. $x = \frac{10}{3}$
7. $x = \frac{17}{4}$
8. (a) all real numbers except $x \neq 5, x \neq 1$, (b) $x = \frac{21}{4}$, (c) $\left(\frac{21}{4}, 4\right)$

Chapter 9

9.1 Solve Quadratic Equations Using the Square Root Property

1. $x = 6, x = -6$
2. $x = 5\sqrt{3}, x = -5\sqrt{3}$
3. $x = 4i, x = -4i$
4. $x = 2i\sqrt{3}, x = -2i\sqrt{3}$
5. $x = \frac{\sqrt{5}}{5}, x = -\frac{\sqrt{5}}{5}$
6. $x = 3\sqrt{2}, x = -3\sqrt{2}$
7. $x = 9, x = -11$
8. $x = 4 + 2\sqrt{5}, x = 4 - 2\sqrt{5}$
9. $x = -\frac{1}{5} + \frac{\sqrt{3}}{5}, x = -\frac{1}{5} - \frac{\sqrt{3}}{5}$
10. $x = -\frac{3}{2} + i\sqrt{2}, x = -\frac{3}{2} - i\sqrt{2}$

9.2 Solve Quadratic Equations by Completing the Square

1. $x = 4, x = -2$
2. $x = -3 + \sqrt{7}, x = -3 - \sqrt{7}$
3. $x = 5 + \sqrt{13}, x = 5 - \sqrt{13}$
4. $x = 2 + 3\sqrt{2}, x = 2 - 3\sqrt{2}$
5. $x = -1 + i\sqrt{19}, x = -1 - i\sqrt{19}$
6. $x = -\frac{3}{2} + \sqrt{2}, x = -\frac{3}{2} - \sqrt{2}$
7. $x = 6, x = 1$
8. $x = 6 + 2i, x = 6 - 2i$
9. $x = \frac{2}{3} + \frac{\sqrt{13}}{3}, x = \frac{2}{3} - \frac{\sqrt{13}}{3}$
10. $x = 2, x = \frac{2}{3}$

9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 1)

1. $x = 2, x = -7$

2. $x = 1, x = -\frac{2}{3}$

3. $x = -4 + 3\sqrt{2}, x = -4 - 3\sqrt{2}$

4. $x = 2 + 3\sqrt{2}, x = 2 - 3\sqrt{2}$

5. $x = \frac{-5+\sqrt{3}}{2}, x = \frac{-5-\sqrt{3}}{2}$

6. $x = \frac{1+i\sqrt{31}}{4}, x = \frac{1-i\sqrt{31}}{4}$

7. $x = -\frac{1}{5} + \frac{\sqrt{19}}{5}i, x = -\frac{1}{5} - \frac{\sqrt{19}}{5}i$

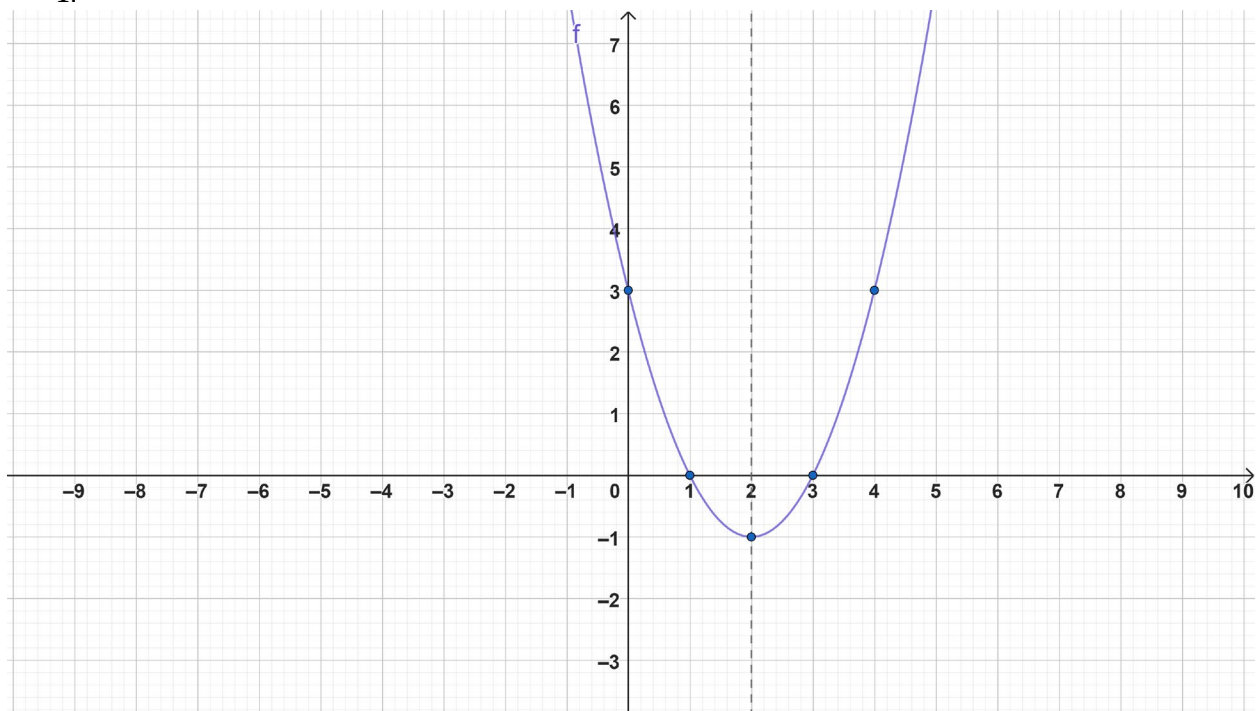
8. $x = -4$

9.3 Solve Quadratic Equations Using the Quadratic Formula (Part 2)

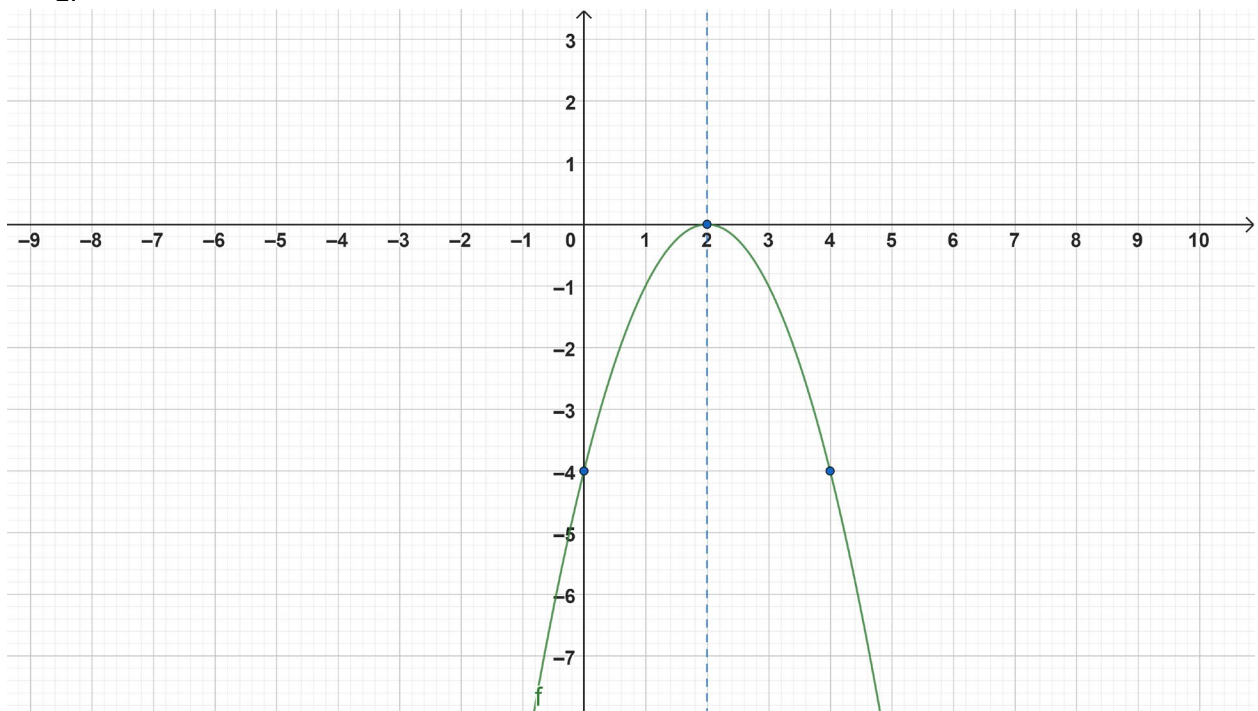
1. Two real solutions
2. One real solution
3. Two complex solutions
4. Two real solutions
5. Two complex solutions
6. One real solution
7. Factoring
8. Square root property
9. Square root property
10. Quadratic formula
11. Factoring
12. Quadratic formula

9.6 Graph Quadratic Equations Using Properties

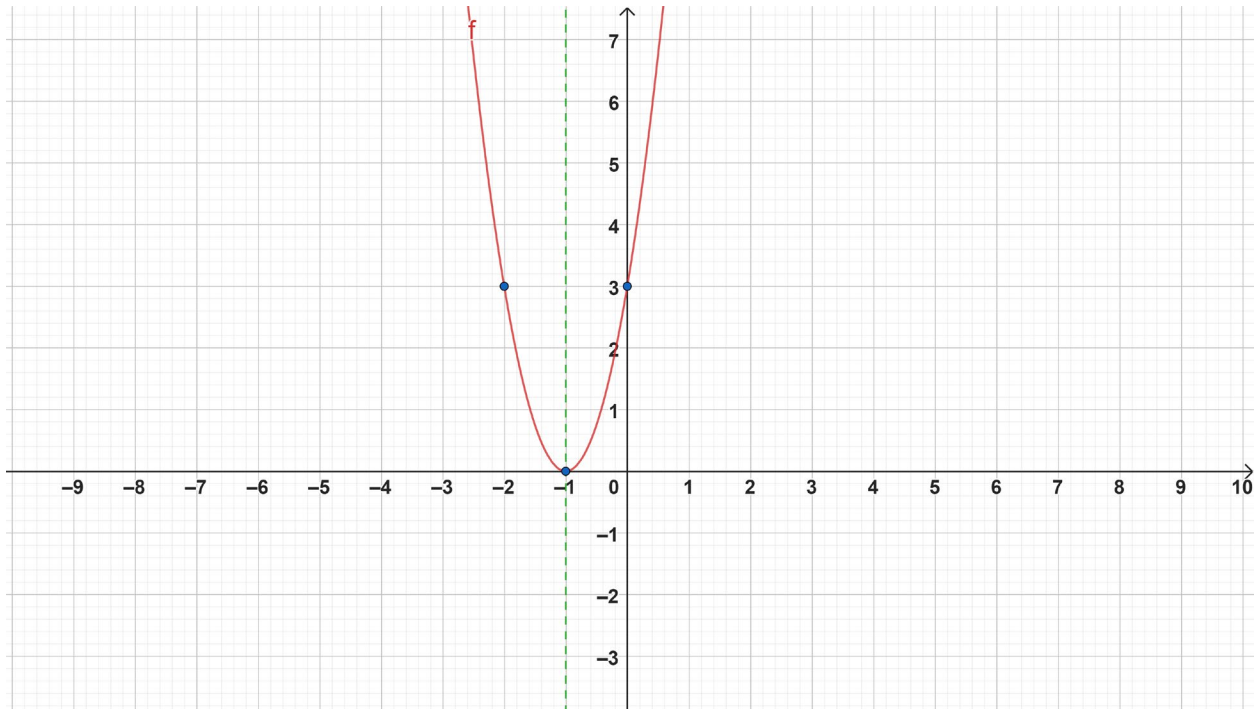
1.



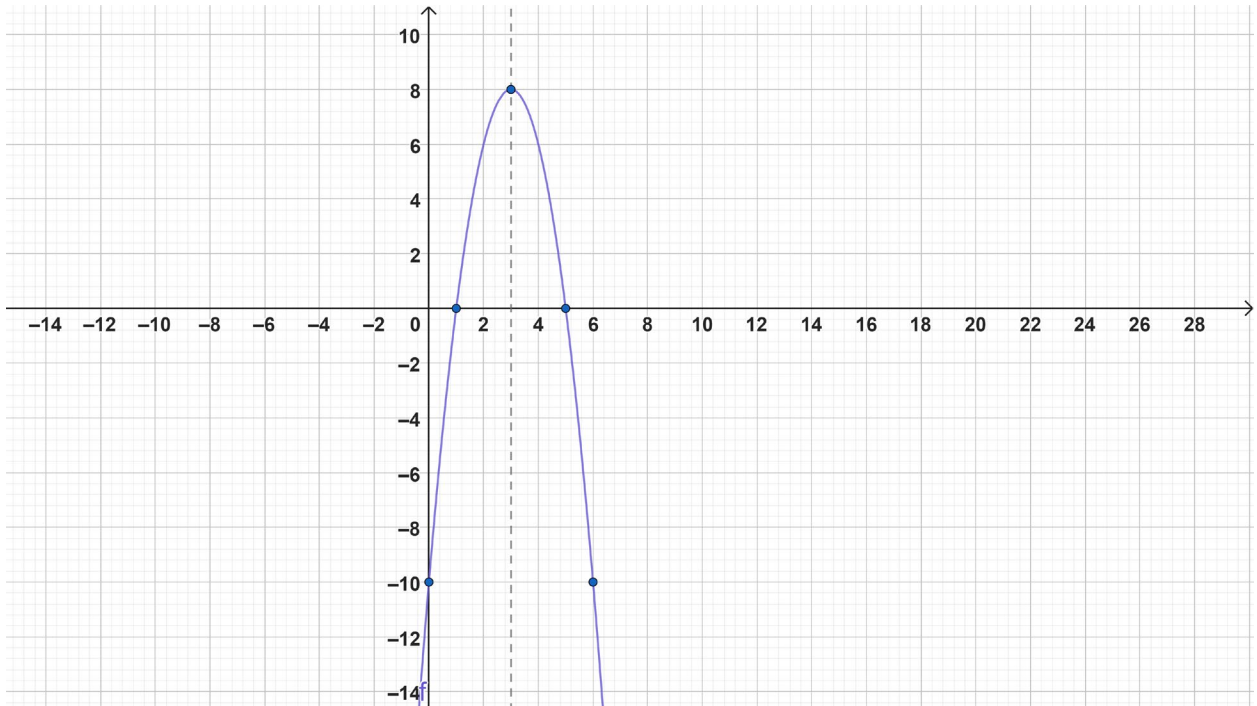
2.



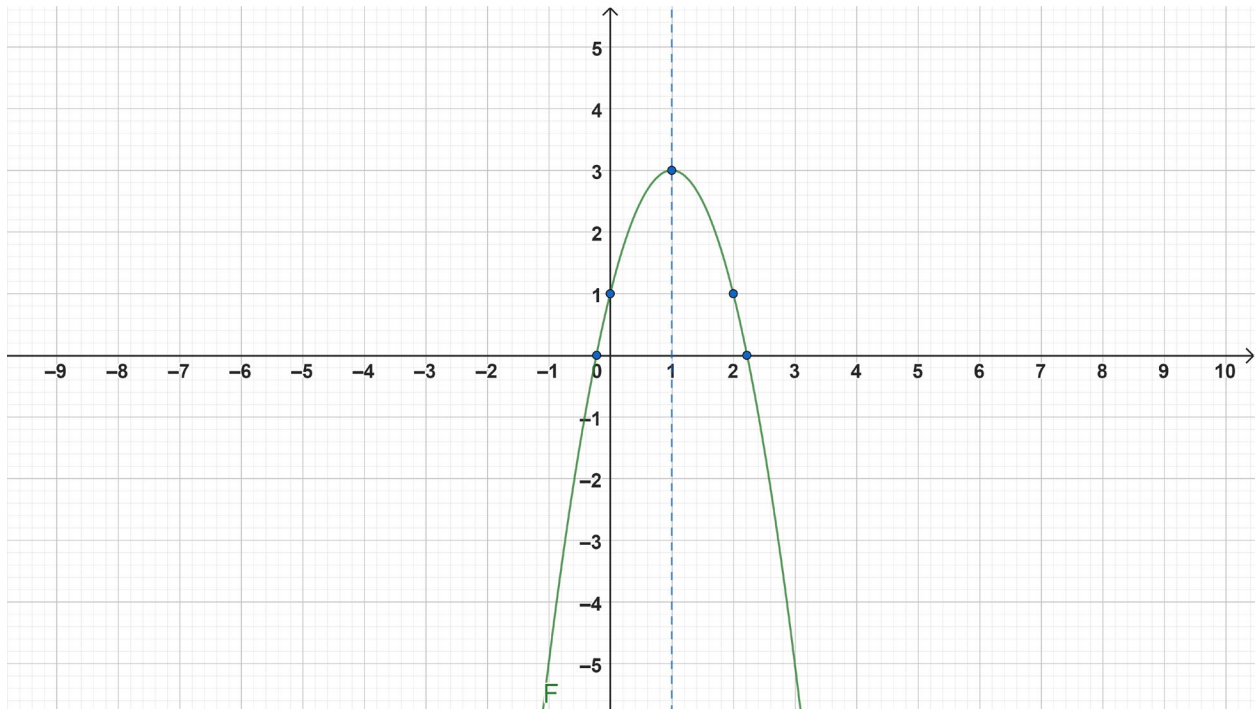
3.



4.



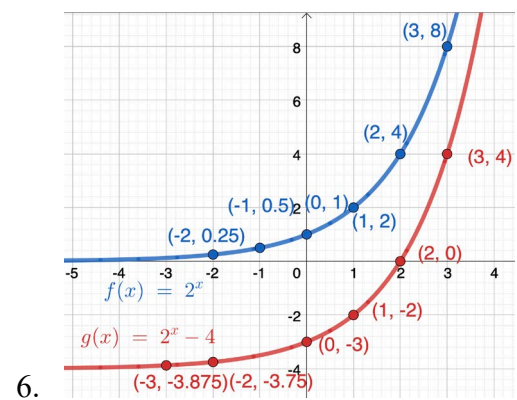
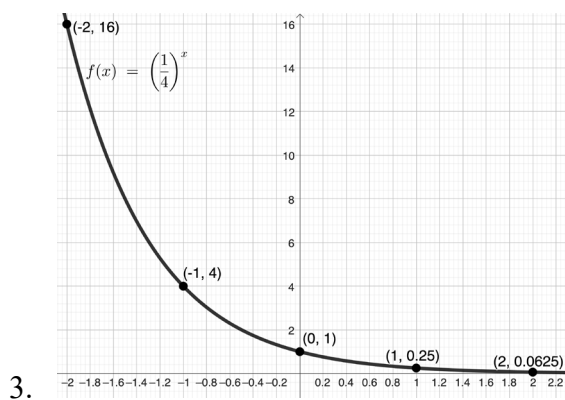
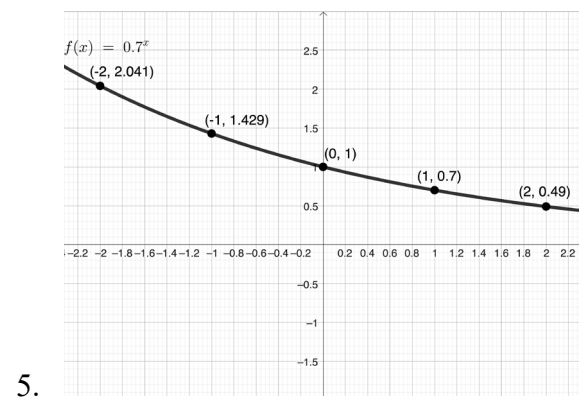
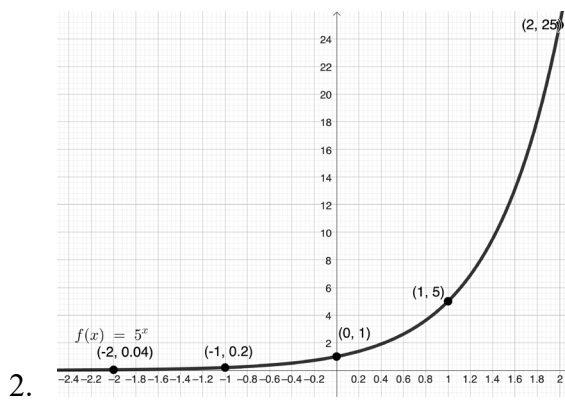
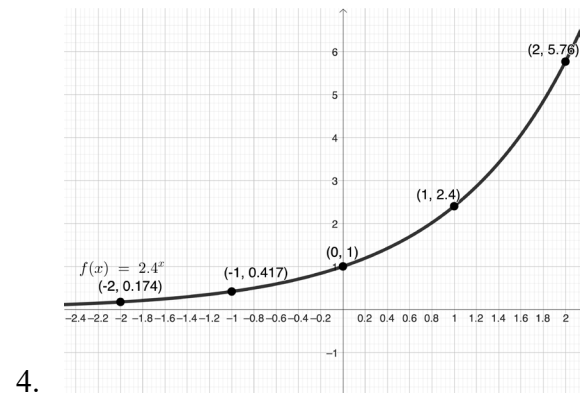
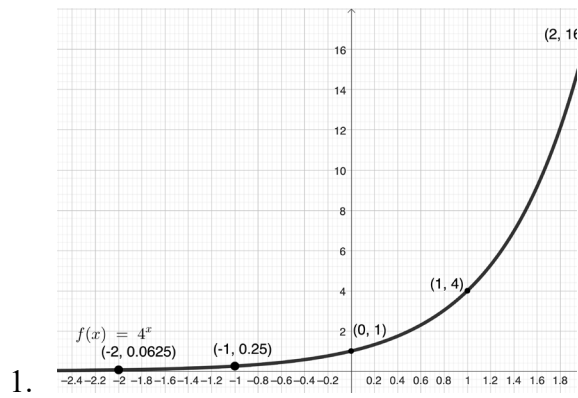
5.

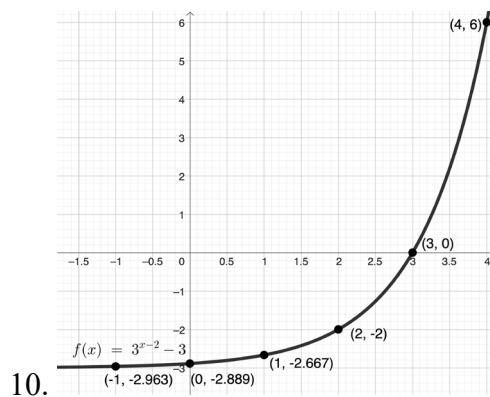
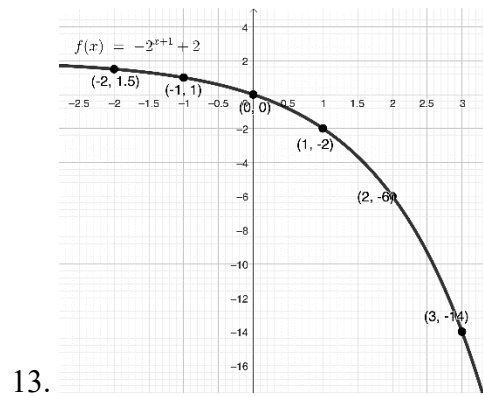
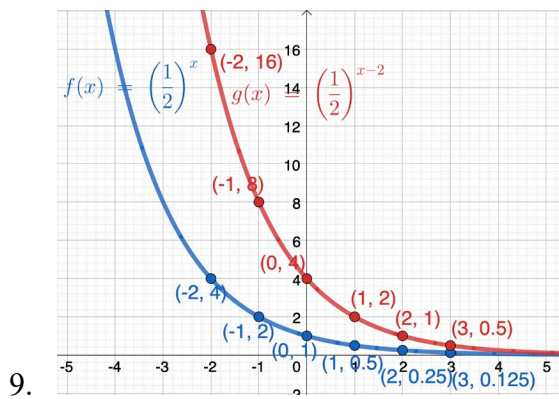
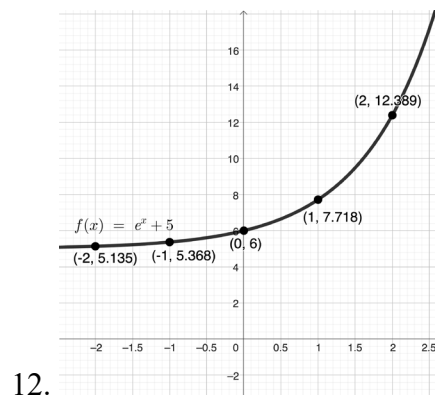
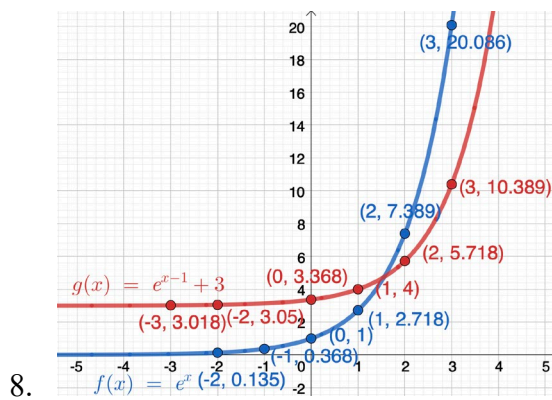
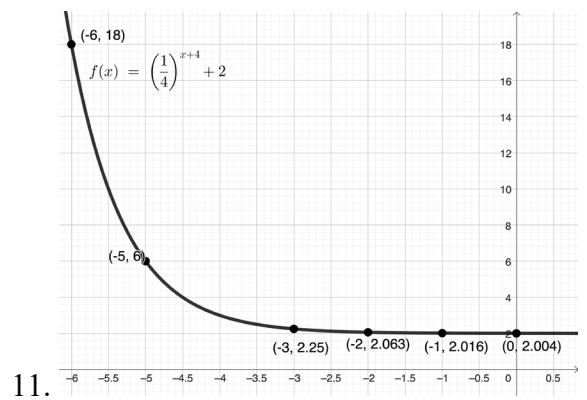
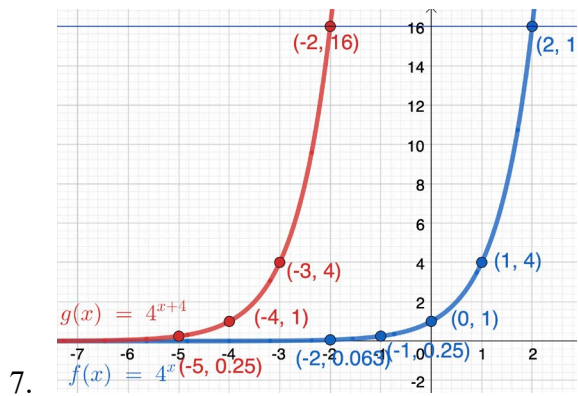


6. The minimum value is -21 when $x = -4$.
7. The maximum value is 54 when $x = 3$.
8. Bobby reached his maximum height of 484 feet after $\frac{1}{2}$ second.
9. The retailer would make a maximum revenue of \$225 if they sell 15 t-shirts.

Chapter 10

10.2 Evaluate and Graph Exponential Functions





10.3 Evaluate and Graph Logarithmic Functions

1. $125 = 3$

2. $\frac{1}{32} = -5$

3. $\sqrt[3]{19} = x$

4. $\sqrt[3]{10} = \frac{1}{3}$

5. $\frac{1}{216} = 3$

6. $x = 4$

7. $4^4 = 256$

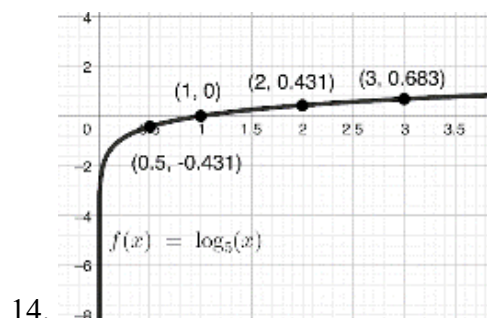
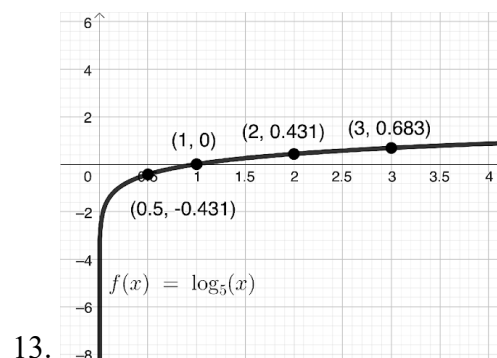
8. $x^7 = 128$

9. $9^0 = 1$

10. $5^2 = 25$

11. $10^{-5} = \frac{1}{100,000}$

12. $e^7 = x$



15.

